Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir
Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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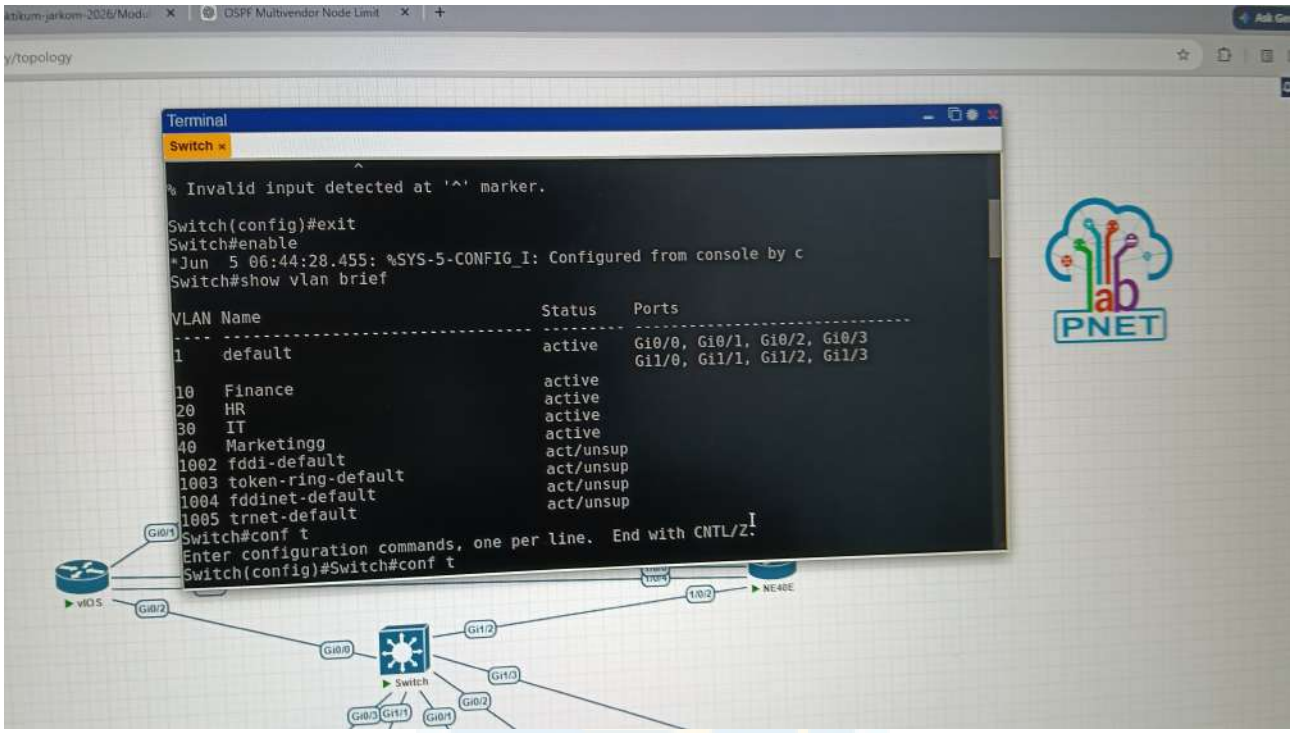
1 Langkah-Langkah Percobaan

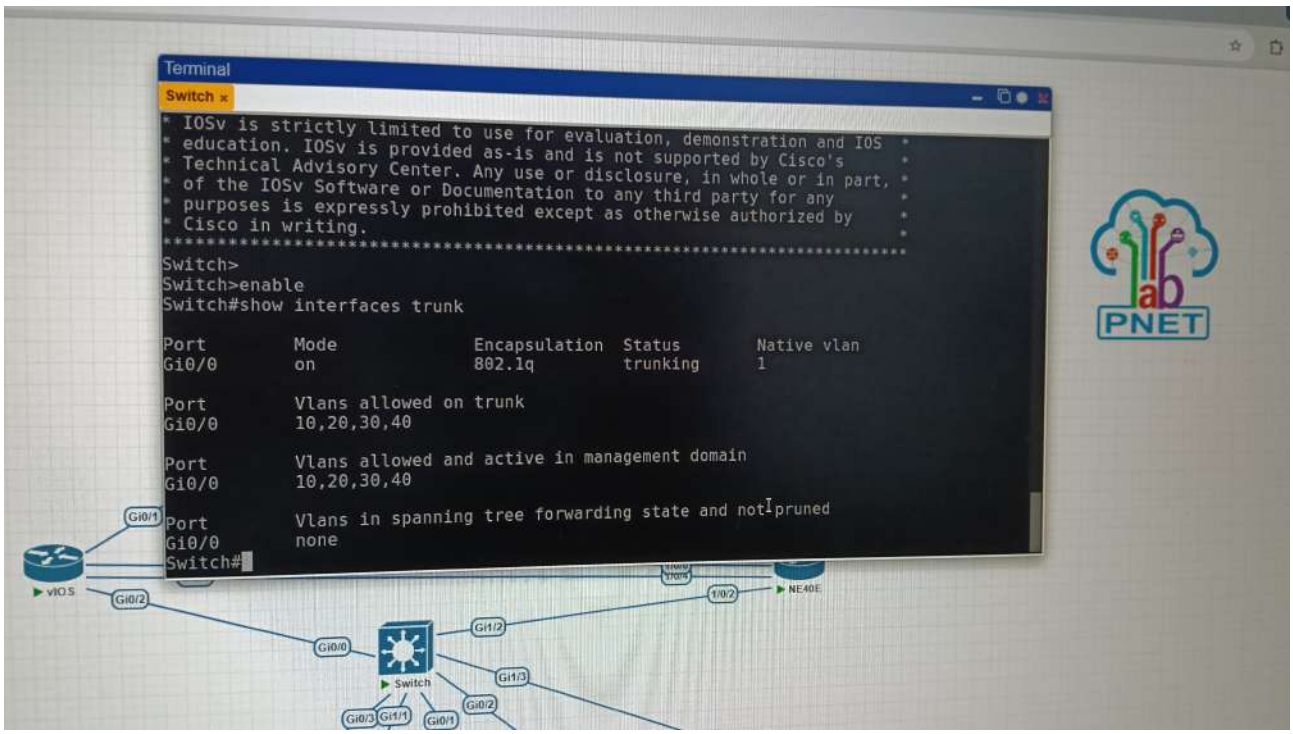
1.1 Praktikum 1: Konfigurasi Cisco Switch

Adapun langkah-langkah yang dilakukan adalah sebagai berikut:

1. Masuk ke mode konfigurasi pada Cisco Switch melalui terminal.
2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing).
3. Memverifikasi VLAN yang telah dibuat menggunakan perintah `show vlan brief`.



4. Mengkonfigurasi port yang terhubung ke PC sebagai access port sesuai VLAN masing-masing.
5. Mengkonfigurasi port yang terhubung ke Cisco Router sebagai trunk port menggunakan encapsulation IEEE 802.1Q.
6. Mengatur VLAN yang diizinkan melewati trunk yaitu VLAN 10, 20, 30, dan 40.
7. Memverifikasi konfigurasi trunk menggunakan perintah `show interfaces trunk`.

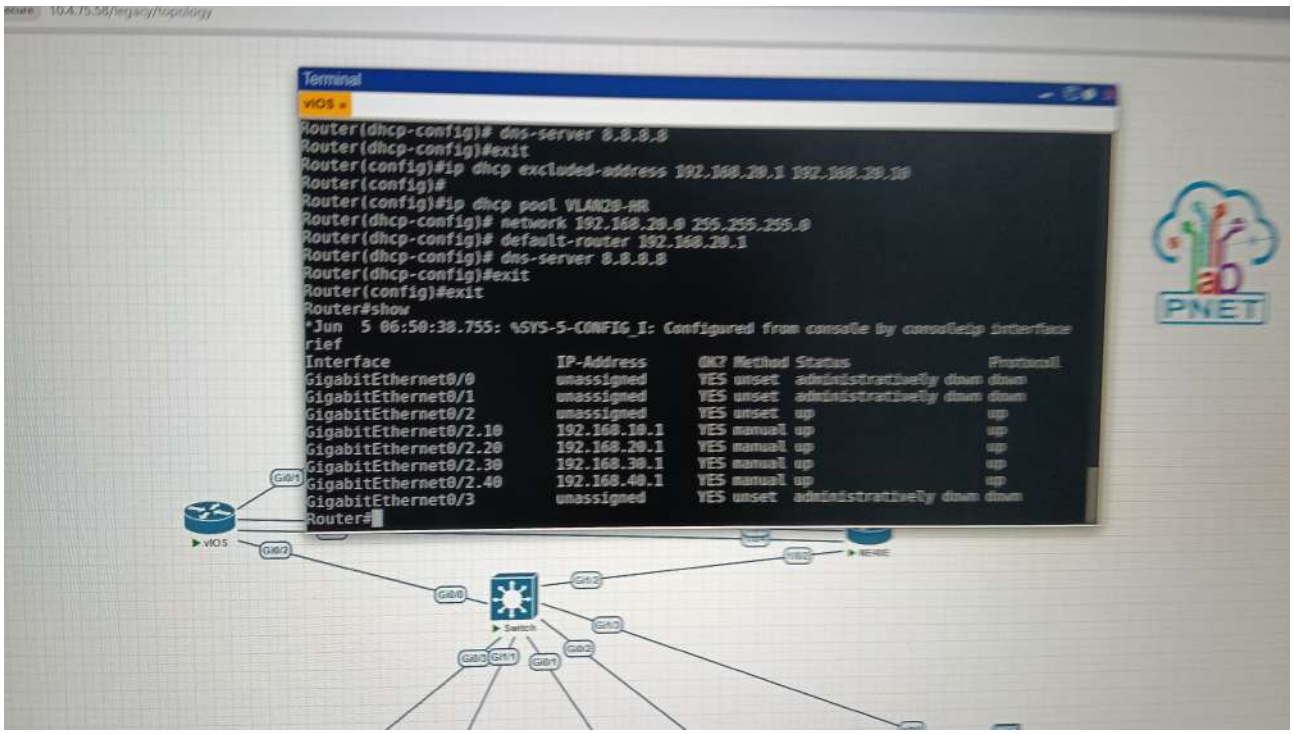


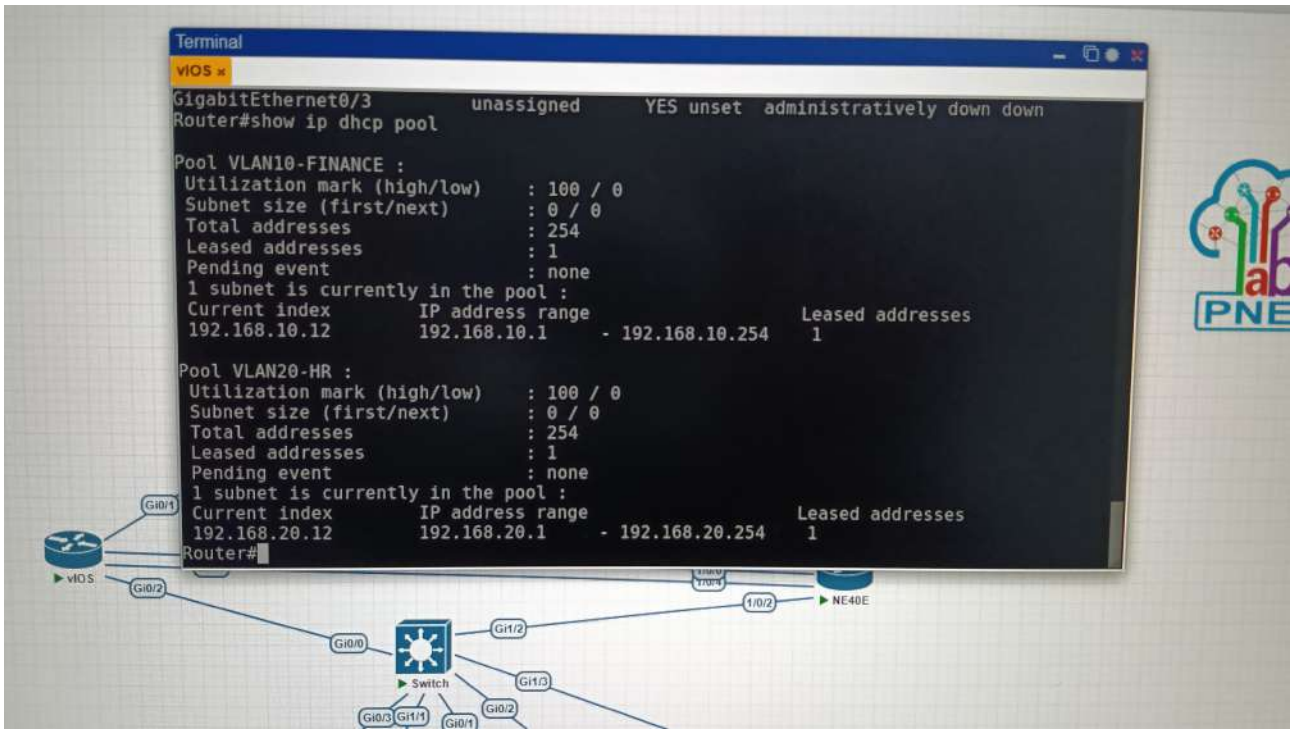
8. Menyimpan konfigurasi menggunakan perintah `write memory`.

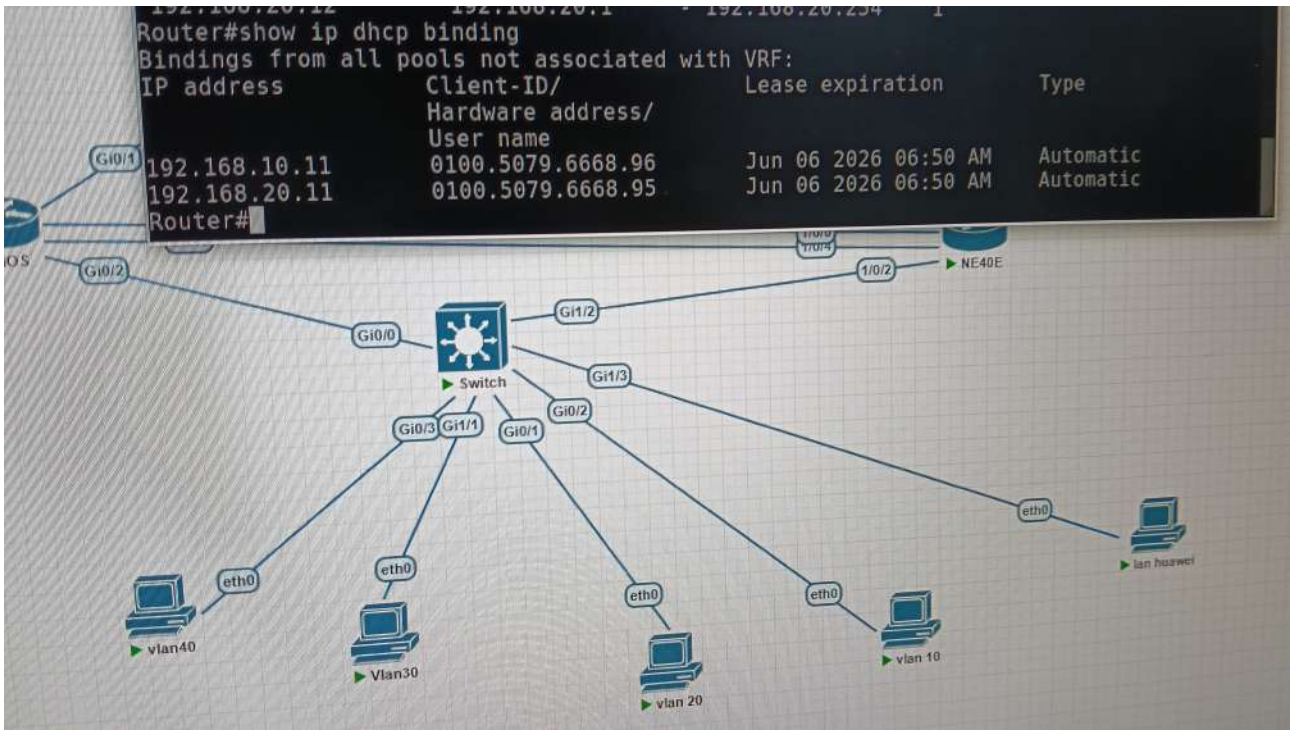
1.2 Praktikum 2: Konfigurasi Cisco Router

Adapun langkah-langkah yang dilakukan adalah sebagai berikut:

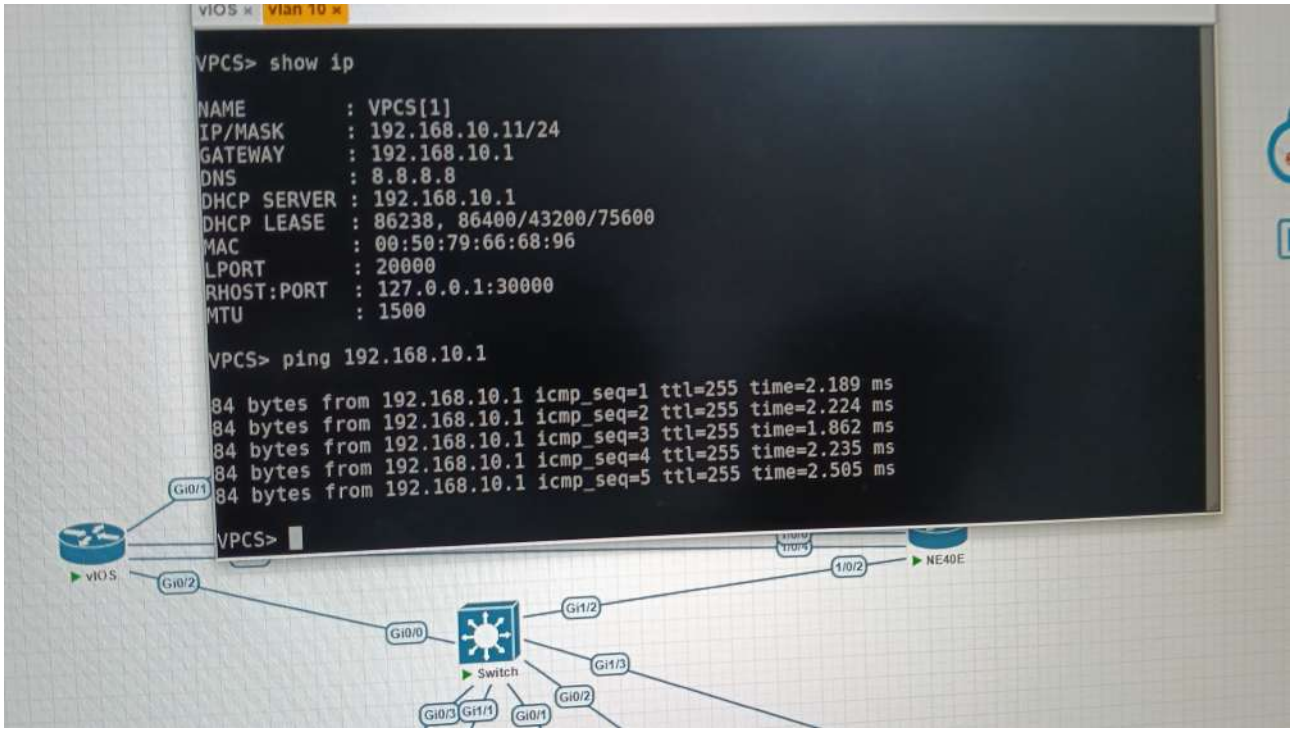
1. Masuk ke mode konfigurasi pada Cisco Router.
2. Mengkonfigurasi interface yang terhubung ke Cisco Switch sebagai trunk.
3. Membuat subinterface VLAN 10 dan memberikan alamat IP gateway 192.168.10.1/24.
4. Membuat subinterface VLAN 20 dan memberikan alamat IP gateway 192.168.20.1/24.
5. Membuat subinterface VLAN 30 dan memberikan alamat IP gateway 192.168.30.1/24.
6. Membuat subinterface VLAN 40 dan memberikan alamat IP gateway 192.168.40.1/24.
7. Mengkonfigurasi DHCP Server untuk VLAN 10.
8. Mengkonfigurasi DHCP Server untuk VLAN 20.
9. Mengatur IP static pada client VLAN 30 dan VLAN 40.
10. Melakukan pengujian DHCP pada VLAN 10 dan VLAN 20.

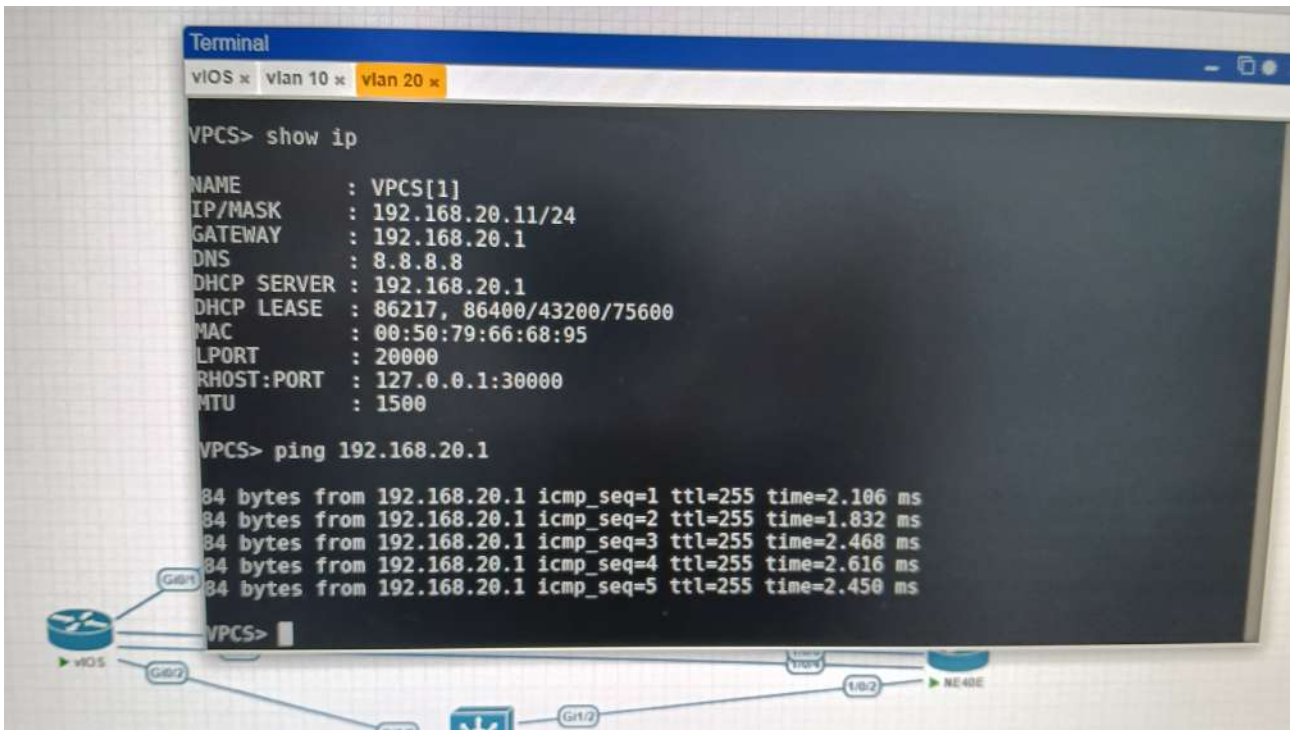

Terminal
vIOS
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#show
*Jun 5 06:50:38.755: %SYS-5-CONFIG_I: Configured from console by consoleip interface
rief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 unassigned YES unset administratively down down
Router#
vIOS
Switch
NE40E

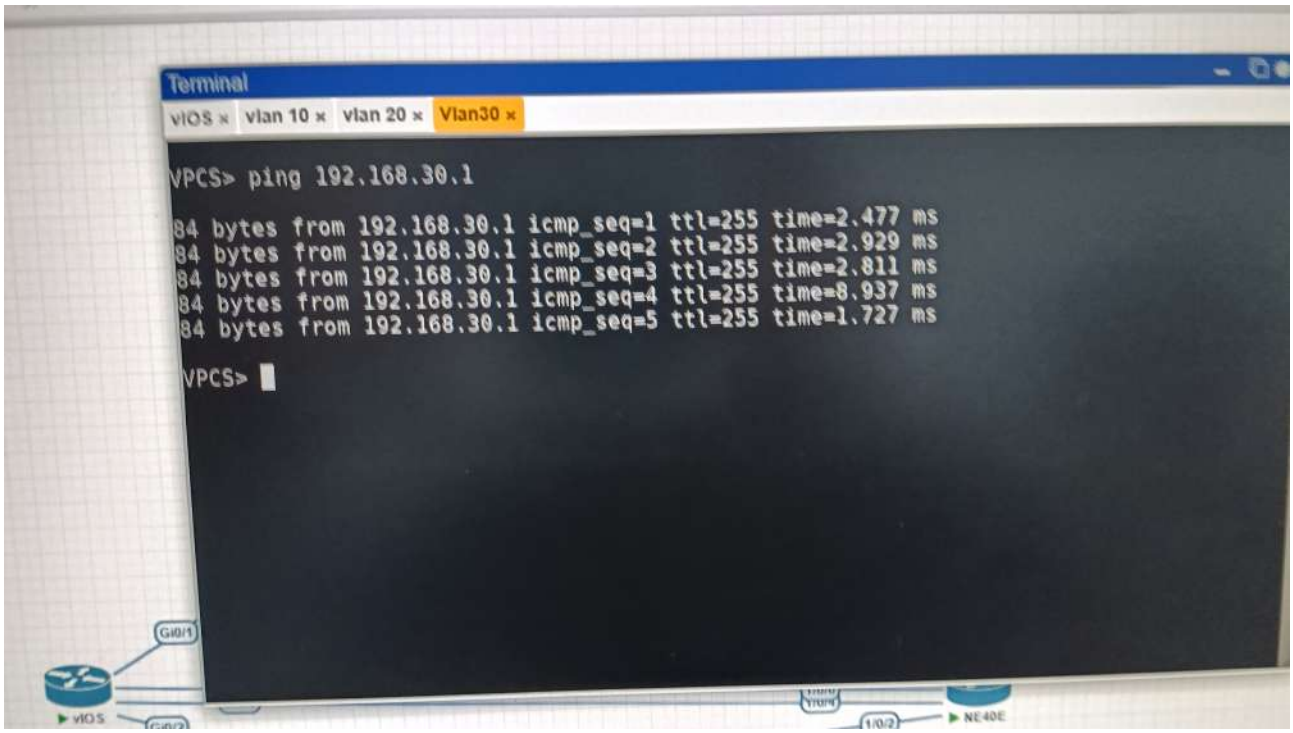

Terminal
vIOS
GigabitEthernet0/3 unassigned YES unset administratively down down
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#
vIOS
NE40E
Switch

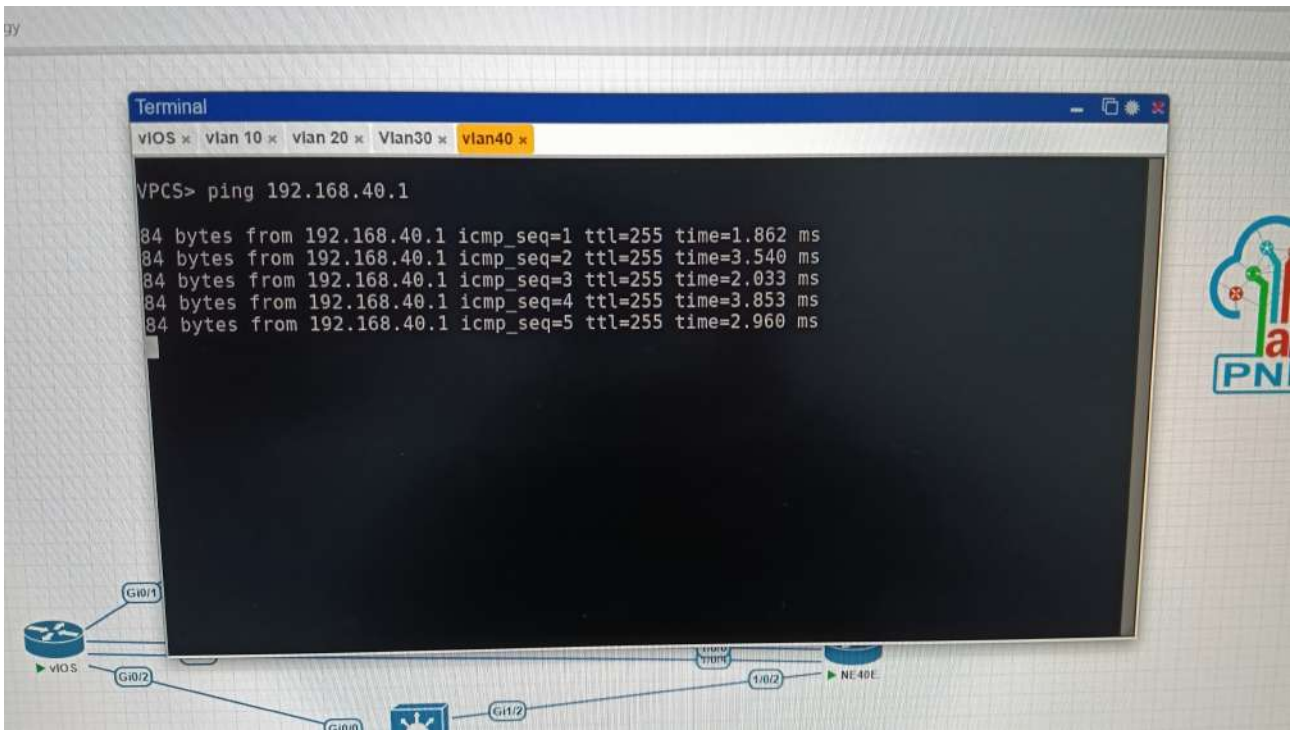


11. Melakukan pengujian konektivitas ke gateway masing-masing VLAN.

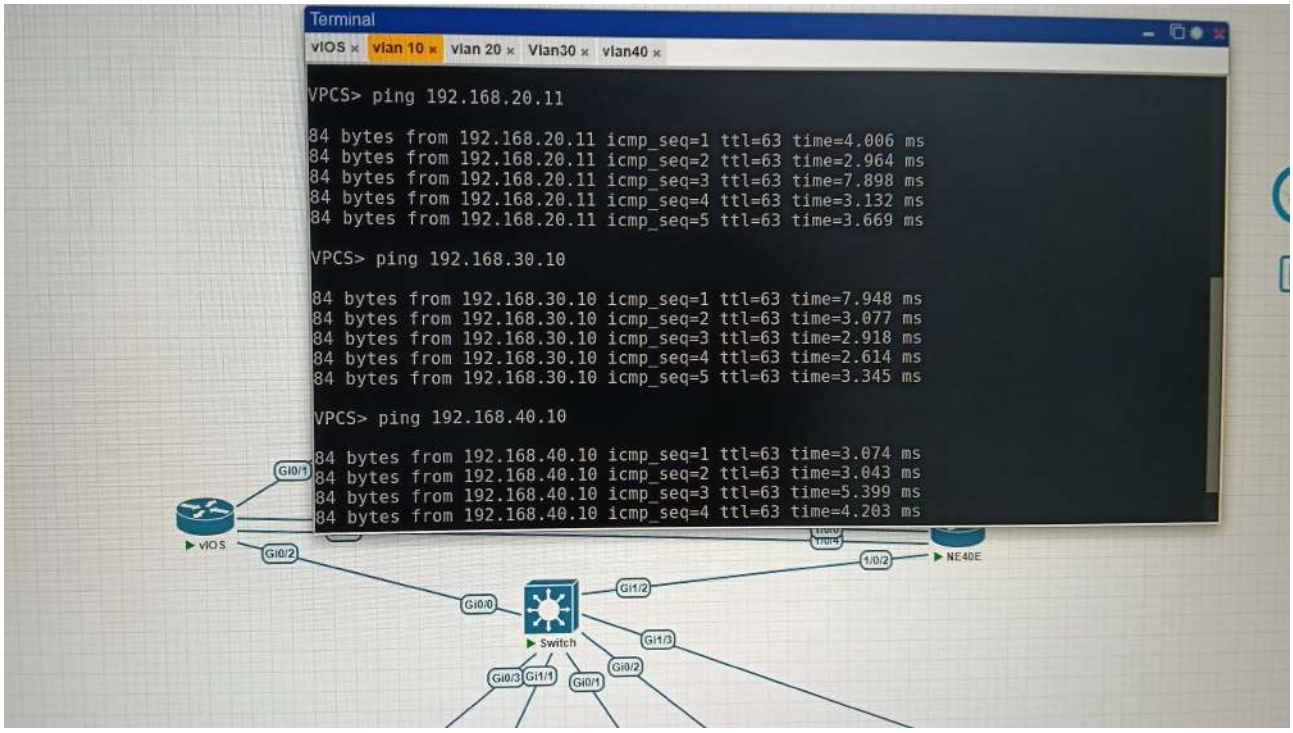



Terminal
vIOS × vlan 10 × vlan 20 ×
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.20.11/24
GATEWAY : 192.168.20.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86217, 86400/43200/75600
MAC : 00:50:79:66:68:95
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.106 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.832 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.468 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.616 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.450 ms
VPCS>
vIOS
NE40E


Terminal
vIOS × vlan 10 × vlan 20 × Vlan30 ×
VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.477 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=2.929 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.811 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=8.937 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.727 ms
VPCS>
vIOS
NE40E



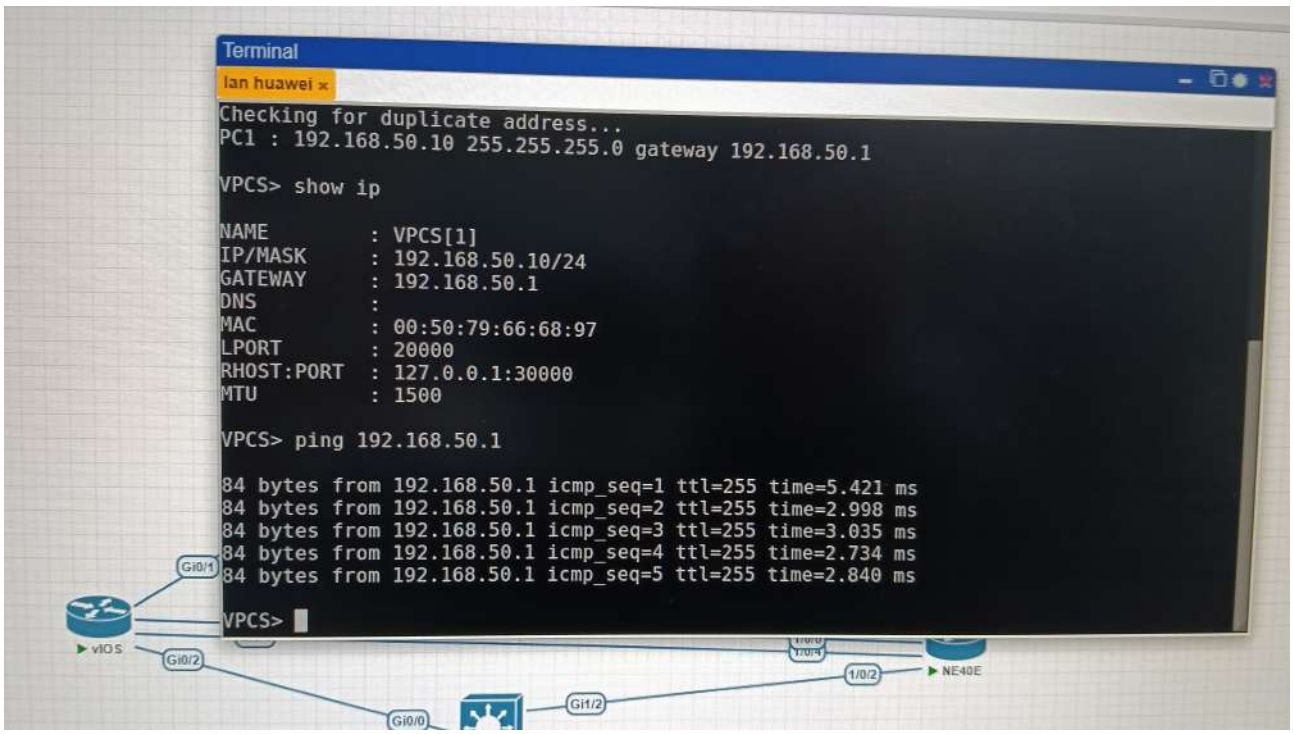
12. Melakukan pengujian komunikasi antar VLAN menggunakan perintah `ping`.



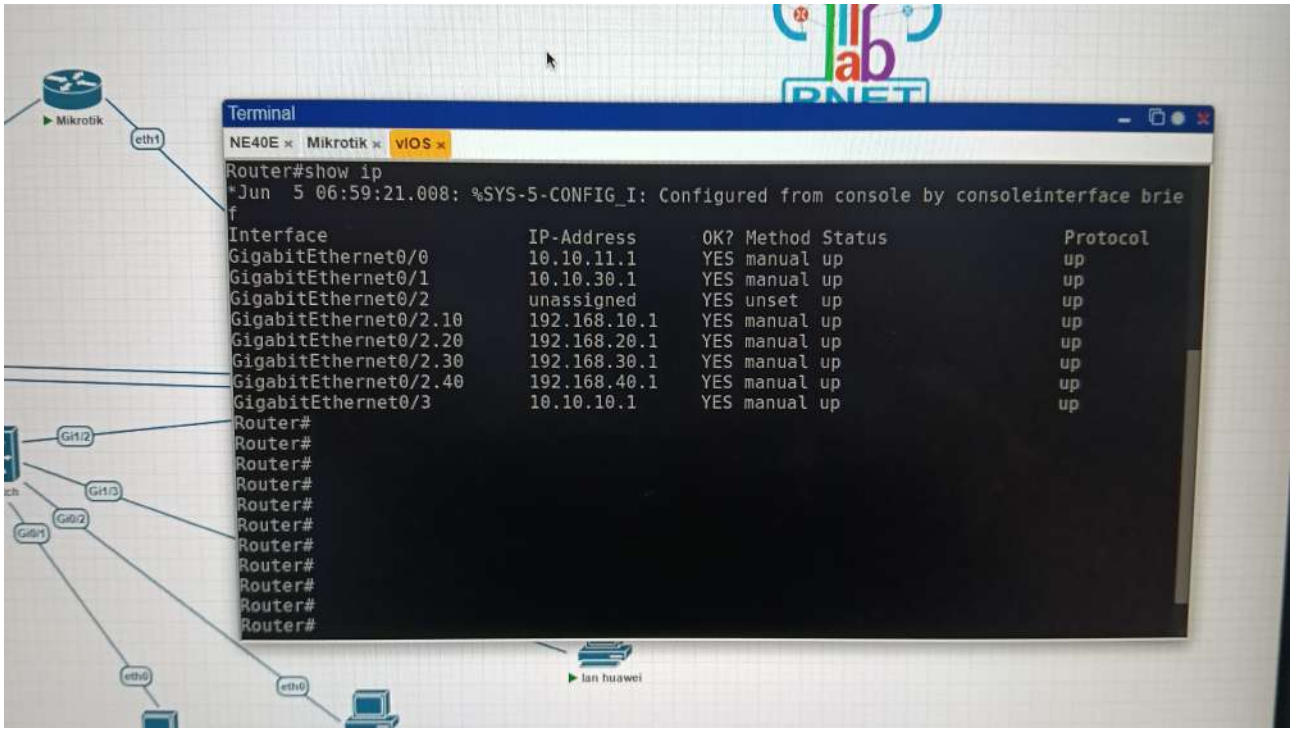
1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antar Router

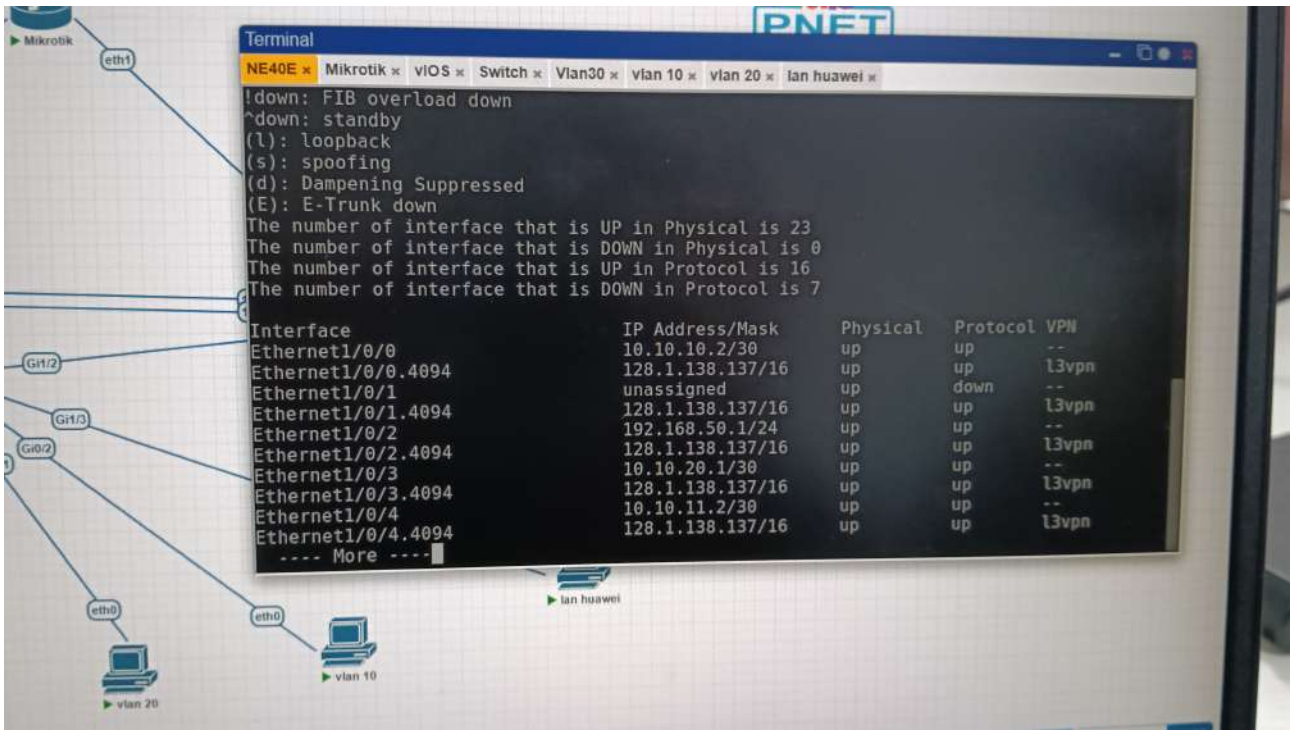
Adapun langkah-langkah yang dilakukan adalah sebagai berikut:

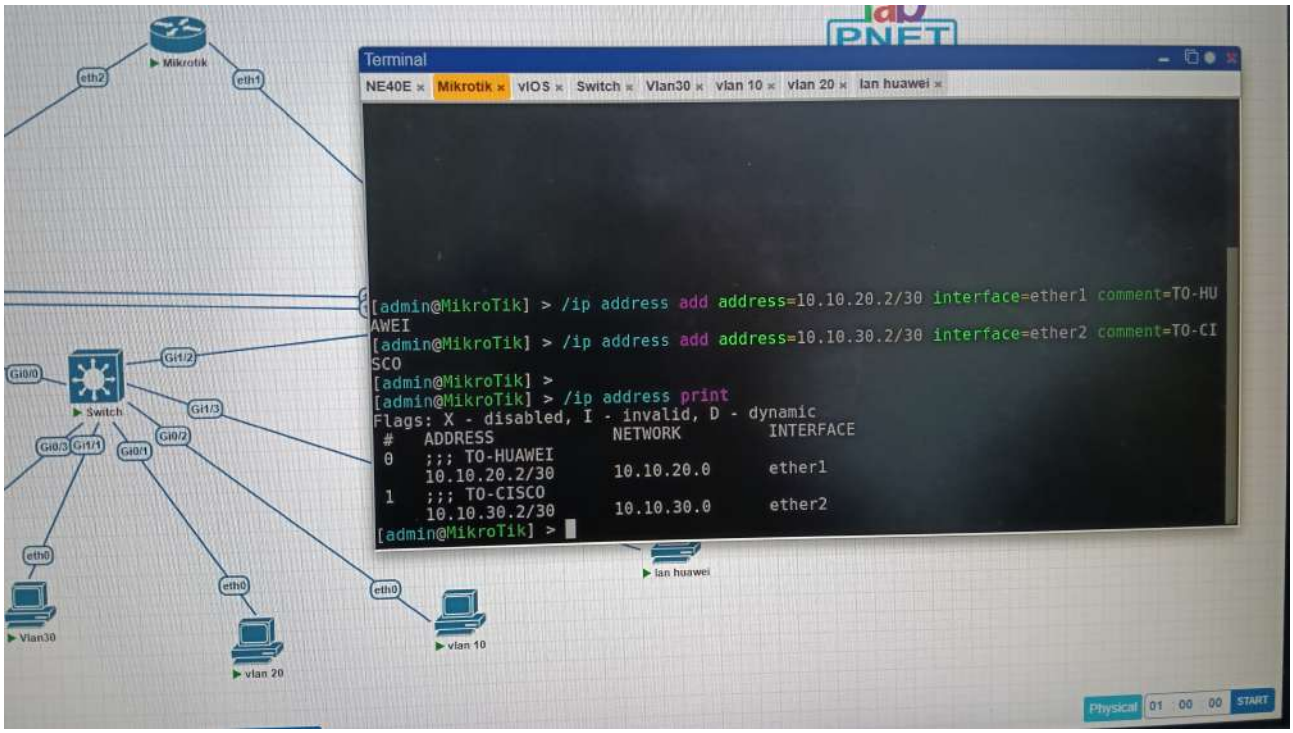
1. Mengkonfigurasi LAN pada Huawei Router dengan network 192.168.50.0/24.

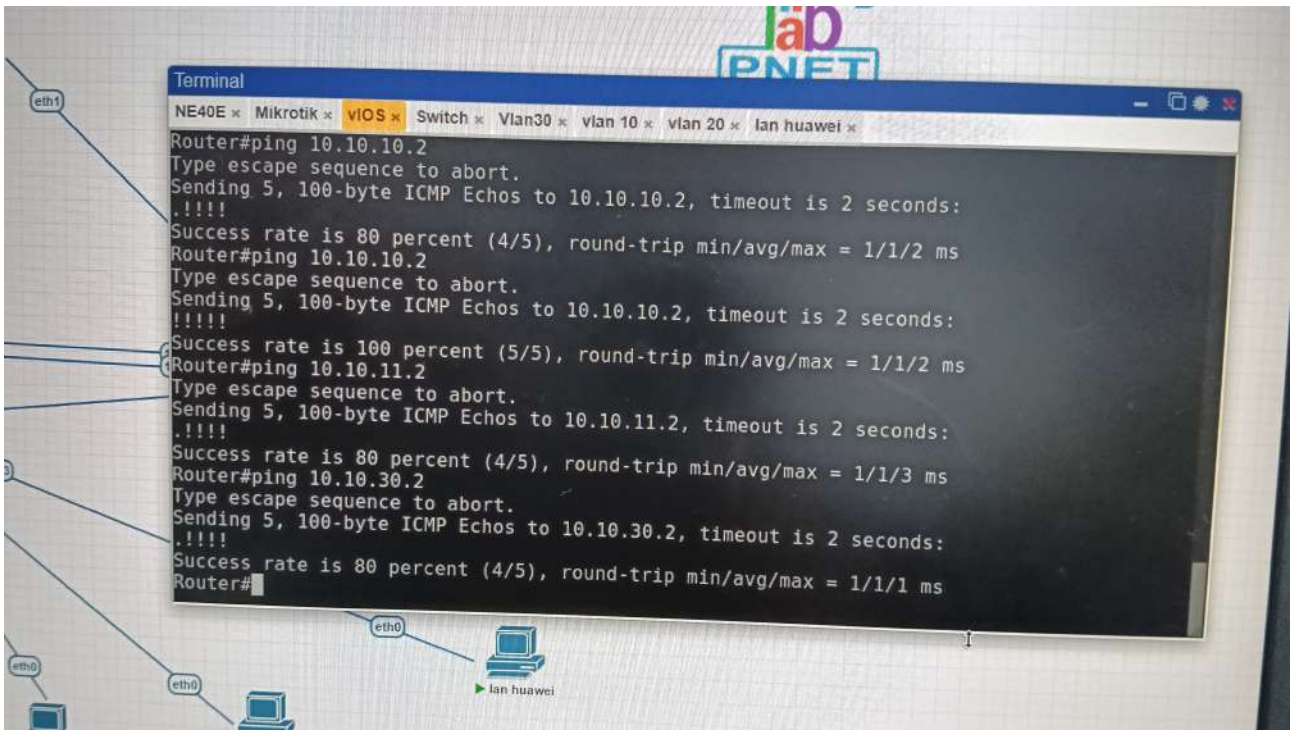


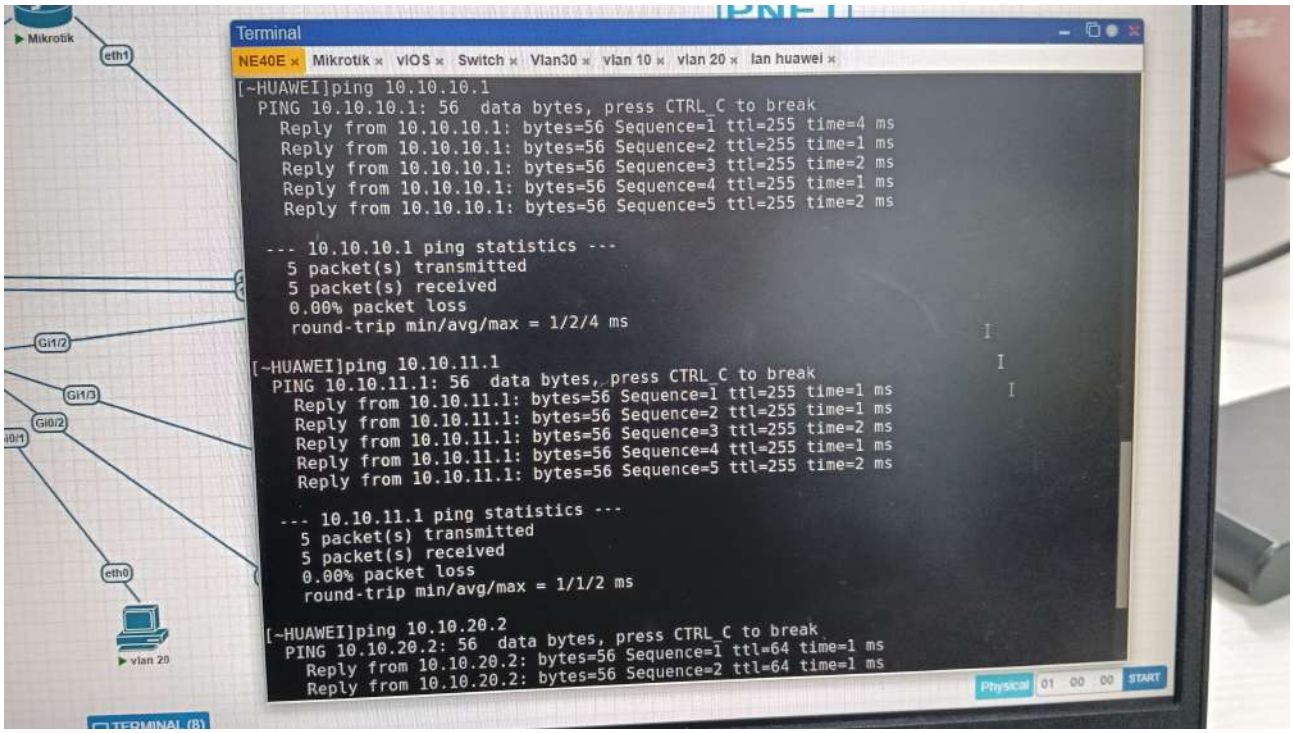
2. Mengkonfigurasi alamat IP pada PC LAN Huawei.
3. Mengkonfigurasi link Cisco Router dan Huawei Router pada network 10.10.10.0/30.
4. Mengkonfigurasi link Cisco Router dan Huawei Router pada network 10.10.11.0/30.
5. Mengkonfigurasi link Huawei Router dan MikroTik pada network 10.10.20.0/30.
6. Mengkonfigurasi link Cisco Router dan MikroTik pada network 10.10.30.0/30.
7. Memverifikasi status interface pada Cisco Router, Huawei Router, dan MikroTik.



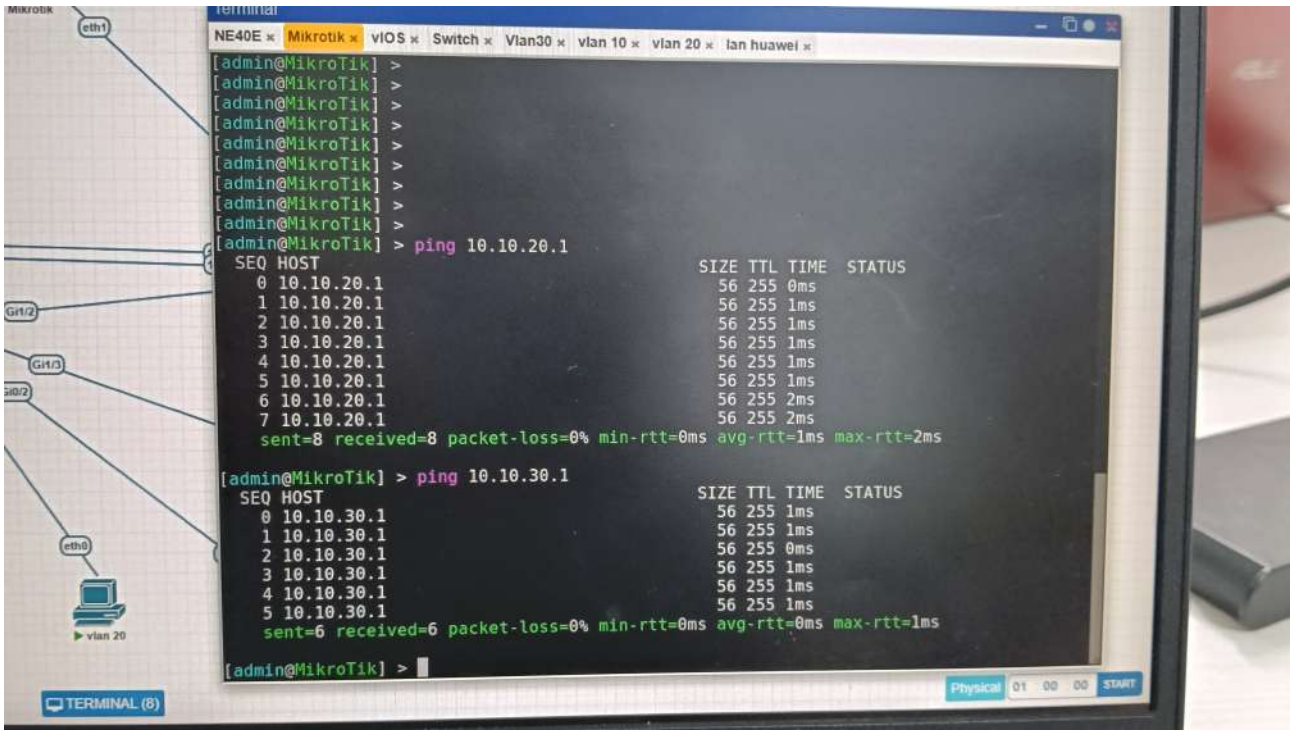

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7
Interface IP Address/Mask Physical Protocol VPN
Ethernet1/0/0 10.10.10.2/30 up up --
Ethernet1/0/0.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/1 unassigned up down --
Ethernet1/0/1.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/2 192.168.50.1/24 up up --
Ethernet1/0/2.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/3 10.10.20.1/30 up up --
Ethernet1/0/3.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/4 10.10.11.2/30 up up --
Ethernet1/0/4.4094 128.1.138.137/16 up up l3vpn
---- More ----
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
eth0
lan huawei
vlan 10
vlan 20


Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] >
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
ADDRESS NETWORK INTERFACE
0 ;;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >
Mikrotik
eth2
eth1
Switch
Gi0/0
Gi1/2
Gi1/3
Gi0/3
Gi1/1
Gi0/1
Gi0/2
eth0
Vlan30
vlan 20
vlan 10
lan huawei
Physical 01 00 00 START





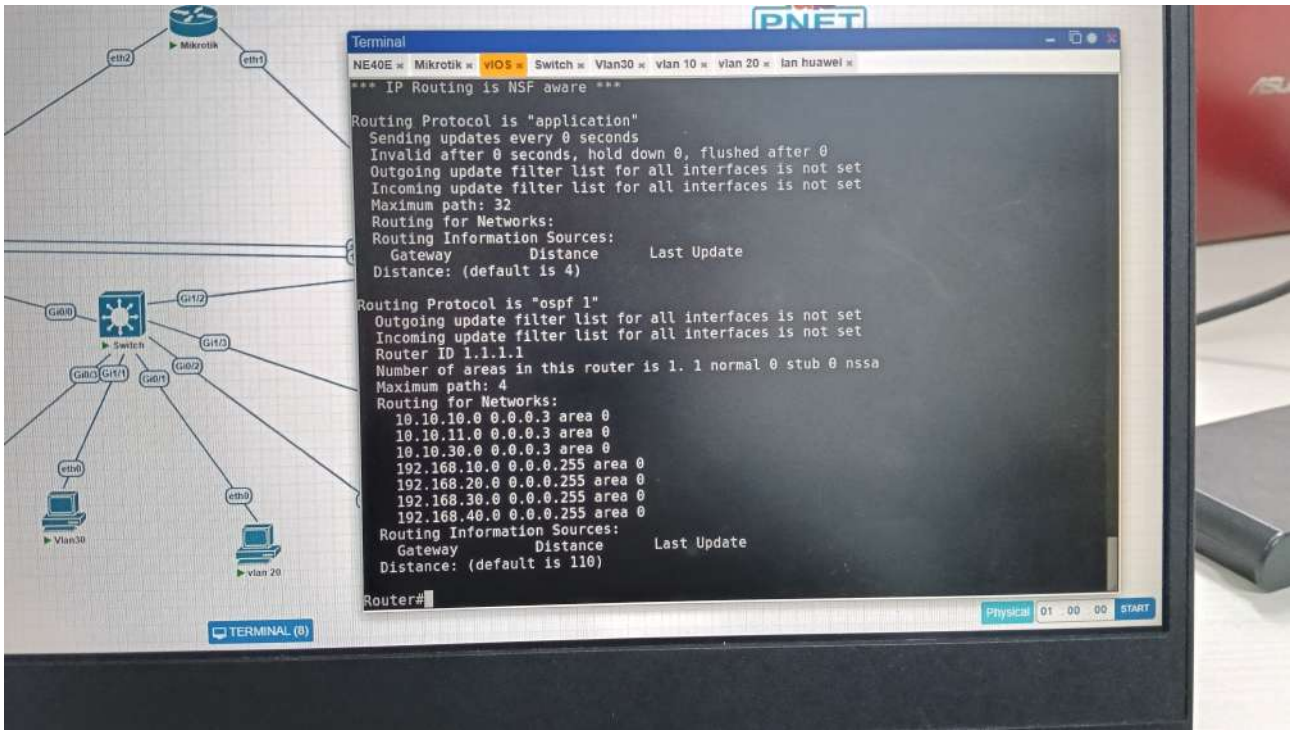
8. Melakukan pengujian konektivitas antar router menggunakan perintah `ping`.



1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Adapun langkah-langkah yang dilakukan adalah sebagai berikut:

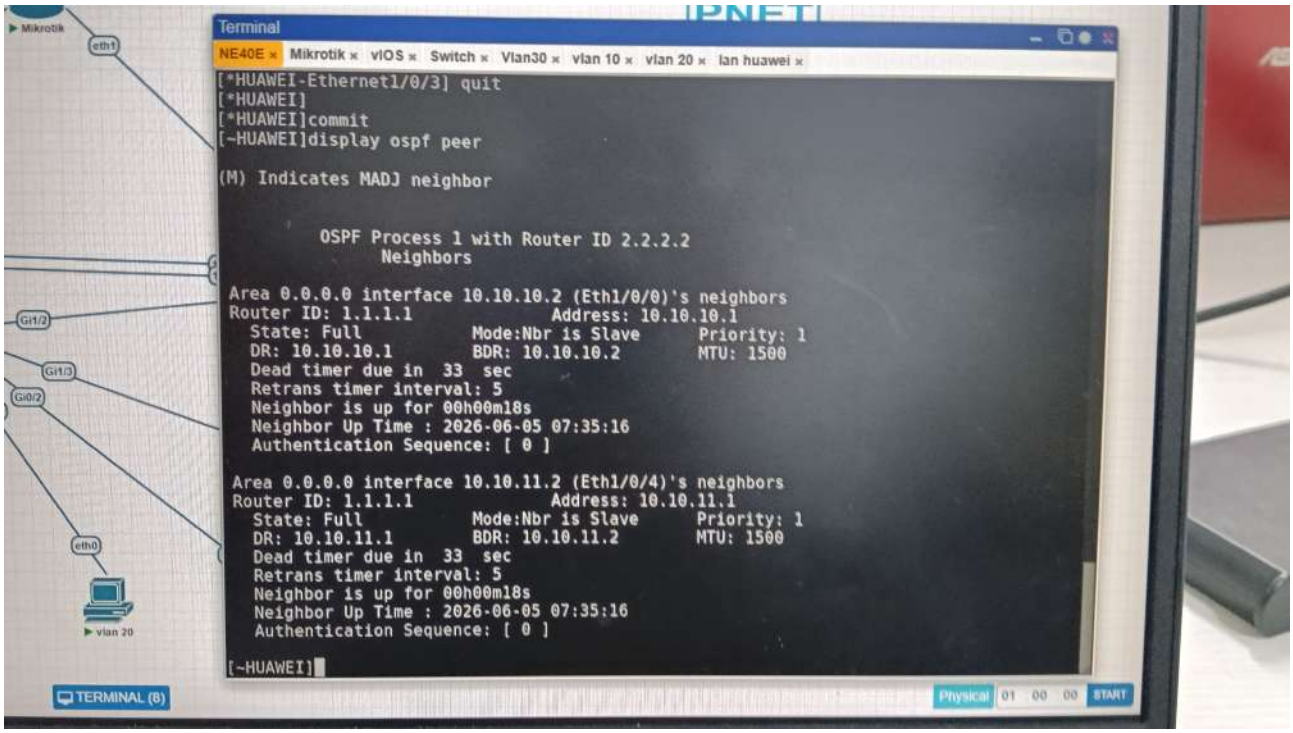
1. Mengkonfigurasi OSPF pada Cisco Router dengan router-id yang telah ditentukan.



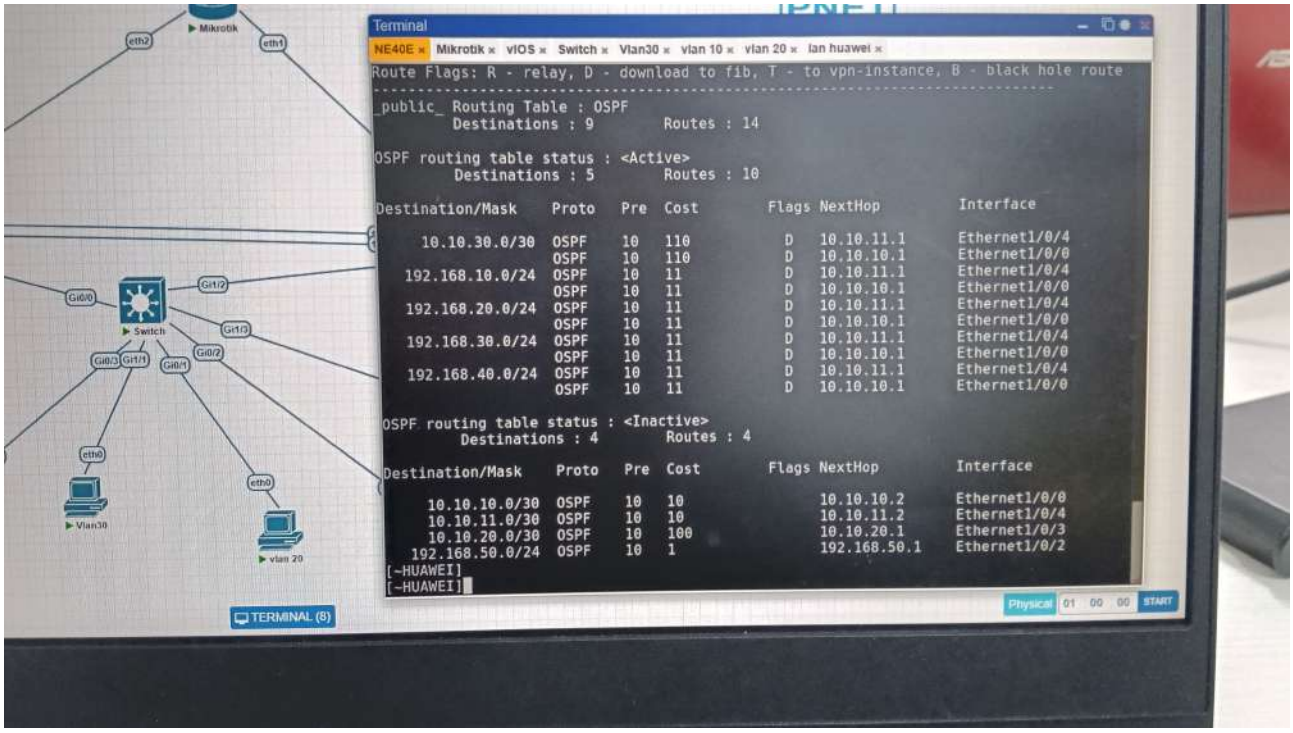
2. Mengiklankan seluruh network Cisco ke area OSPF.

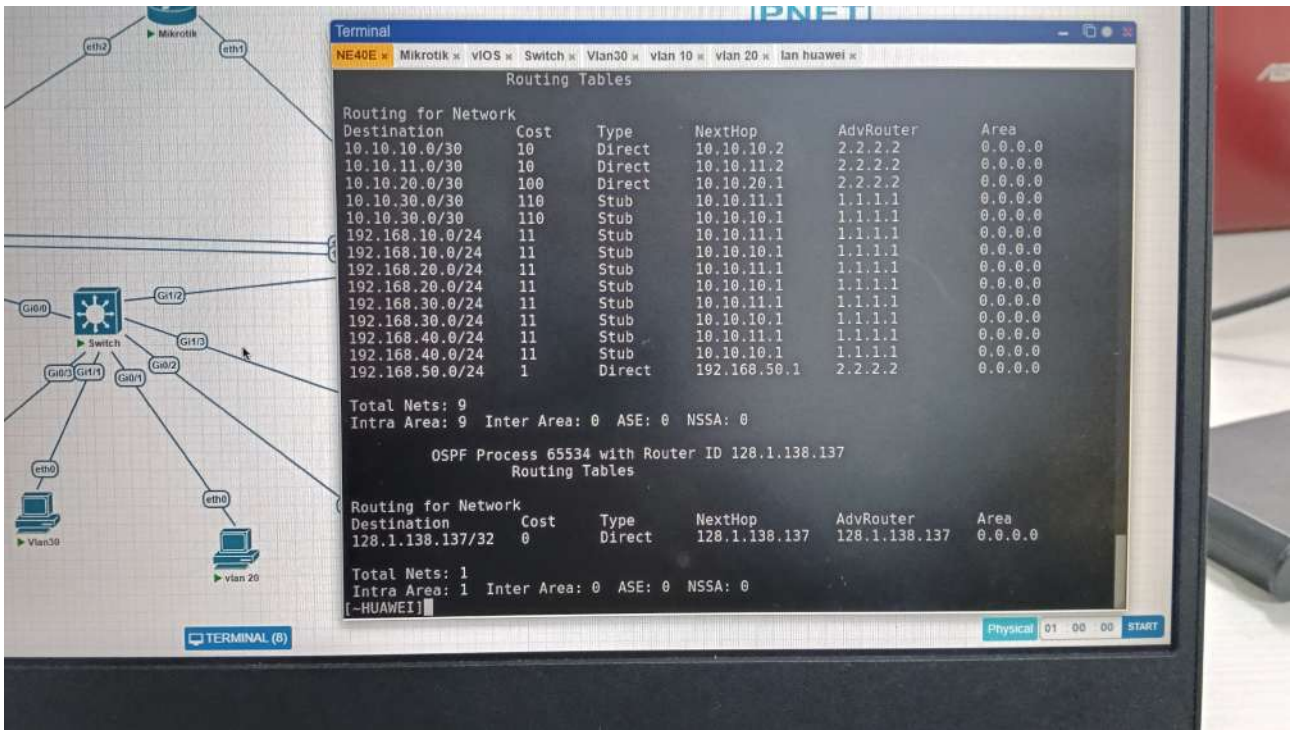
3. Mengatur nilai OSPF cost pada setiap interface Cisco Router.

4. Mengkonfigurasi OSPF pada Huawei Router dan mengiklankan seluruh network yang terhubung.



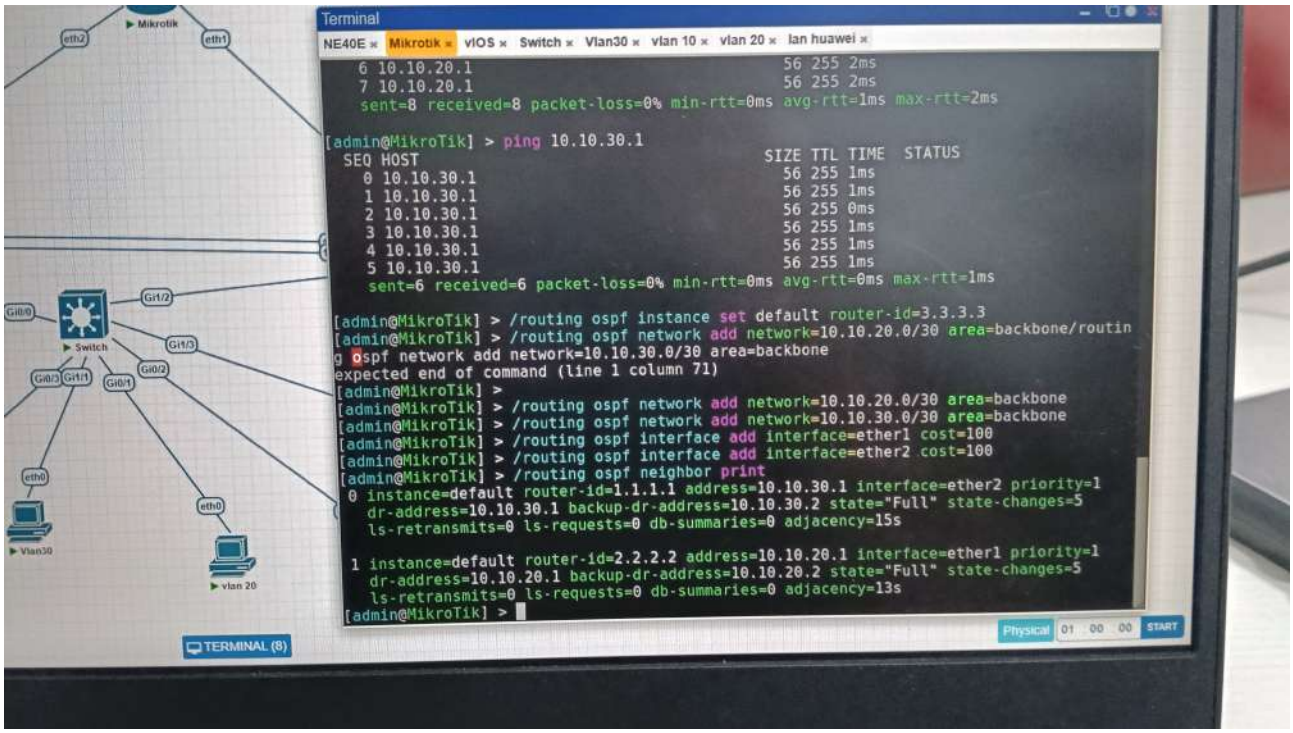
5. Mengatur nilai OSPF cost pada interface Huawei Router.

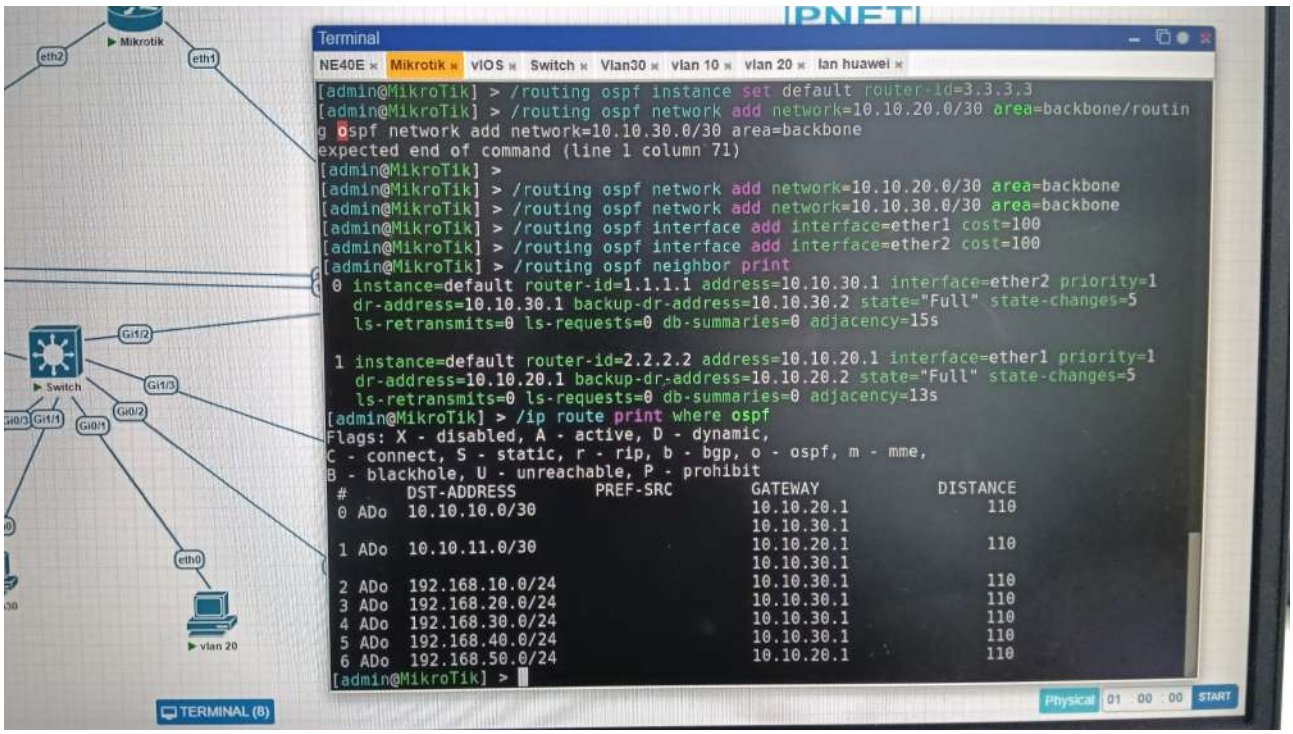




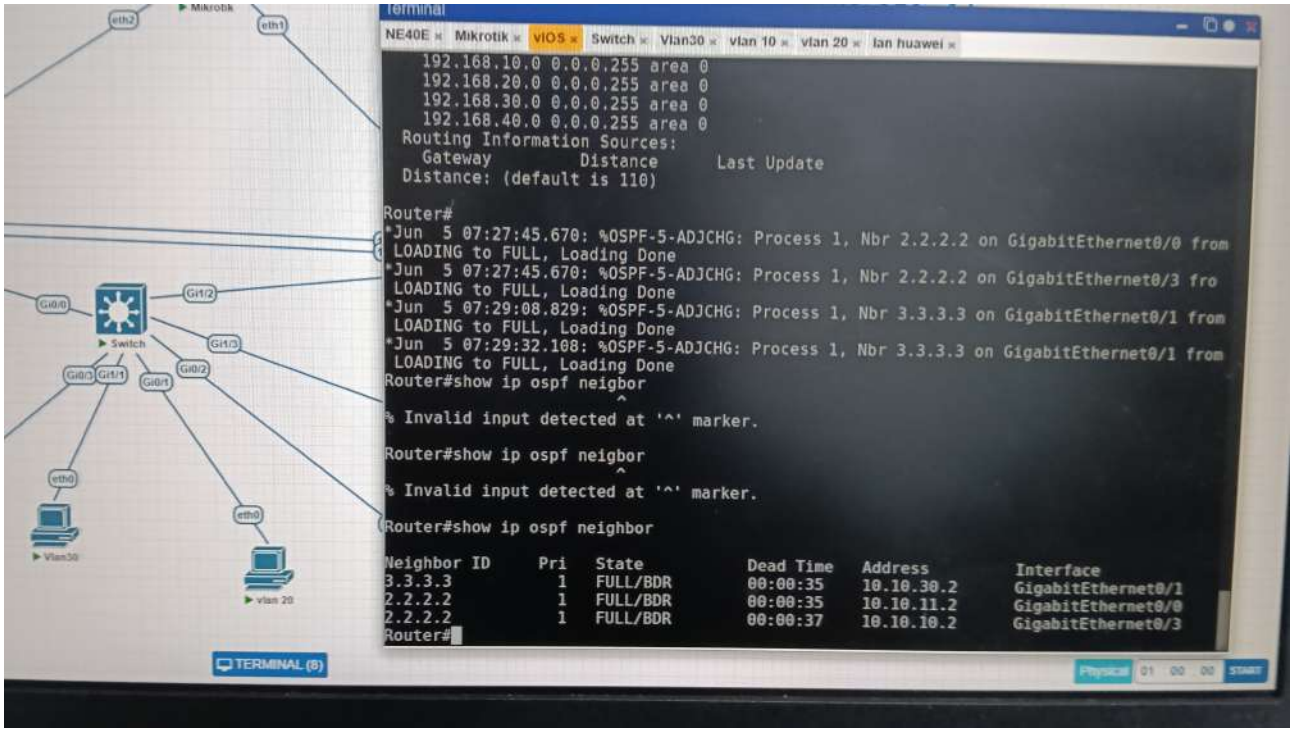
6. Mengkonfigurasi OSPF pada MikroTik Router.

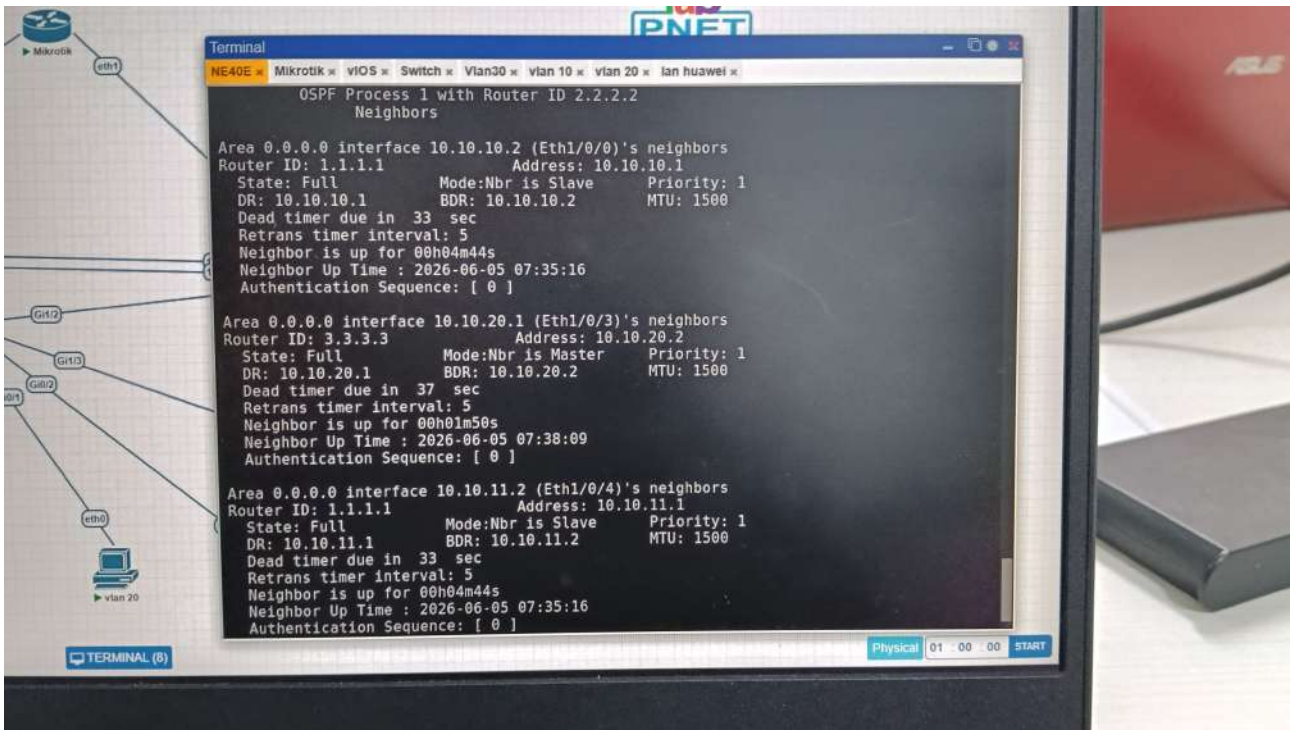
7. Mengatur router-id dan network OSPF pada MikroTik.

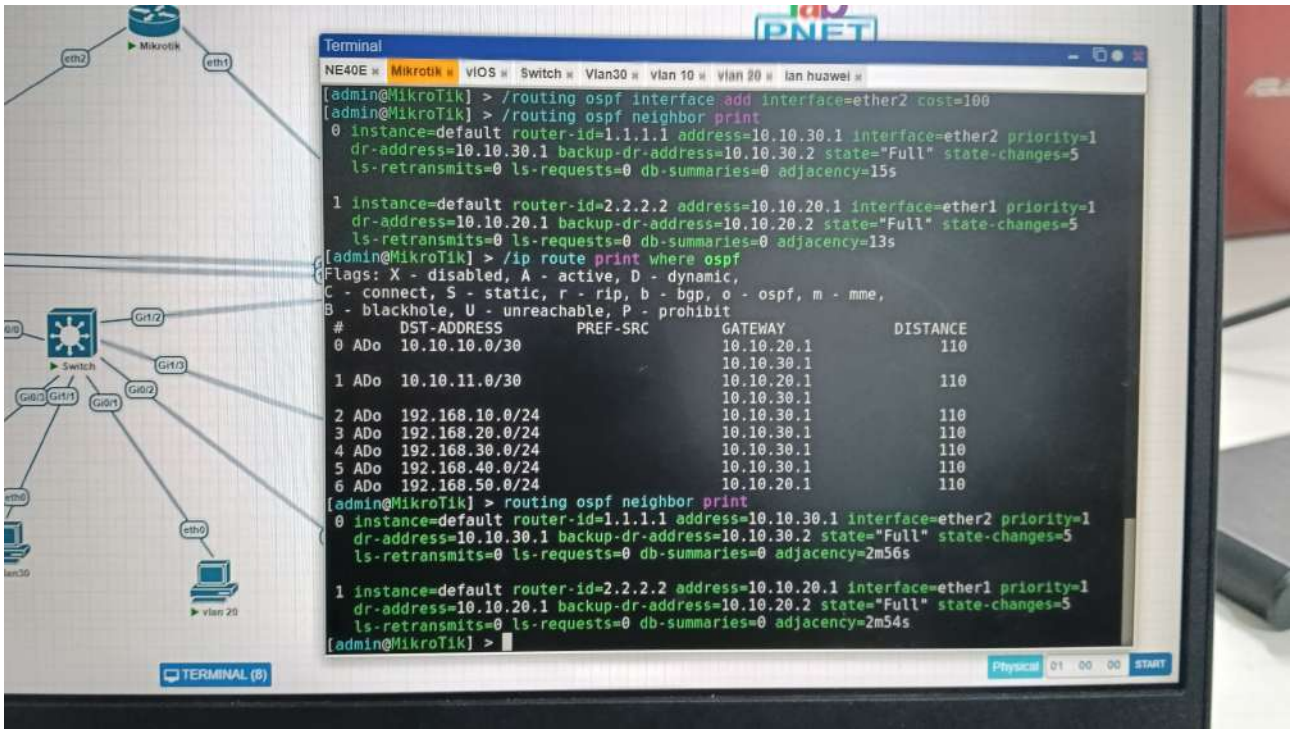




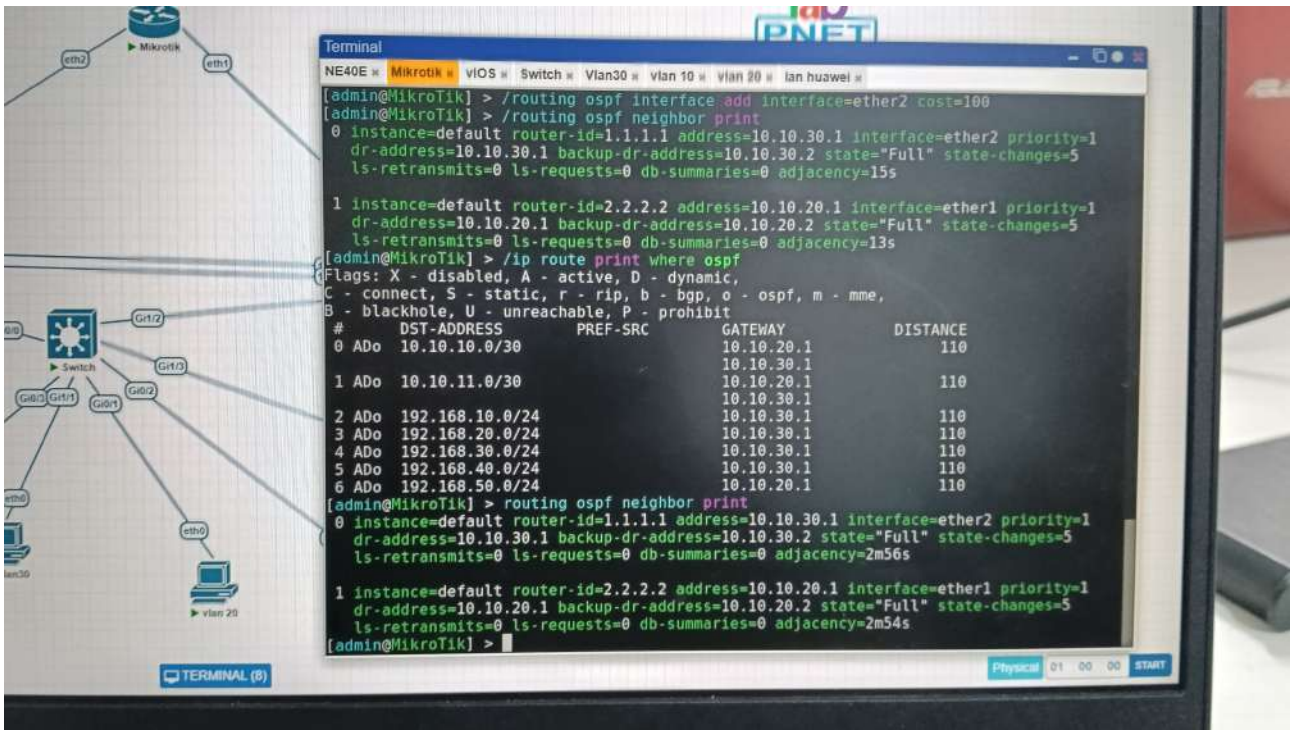
8. Memverifikasi neighbor OSPF pada Cisco Router, Huawei Router, dan MikroTik.

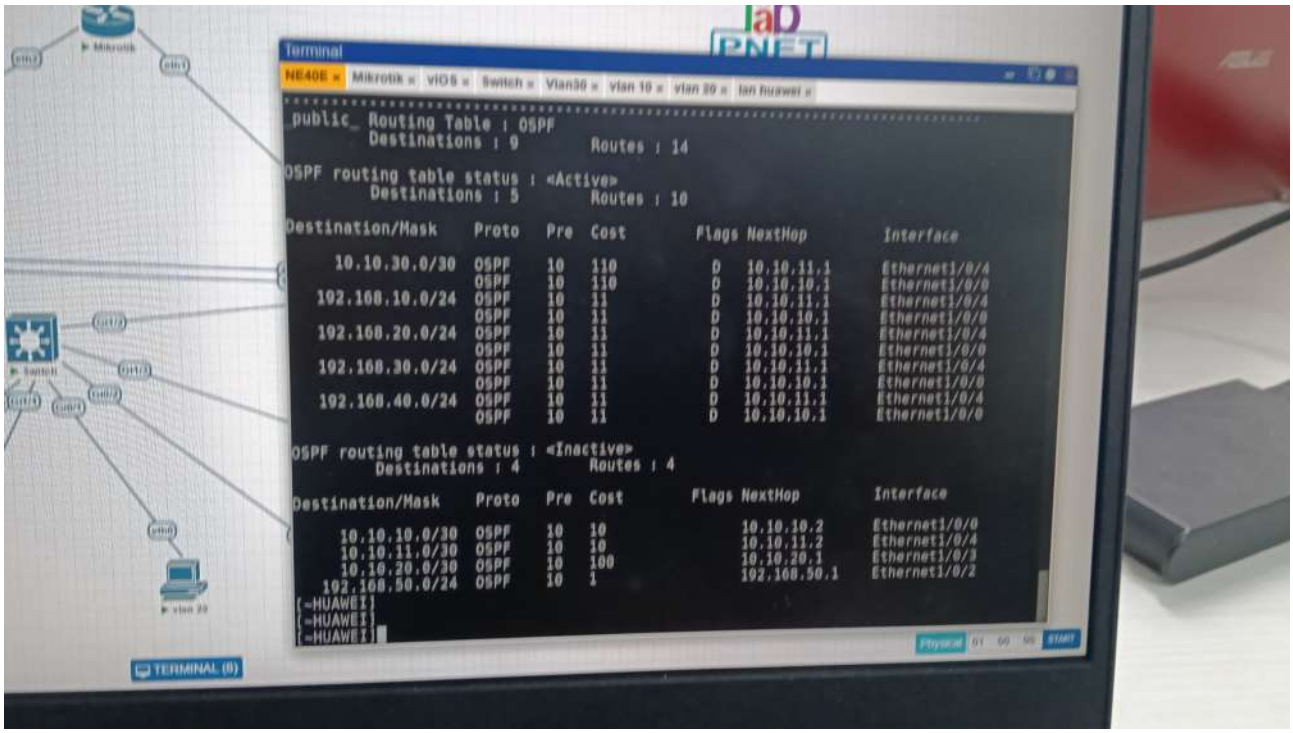


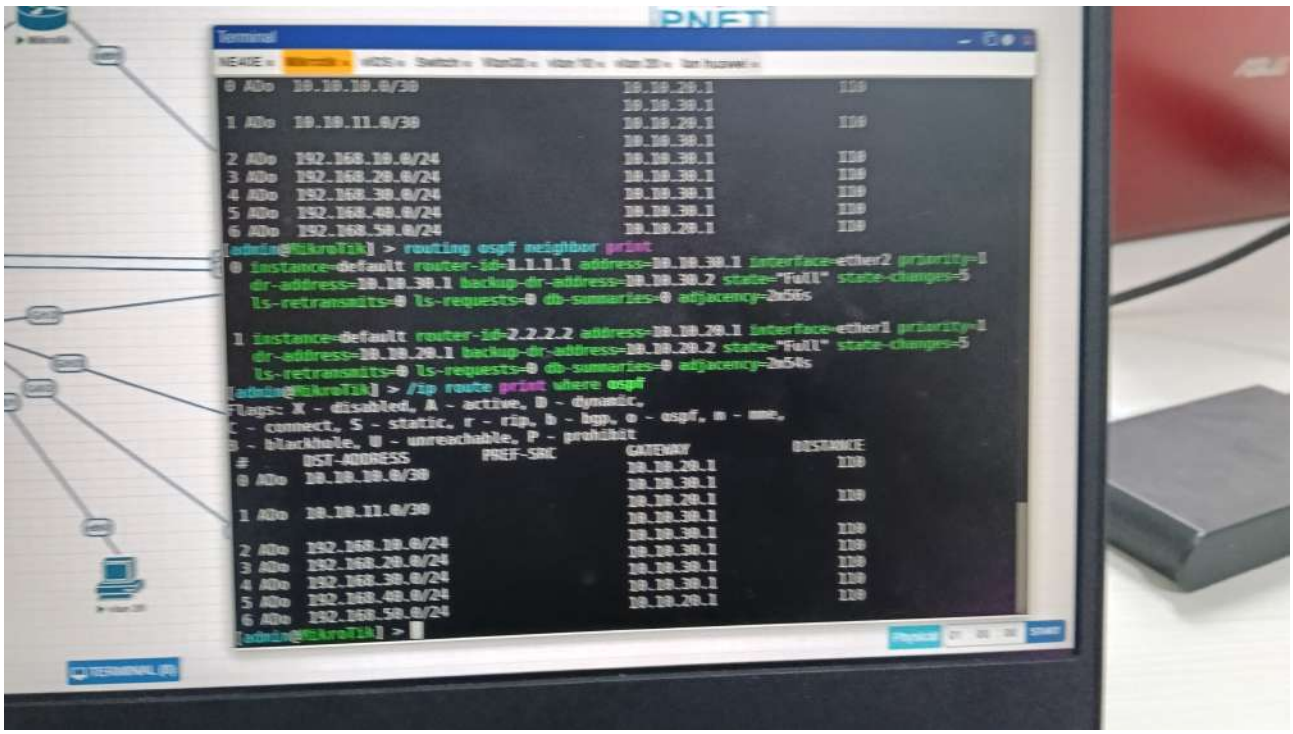




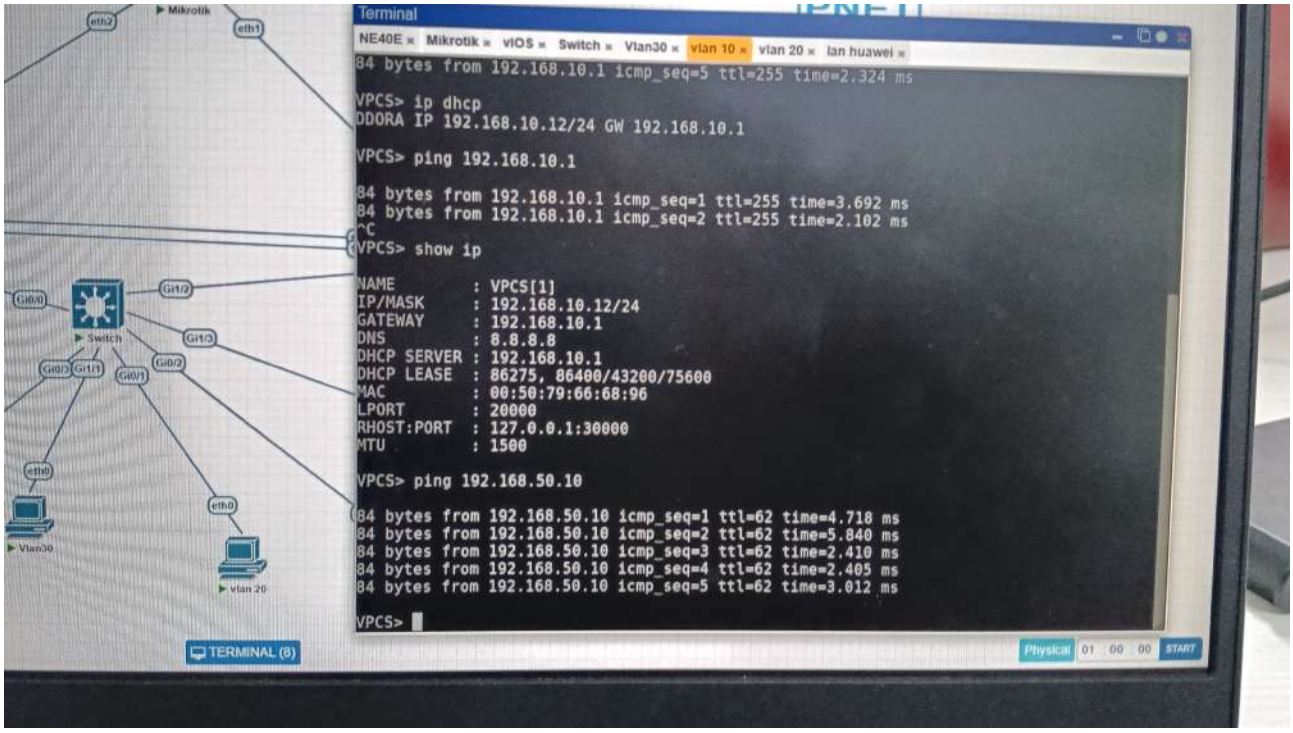
9. Memverifikasi routing table OSPF pada setiap router.

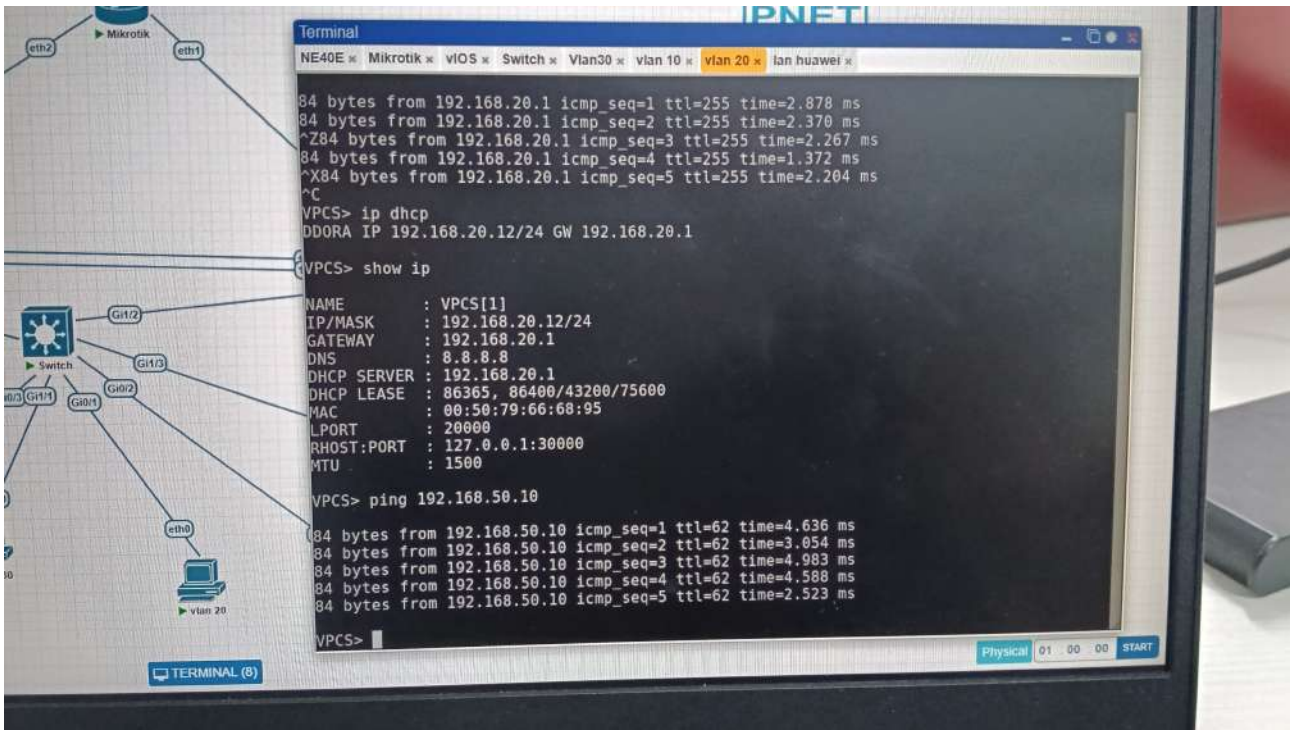

Terminal
NE40E
Mikrotik
vIOS
Switch
Vlan30
vlan 10
vlan 20
lan huawei
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=15s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=13s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 ADo 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m56s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m54s
[admin@MikroTik] >
TERMINAL (8)

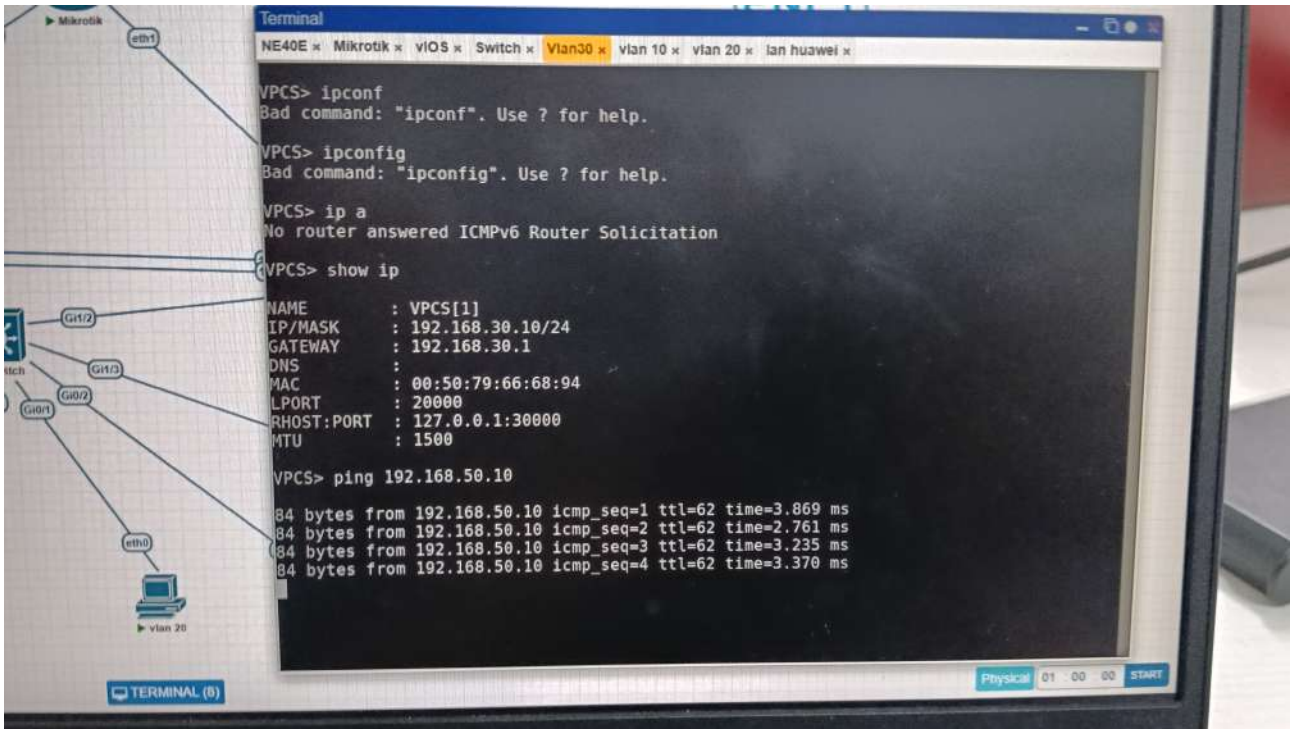

Terminal
NE40E
public_ Routing Table : OSPF
Destinations : 9 Routes : 14
OSPF routing table status : <Active>
Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
TERMINAL (8)

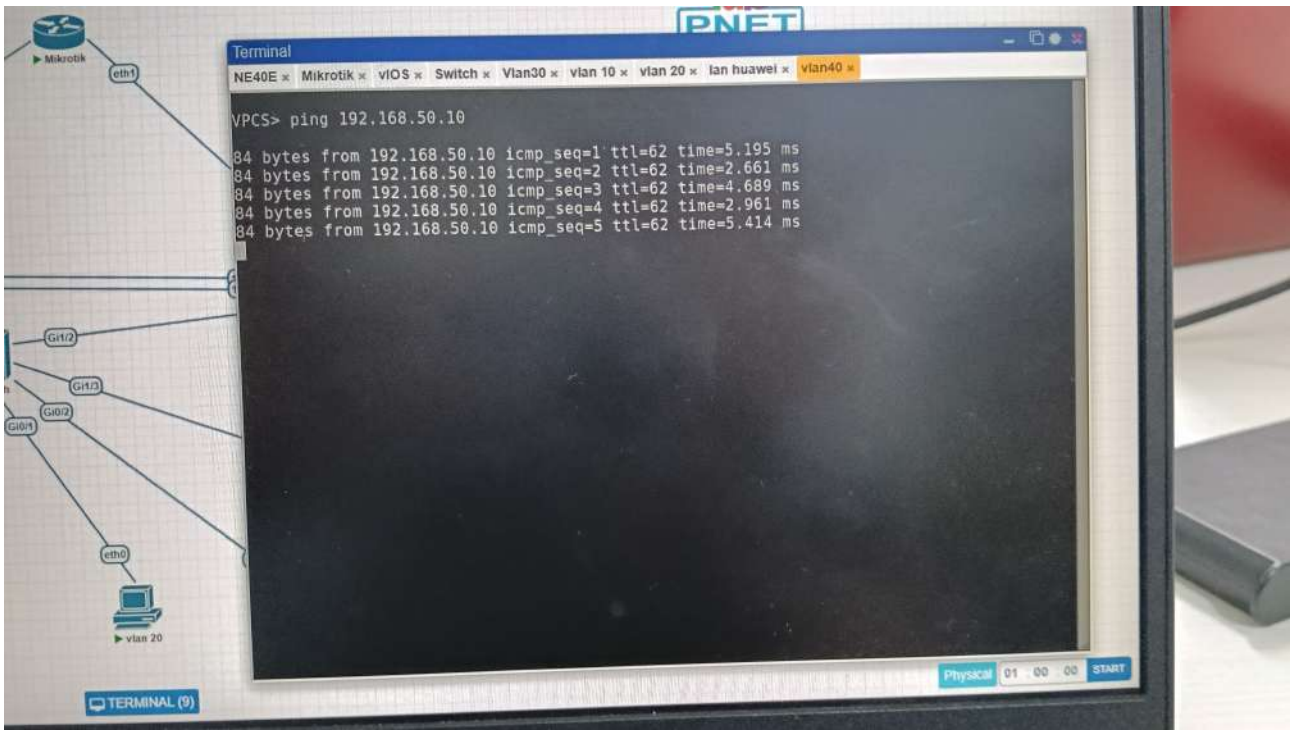


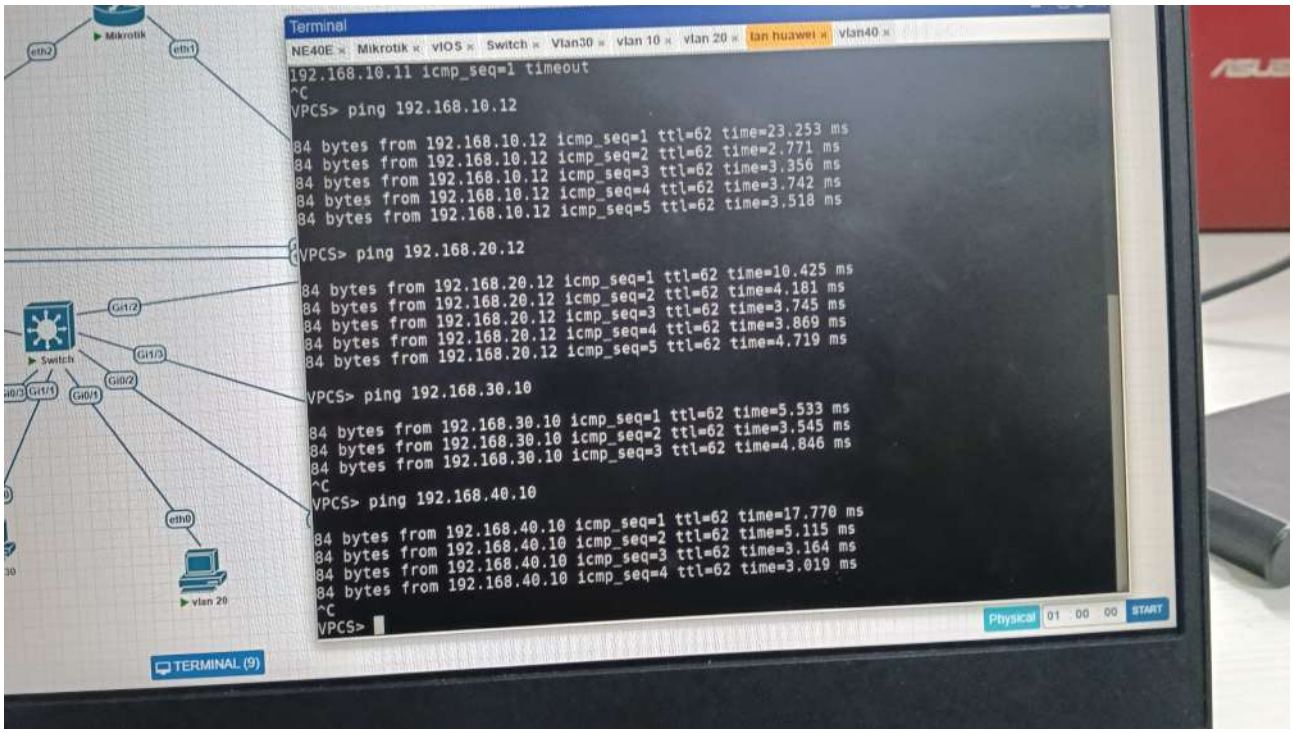
10. Melakukan pengujian konektivitas end-to-end antara jaringan VLAN Cisco dan LAN Huawei.




Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.878 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.370 ms
^Z84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.267 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.372 ms
^X84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.204 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.20.12/24 GW 192.168.20.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.20.12/24
GATEWAY : 192.168.20.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86365, 86400/43200/75600
MAC : 00:50:79:66:68:95
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.054 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.983 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.588 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.523 ms
VPCS>
Mikrotik
eth2
eth1
Switch
Gi1/2
Gi1/3
Gi0/2
Gi0/1
Gi1/1
eth0
vlan 20
TERMINAL (8)
Physical
START


Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
VPCS> ipconf
Bad command: "ipconf". Use ? for help.
VPCS> ipconfig
Bad command: "ipconfig". Use ? for help.
VPCS> ip a
No router answered ICMPv6 Router Solicitation
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.30.10/24
GATEWAY : 192.168.30.1
DNS :
MAC : 00:50:79:66:68:94
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.869 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.761 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.235 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.370 ms
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (8)
Physical
START

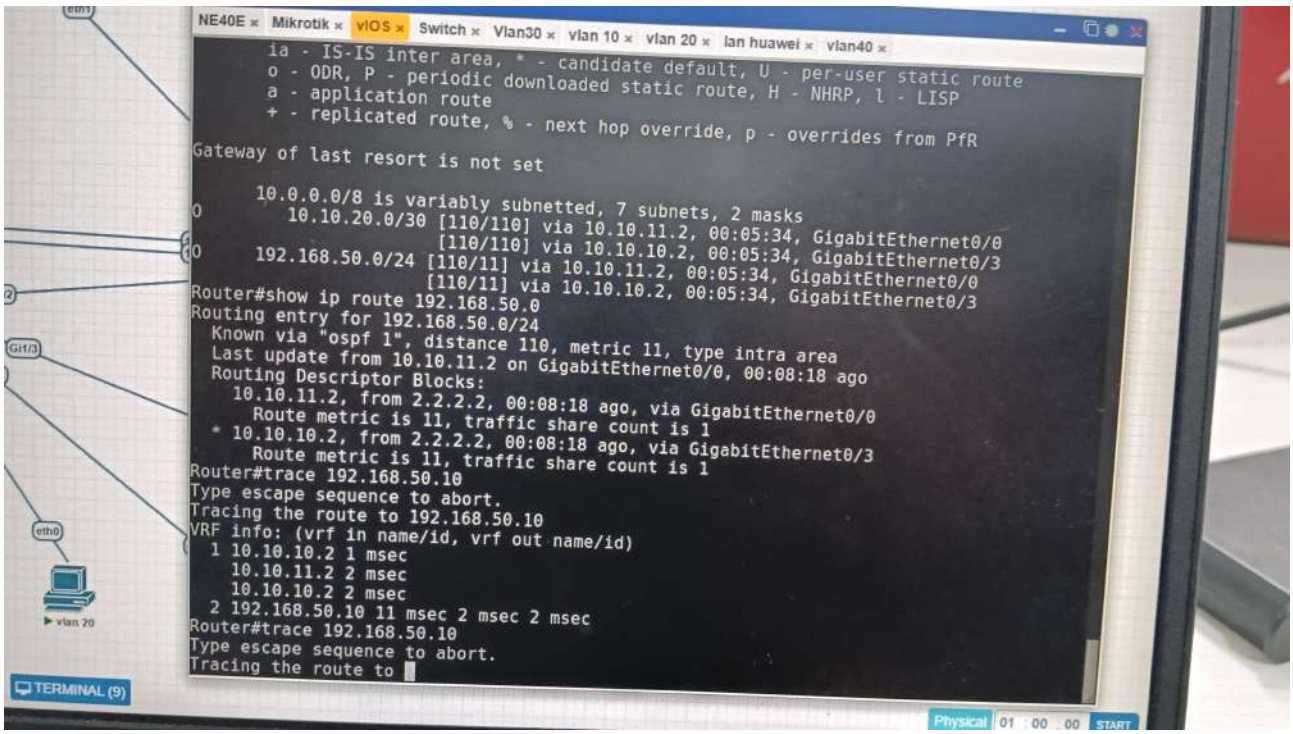




1.5 Praktikum 5: Pengujian Failover OSPF

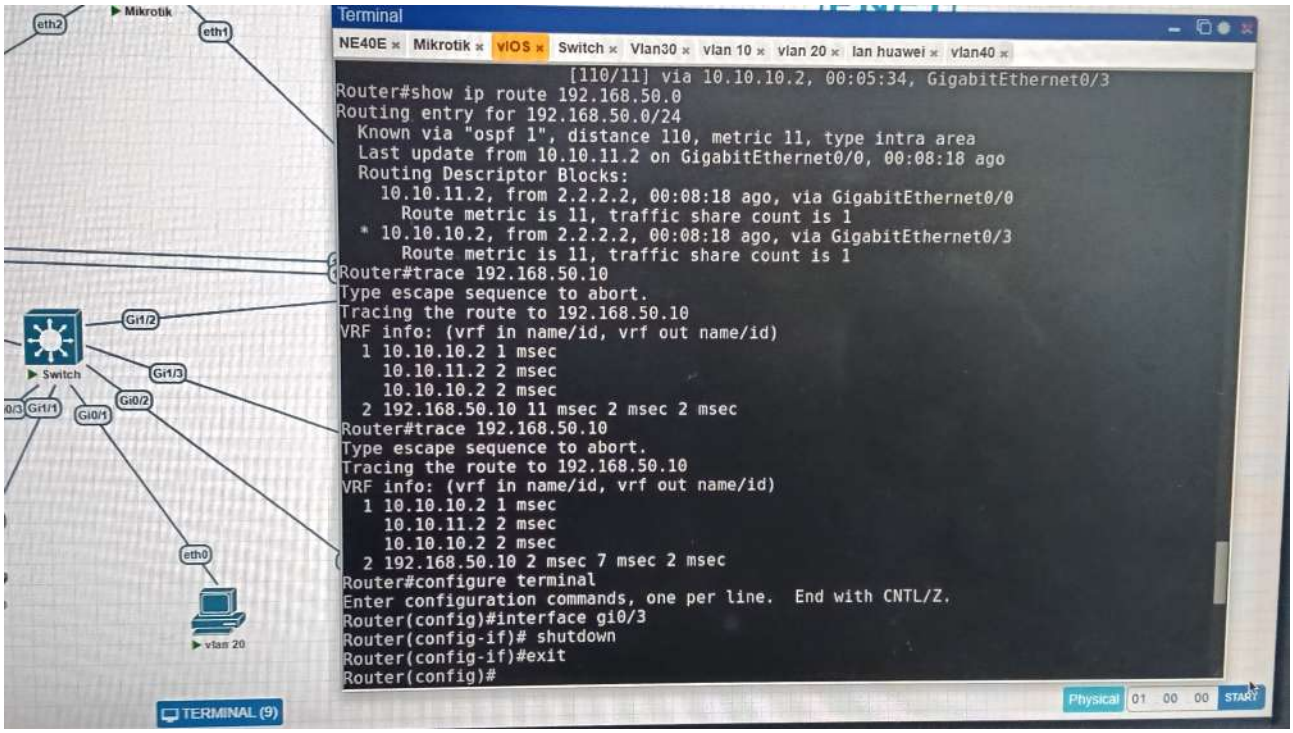
Adapun langkah-langkah yang dilakukan adalah sebagai berikut:

1. Melakukan pengujian routing pada kondisi normal saat kedua link Cisco-Huawei aktif.
2. Melakukan traceroute dari client VLAN menuju LAN Huawei.



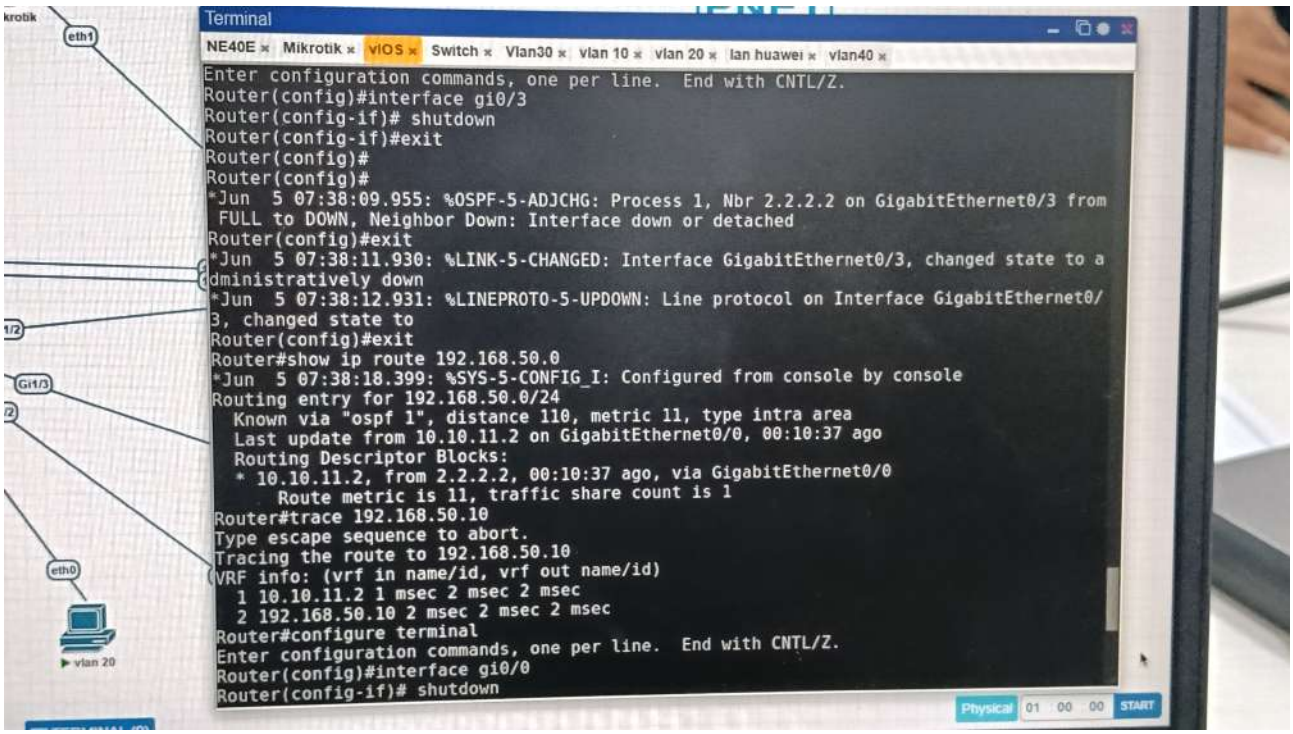
3. Menonaktifkan salah satu link Cisco-Huawei dan mengamati perubahan routing.

4. Melakukan traceroute kembali untuk memastikan jalur utama masih digunakan.



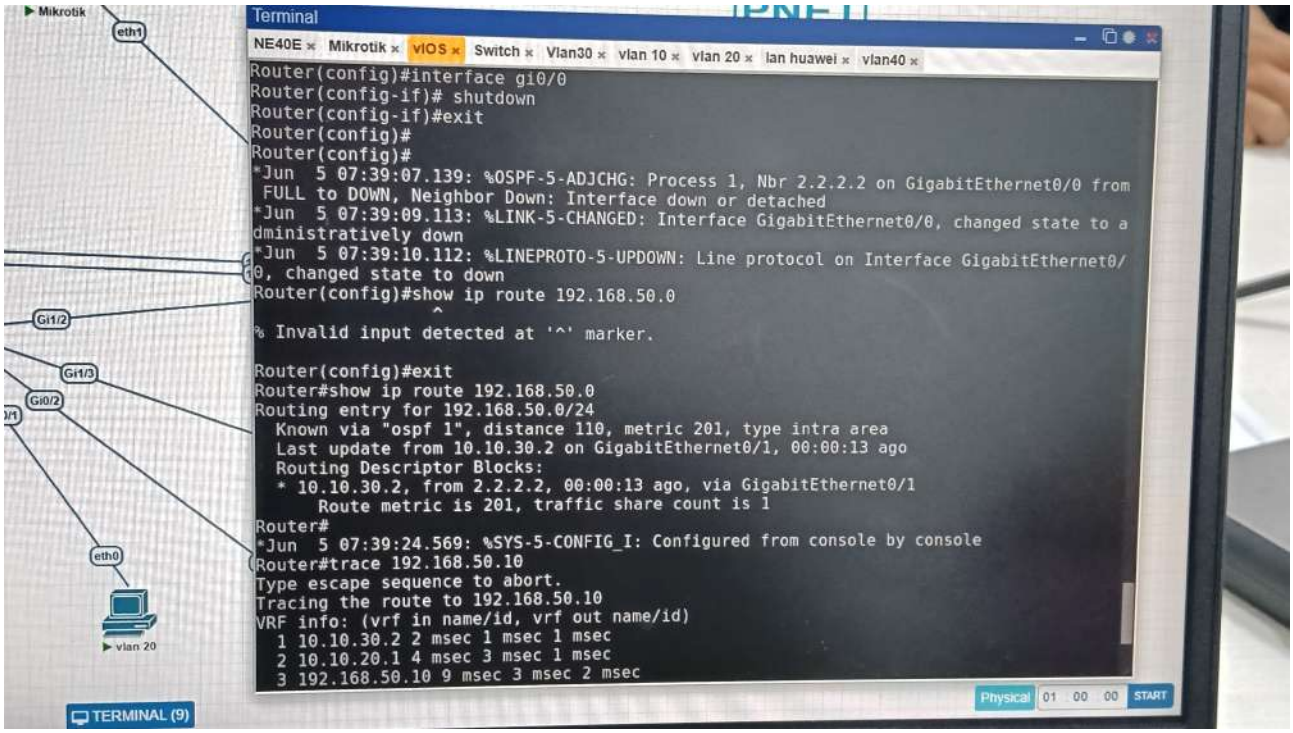
5. Menonaktifkan kedua link Cisco-Huawei sehingga jalur utama terputus.

6. Memverifikasi perubahan route yang berpindah melalui MikroTik sebagai jalur cadangan.

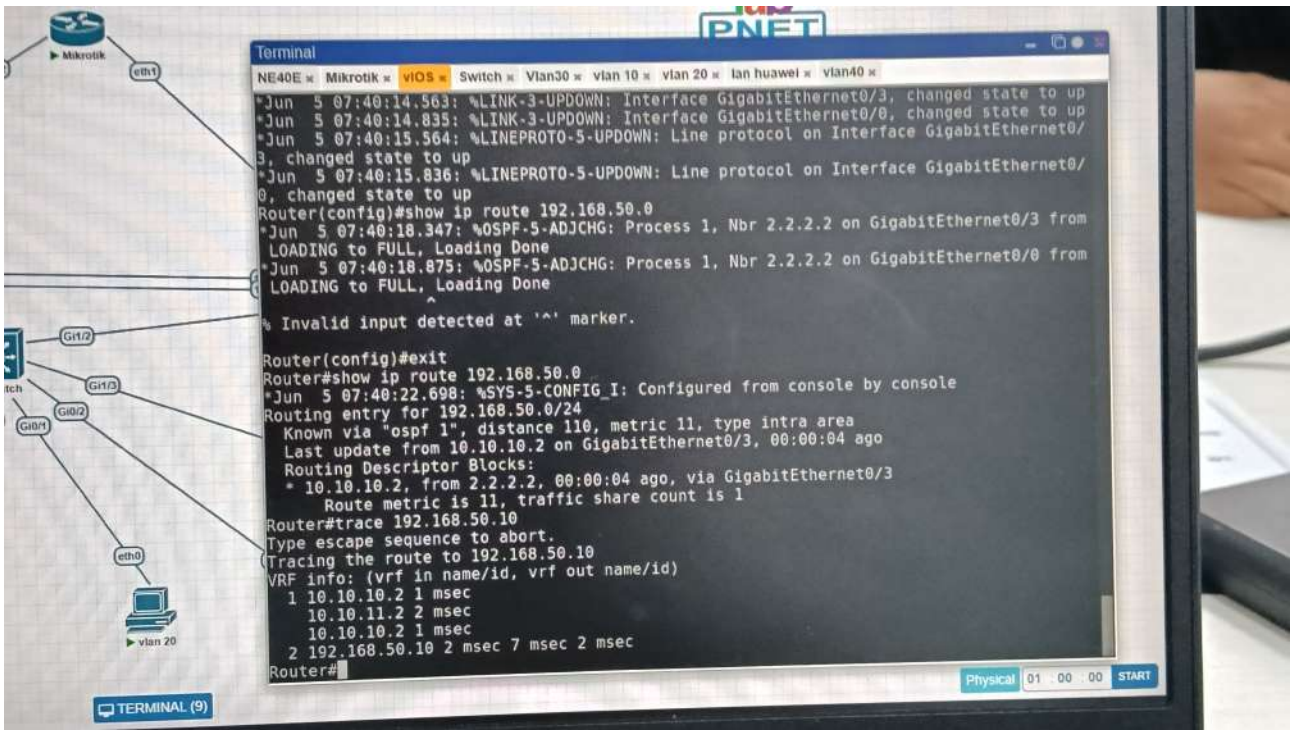


7. Melakukan traceroute untuk mengamati perpindahan jalur routing.

8. Mengaktifkan kembali kedua link Cisco-Huawei.



9. Memverifikasi bahwa jalur routing kembali menggunakan link utama.



2 Analisis Hasil Percobaan

Berdasarkan hasil praktikum yang telah dilakukan, konfigurasi VLAN pada Cisco Switch berhasil memisahkan jaringan ke dalam beberapa segmen logis, yaitu VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing). Hasil verifikasi menggunakan perintah `show vlan brief` menunjukkan bahwa seluruh VLAN telah terbentuk dan port access telah berada pada VLAN yang sesuai. Selain itu, konfigurasi trunk antara Cisco Switch dan Cisco Router berhasil membawa seluruh VLAN yang dibutuhkan menggunakan protokol IEEE 802.1Q. Hasil ini sesuai dengan teori VLAN yang menyatakan bahwa segmentasi jaringan dapat meningkatkan efisiensi pengelolaan jaringan serta mengurangi domain broadcast.

Pada konfigurasi Cisco Router menggunakan metode Router-on-a-Stick (ROAS), setiap VLAN berhasil diberikan gateway melalui subinterface yang berbeda. Penggunaan DHCP Server pada VLAN 10 dan VLAN 20 juga berjalan dengan baik yang ditunjukkan oleh keberhasilan client memperoleh alamat IP secara otomatis sesuai rentang yang telah ditentukan. Pengujian konektivitas menunjukkan bahwa seluruh client dapat melakukan komunikasi dengan gateway masing-masing dan dapat saling berkomunikasi antar VLAN. Hasil tersebut membuktikan bahwa proses routing antar VLAN telah berjalan sesuai teori inter-VLAN routing menggunakan router sebagai perangkat layer 3.

Konfigurasi alamat IP pada Huawei Router, MikroTik Router, serta link antar router juga berhasil dilakukan. Setiap interface yang terhubung antar perangkat menunjukkan status aktif dan mampu melakukan komunikasi menggunakan protokol ICMP. Hasil pengujian ping antar router menunjukkan bahwa seluruh link point-to-point dapat berfungsi dengan baik sehingga siap digunakan sebagai jalur pertukaran informasi routing. Keberhasilan tahap ini sangat penting karena OSPF memerlukan konektivitas layer 3 yang stabil sebelum proses pembentukan neighbor dapat dilakukan.

Implementasi routing dinamis menggunakan OSPF pada perangkat Cisco, Huawei, dan MikroTik berhasil membentuk hubungan neighbor antar router meskipun menggunakan vendor yang berbeda. Seluruh router mampu bertukar informasi routing dan mempelajari jaringan yang tidak terhubung

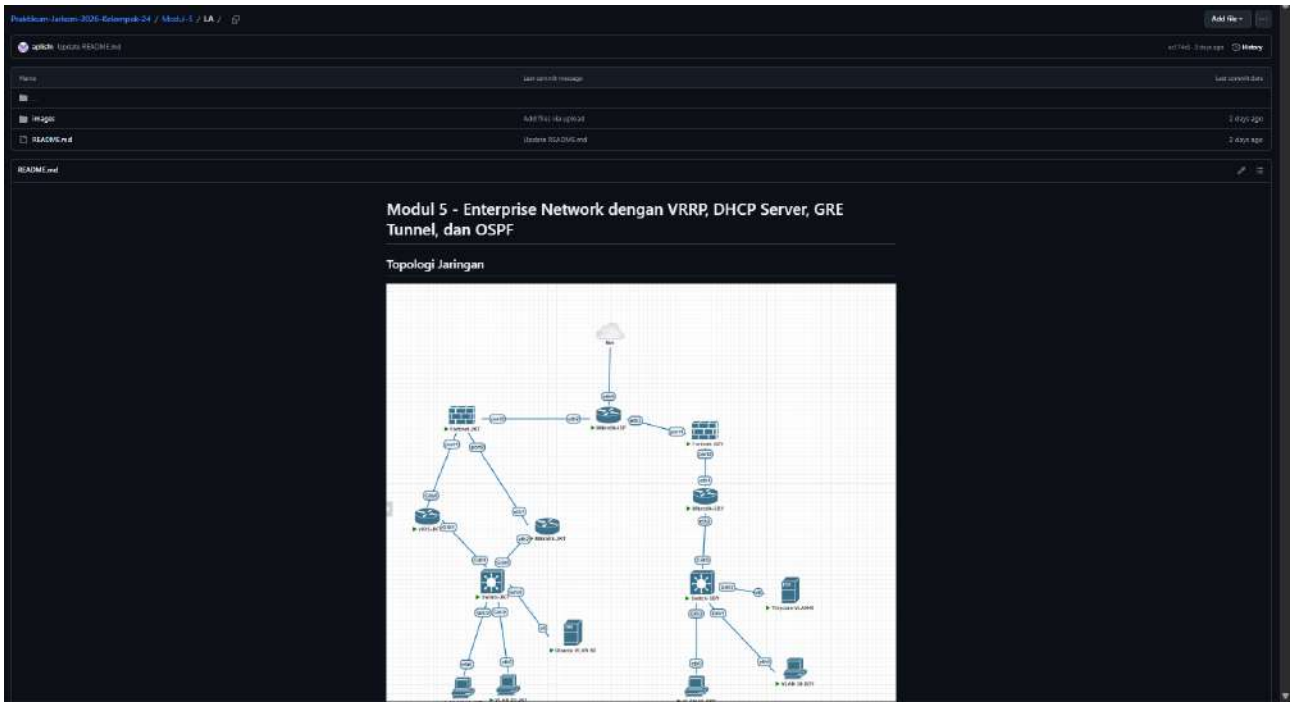
secara langsung. Routing table pada masing-masing perangkat menunjukkan adanya route OSPF menuju jaringan lain dalam topologi. Selain itu, pengaturan nilai cost pada setiap link berhasil memengaruhi pemilihan jalur routing sehingga jalur utama dan jalur cadangan dapat dibedakan sesuai kebutuhan. Hasil ini sesuai dengan teori OSPF yang menggunakan algoritma Shortest Path First (SPF) untuk menentukan jalur terbaik berdasarkan nilai cost.

Pengujian failover OSPF menunjukkan bahwa mekanisme redundansi jaringan bekerja dengan baik. Ketika salah satu link utama antara Cisco dan Huawei dimatikan, konektivitas jaringan tetap terjaga melalui link utama lainnya. Saat kedua link utama dinonaktifkan, OSPF secara otomatis memilih jalur alternatif melalui MikroTik sebagai jalur cadangan. Setelah link utama diaktifkan kembali, routing kembali menggunakan jalur dengan cost yang lebih rendah tanpa memerlukan konfigurasi ulang. Hasil ini membuktikan bahwa OSPF mampu melakukan konvergensi secara otomatis ketika terjadi perubahan topologi jaringan sehingga ketersediaan layanan jaringan tetap terjaga.

Secara keseluruhan, seluruh tujuan praktikum berhasil dicapai. Implementasi VLAN, inter-VLAN routing, konfigurasi IP antar router, routing dinamis OSPF multivendor, serta mekanisme failover berhasil dijalankan sesuai dengan teori yang dipelajari.

3 Hasil Tugas Modul

1. Hasil tugas modul 5 berada di Github folder Modul-5/LA/README.md
   https://github.com/aplisfn/Praktikum-Jarkom-2026-Kelompok-24/blob/main/Modul-5/LA/README.md



4 Kesimpulan

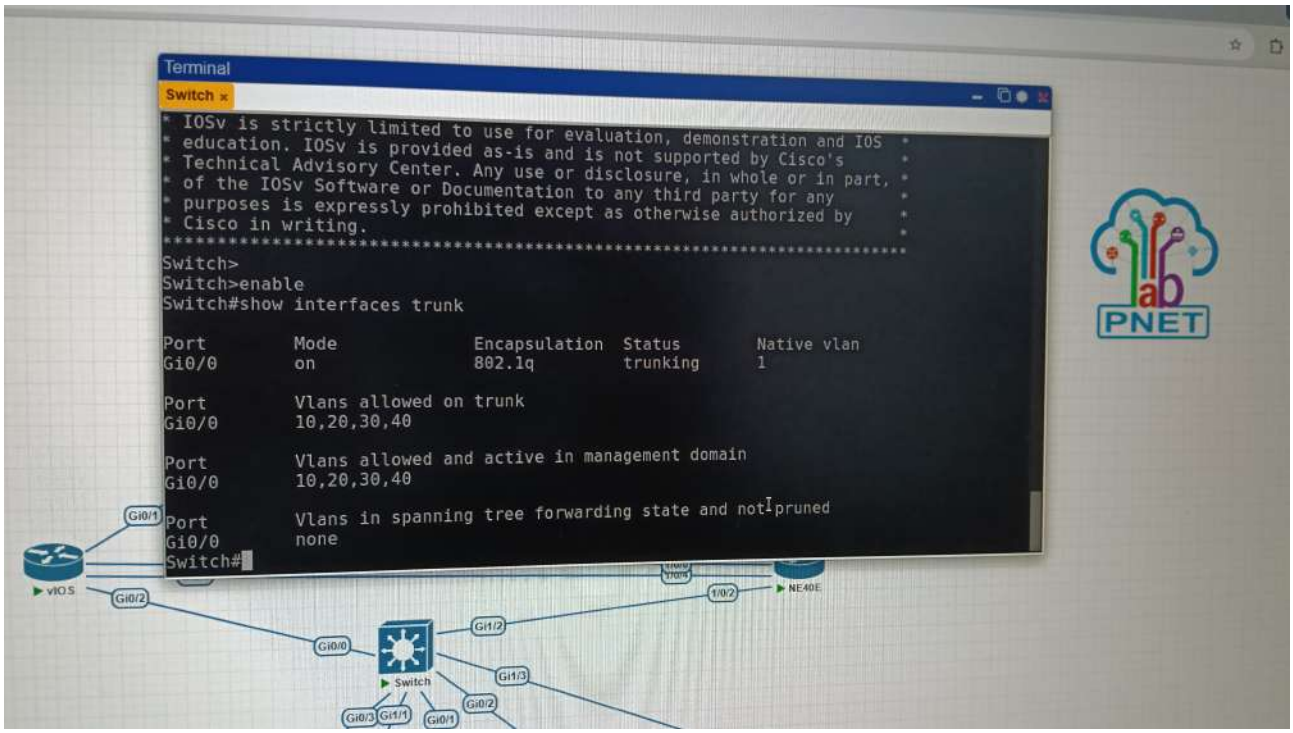
Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa seluruh konfigurasi pada modul ini berhasil dijalankan dengan baik. VLAN yang dibuat pada Cisco Switch berhasil memisahkan jaringan sesuai kebutuhan, kemudian Cisco Router dapat berfungsi sebagai gateway untuk setiap VLAN sehingga komunikasi antar VLAN dapat dilakukan dengan lancar.

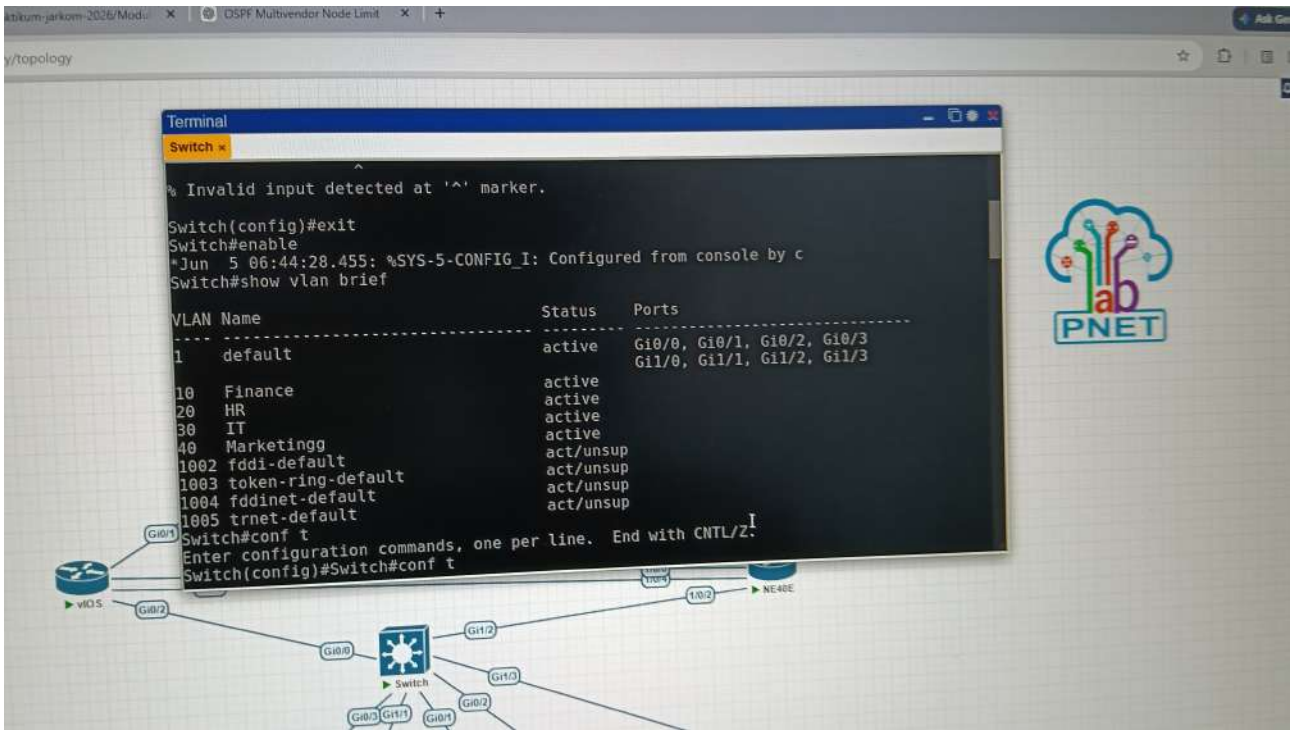
Konfigurasi alamat IP pada Cisco Router, Huawei Router, dan MikroTik Router juga berhasil dilakukan sehingga seluruh perangkat dapat saling terhubung melalui jaringan yang telah dirancang. Setelah konektivitas antar router berhasil dibangun, protokol routing OSPF dapat dijalankan pada ketiga perangkat meskipun berasal dari vendor yang berbeda. Hal ini menunjukkan bahwa OSPF mampu digunakan sebagai solusi routing dinamis pada lingkungan multivendor.

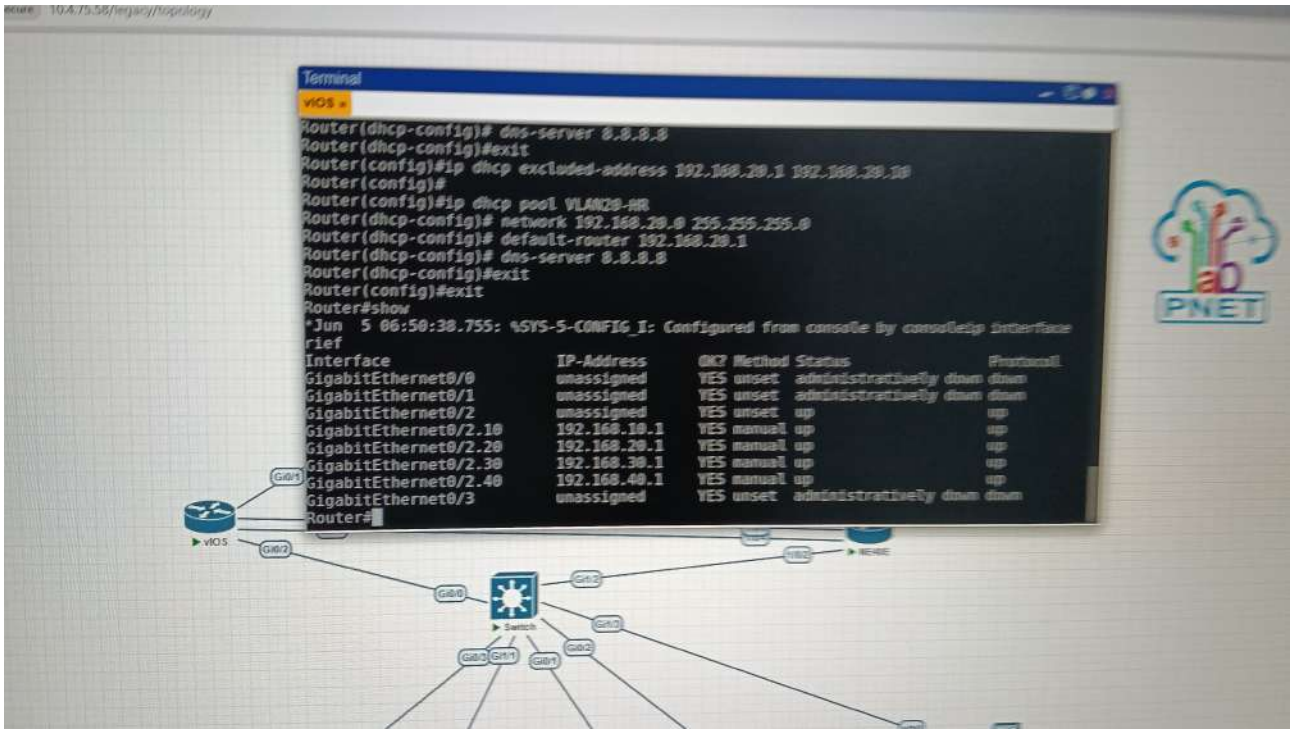
Hasil pengujian juga menunjukkan bahwa setiap router berhasil bertukar informasi routing dan membentuk tabel routing secara otomatis. Selain itu, mekanisme failover OSPF dapat bekerja sesuai harapan. Ketika jalur utama mengalami gangguan, trafik secara otomatis dialihkan ke jalur cadangan melalui MikroTik tanpa perlu melakukan konfigurasi ulang secara manual. Setelah jalur utama kembali aktif, OSPF kembali memilih jalur terbaik berdasarkan nilai cost yang telah ditentukan.

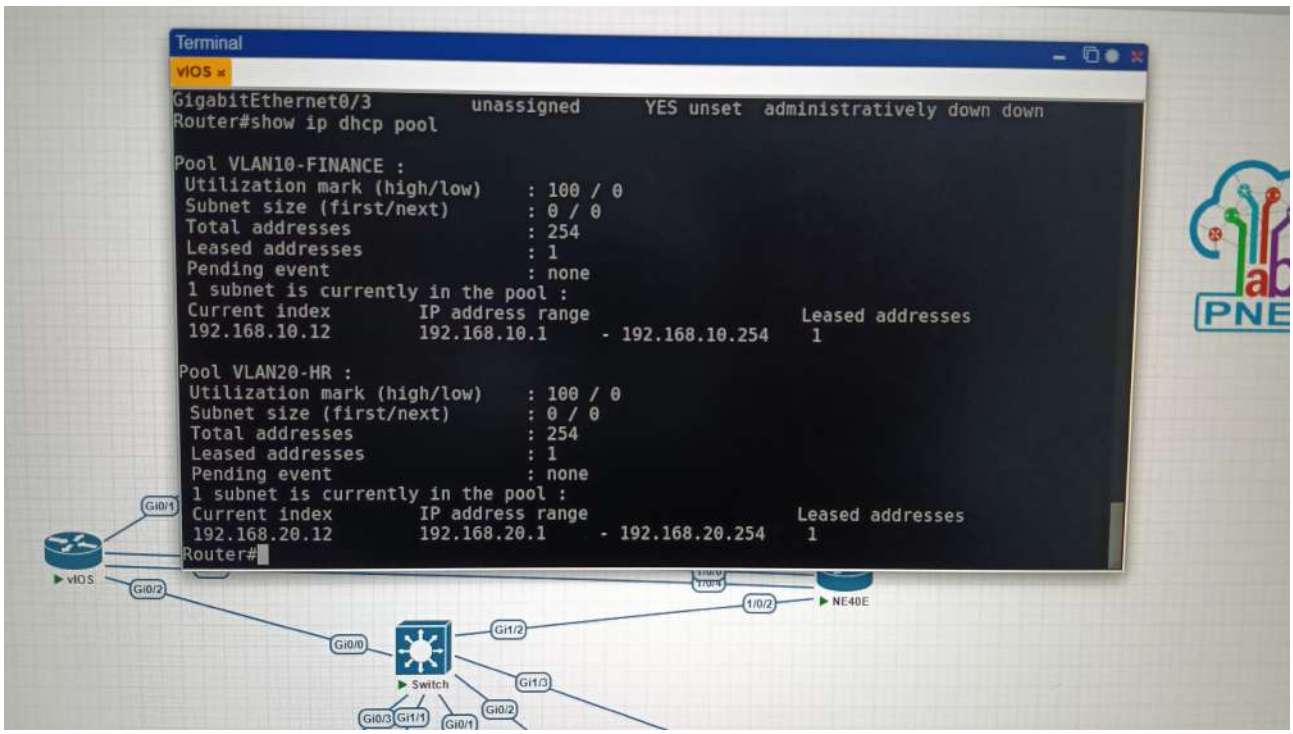
5 Lampiran

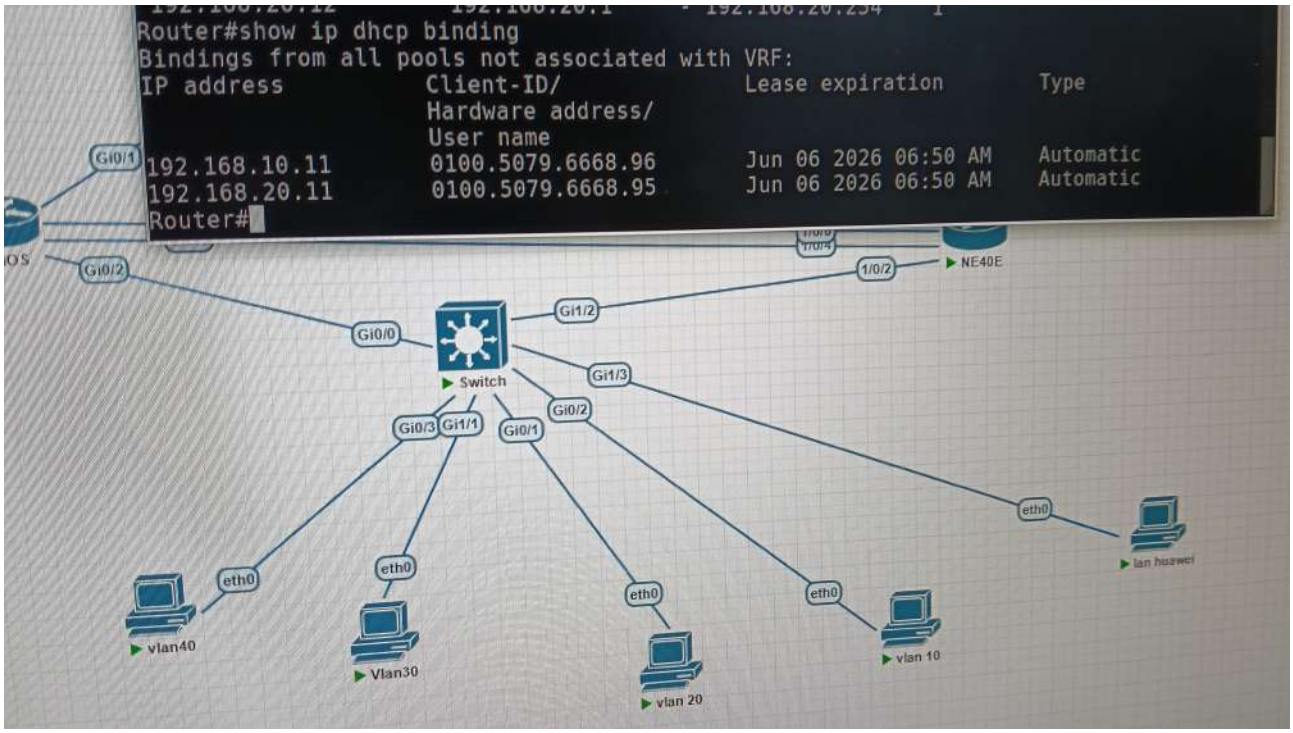
5.1 Dokumentasi saat praktikum

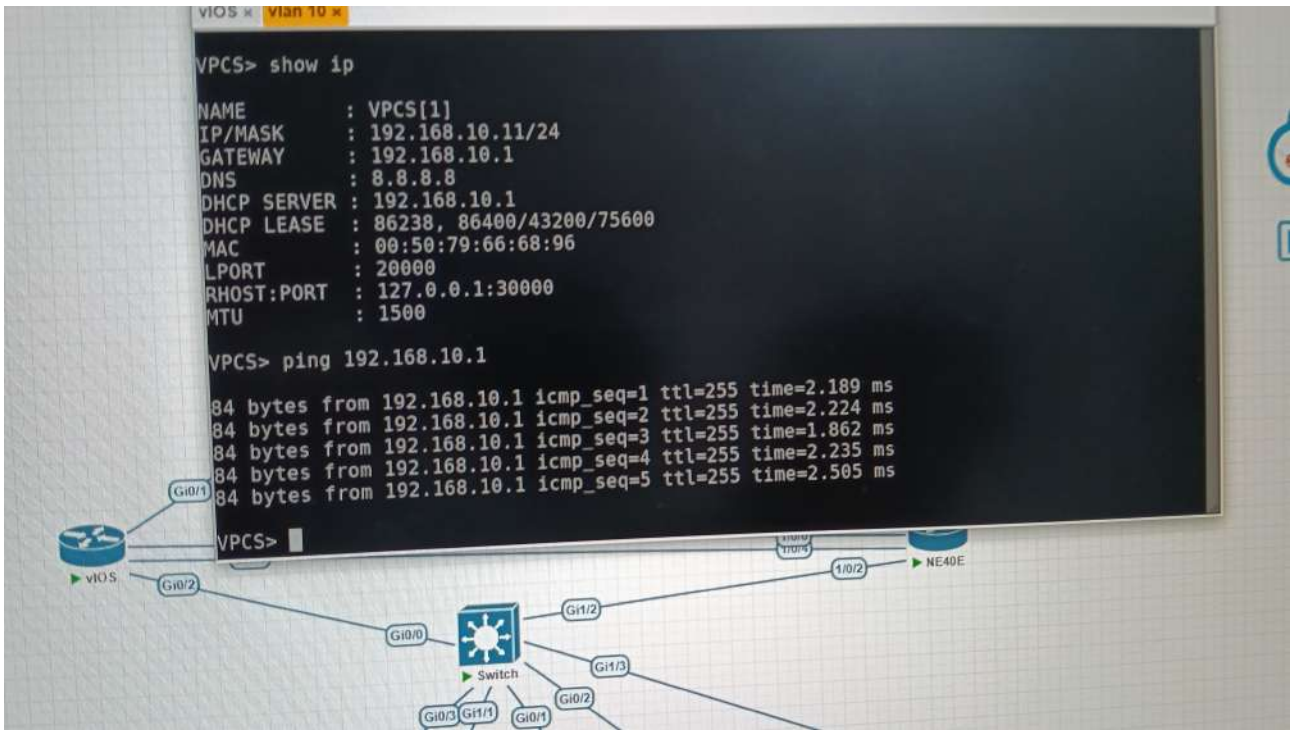


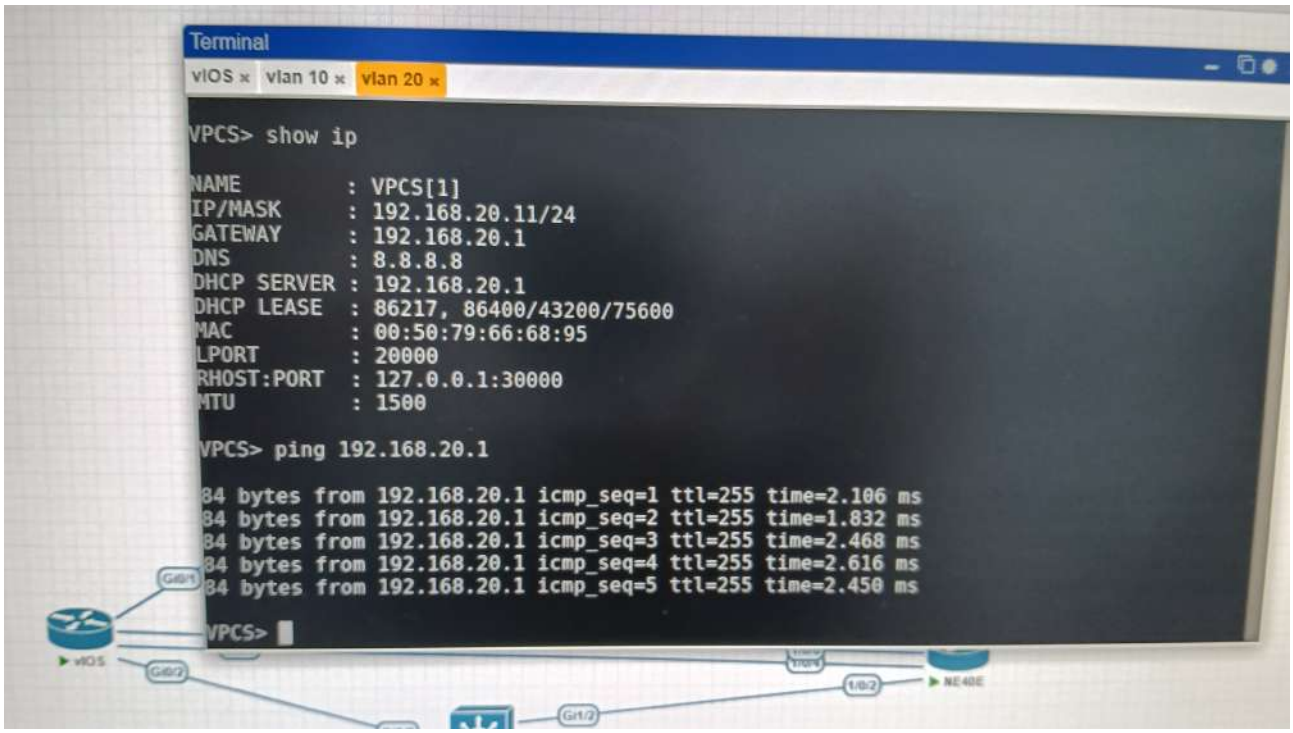

Terminal
Switch
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch#enable
*Jun 5 06:44:28.455: %SYS-5-CONFIG_I: Configured from console by c
Switch#show vlan brief
VLAN Name Status Ports
1 default active Gi0/0, Gi0/1, Gi0/2, Gi0/3
Gi1/0, Gi1/1, Gi1/2, Gi1/3
10 Finance active
20 HR active
30 IT active
40 Marketingg active
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Switch#conf t
PNET

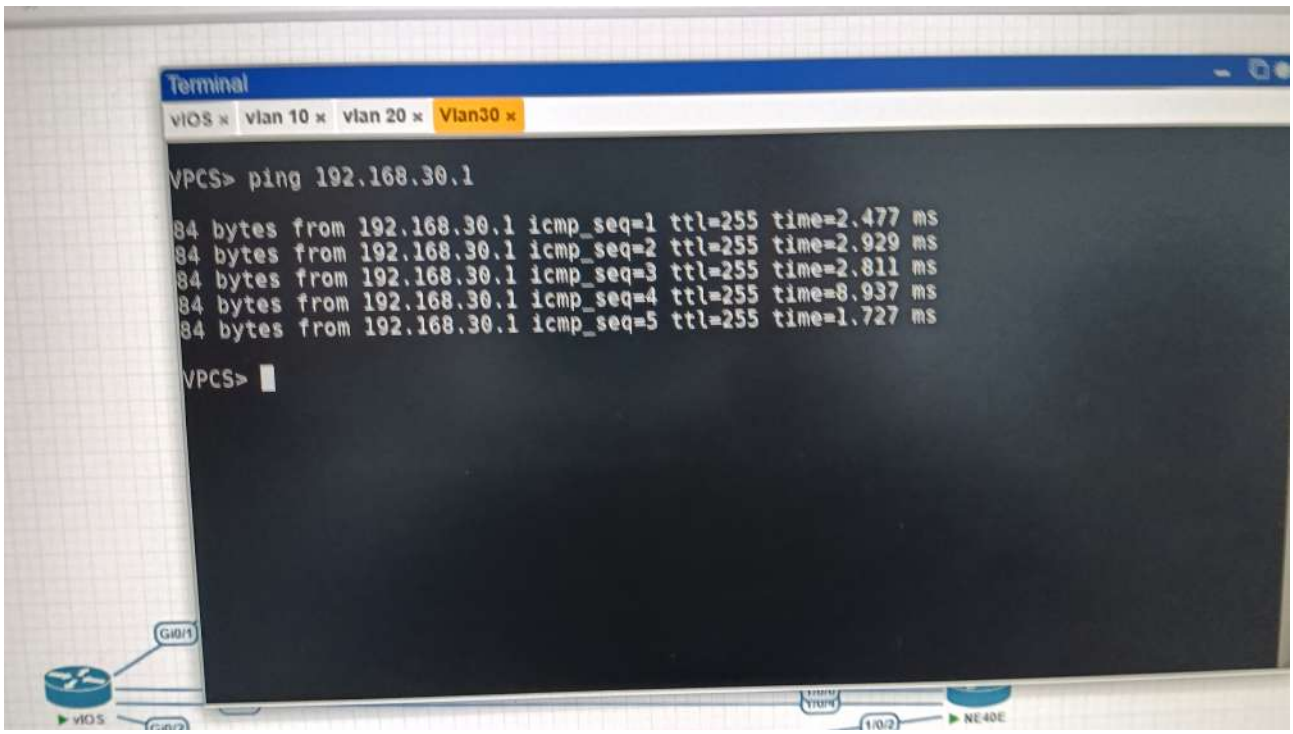

Terminal
vIOS
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#show
*Jun 5 06:50:38.755: %SYS-5-CONFIG_I: Configured from console by consoleip interface
rief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 unassigned YES unset administratively down down
Router#
PNET

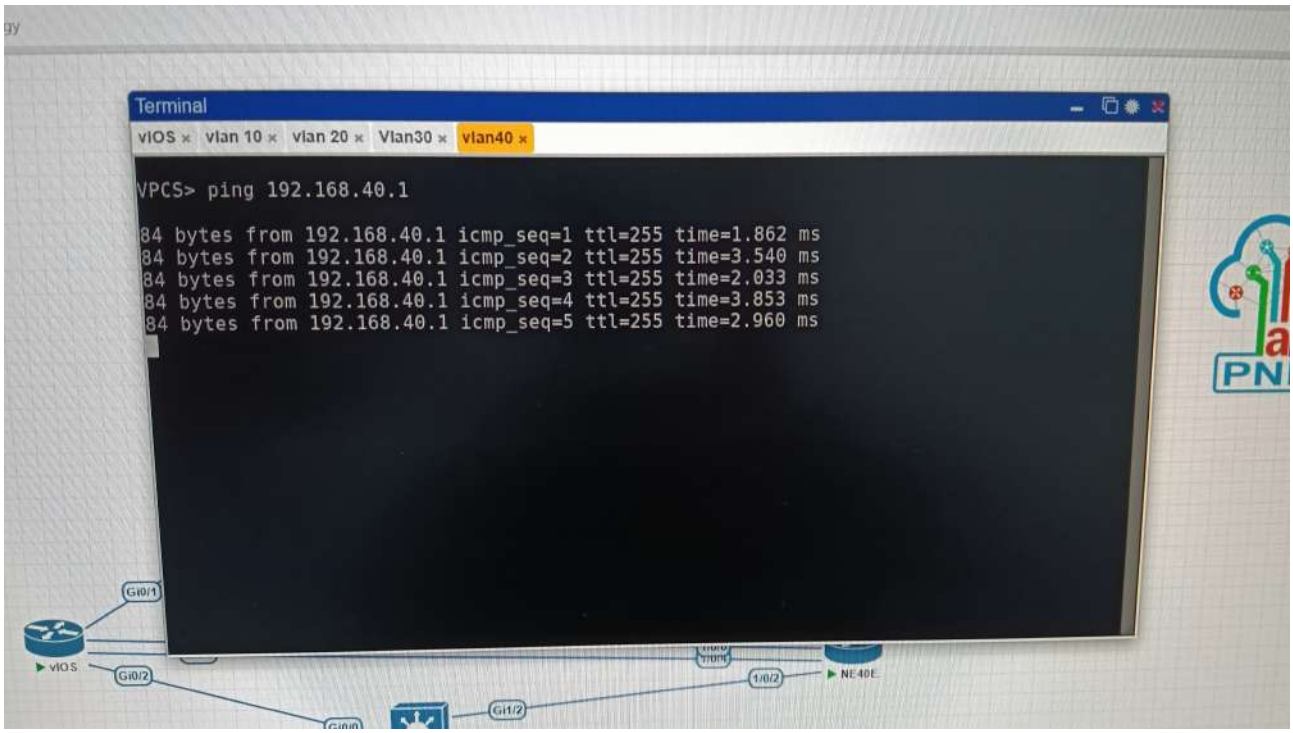

Terminal
vIOS
GigabitEthernet0/3 unassigned YES unset administratively down down
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1
Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#
Gi0/1
vIOS
Gi0/2
Gi0/0
Switch
Gi1/2
1/0/2
NE40E
Gi1/3
Gi0/3
Gi1/1
Gi0/1
Gi0/2

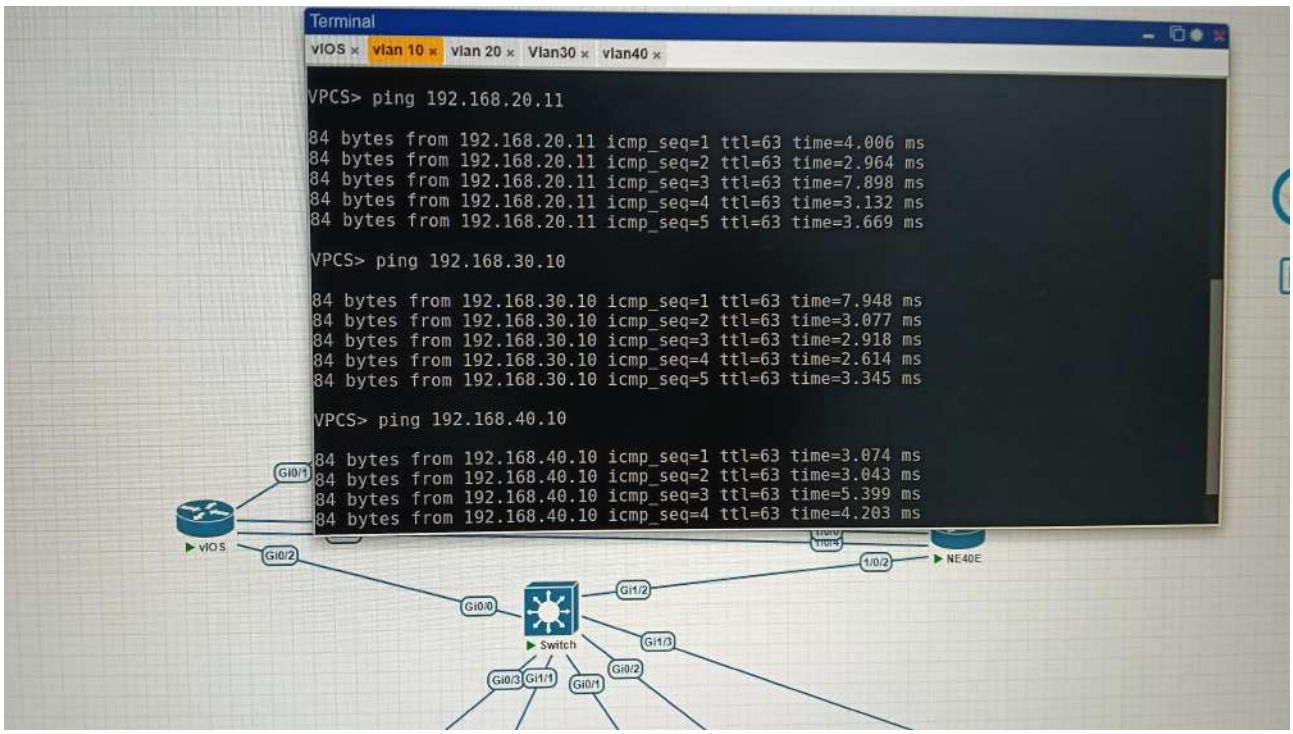

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Lease expiration Type
Hardware address/
User name
192.168.10.11 0100.5079.6668.96 Jun 06 2026 06:50 AM Automatic
192.168.20.11 0100.5079.6668.95 Jun 06 2026 06:50 AM Automatic
Router#
Gi0/1
Gi0/2
Gi0/0
Switch
Gi1/2
1/0/2
NE40E
Gi1/3
Gi0/3
Gi1/1
Gi0/1
Gi0/2
eth0
vlan40
Vlan30
vlan 20
vlan 10
lan huawei

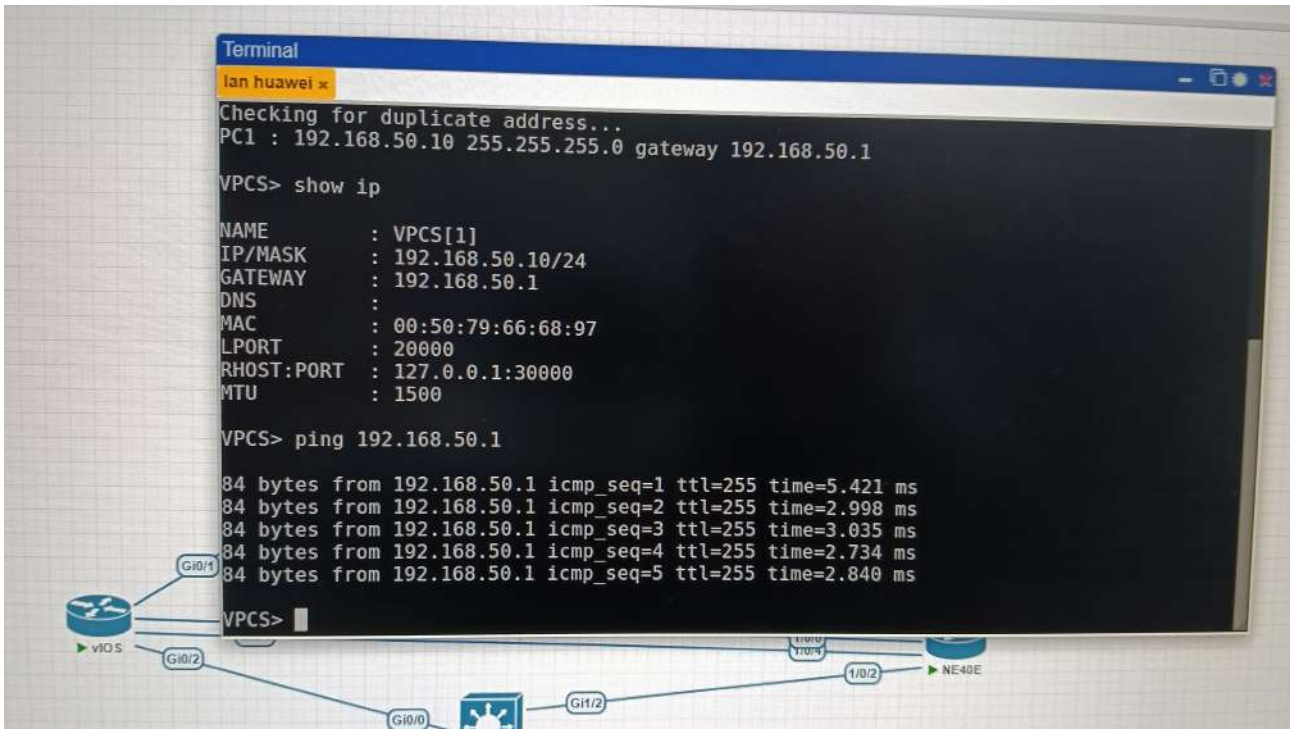

vIOS
vlan 10
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.10.11/24
GATEWAY : 192.168.10.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86238, 86400/43200/75600
MAC : 00:50:79:66:68:96
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.189 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.224 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.862 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=2.235 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.505 ms
VPCS>
Gi0/1
Gi0/2
Gi0/0
Switch
Gi1/2
Gi1/3
Gi0/2
Gi0/3
Gi1/1
Gi0/1
1/0/2
NE40E

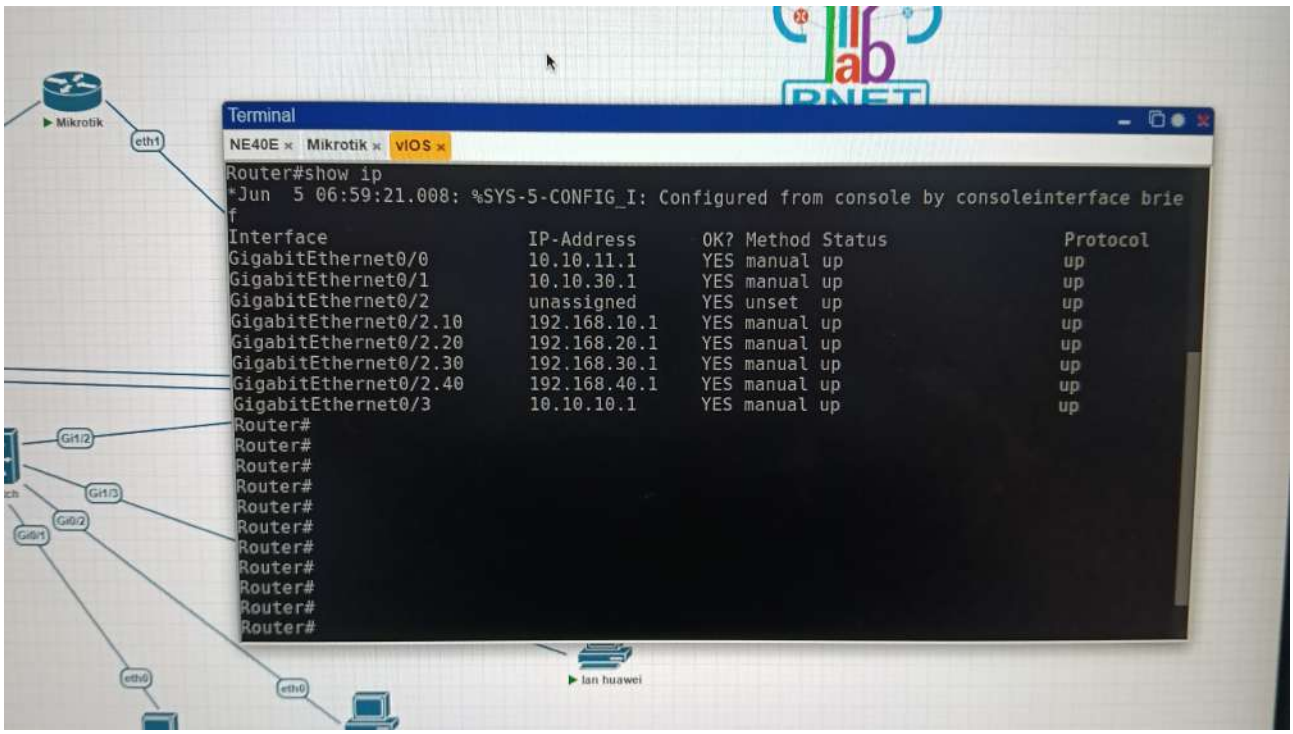

Terminal
vIOS
vlan 10
vlan 20
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.20.11/24
GATEWAY : 192.168.20.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86217, 86400/43200/75600
MAC : 00:50:79:66:68:95
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.106 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.832 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.468 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.616 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.450 ms
VPCS>
vIOS
NE40E

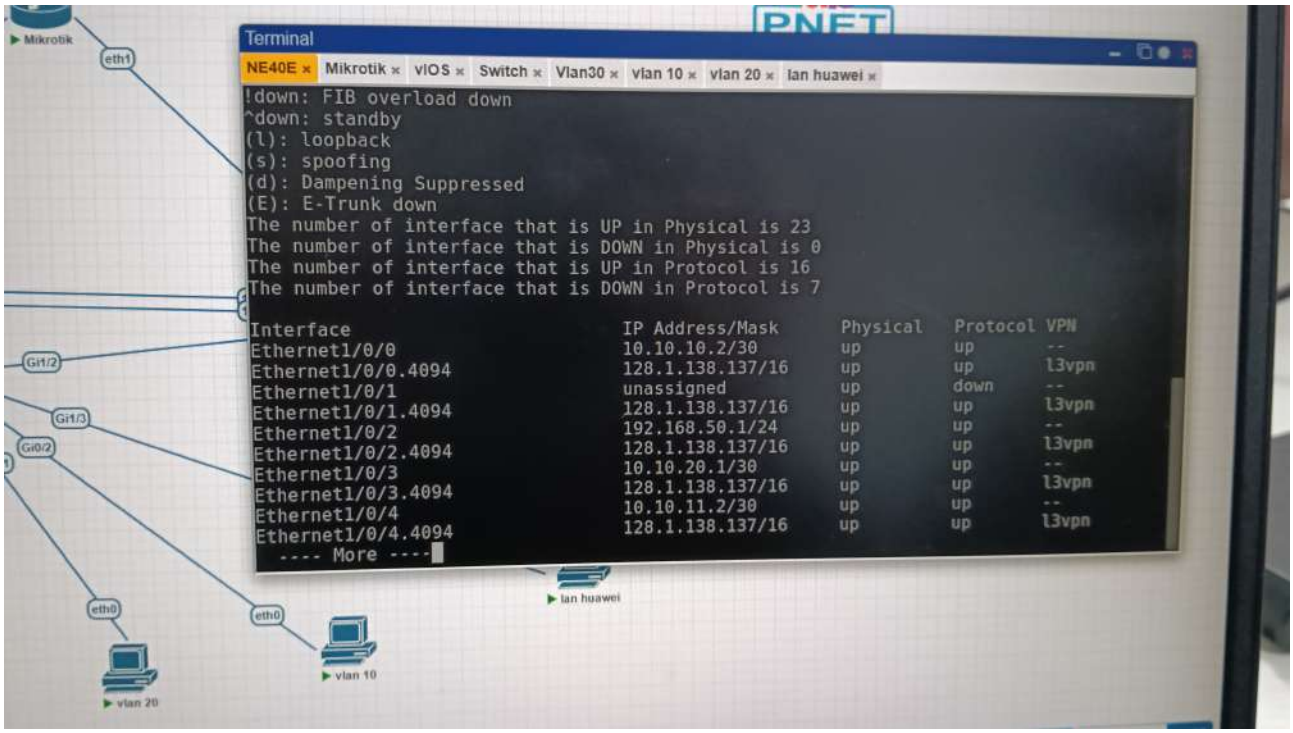

Terminal
vIOS × vlan 10 × vlan 20 × Vlan30 ×
VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.477 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=2.929 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=2.811 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=8.937 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.727 ms
VPCS>
Gi0/1
vIOS
1/0/2
NE40E

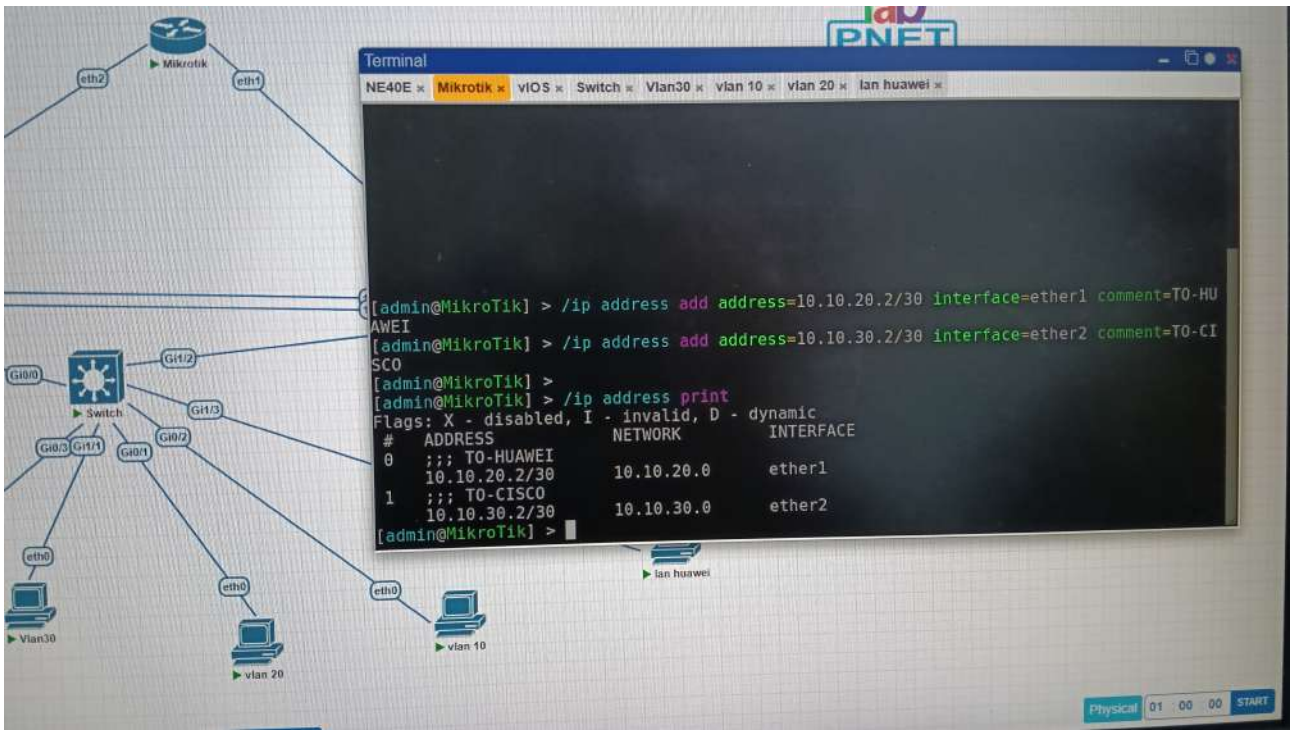

Terminal
vIOS × vlan 10 × vlan 20 × Vlan30 × vlan40 ×
VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.862 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.540 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.033 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.853 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.960 ms
Gi0/1
vIOS
Gi0/2
Gi1/2
1/0/2
NE40E

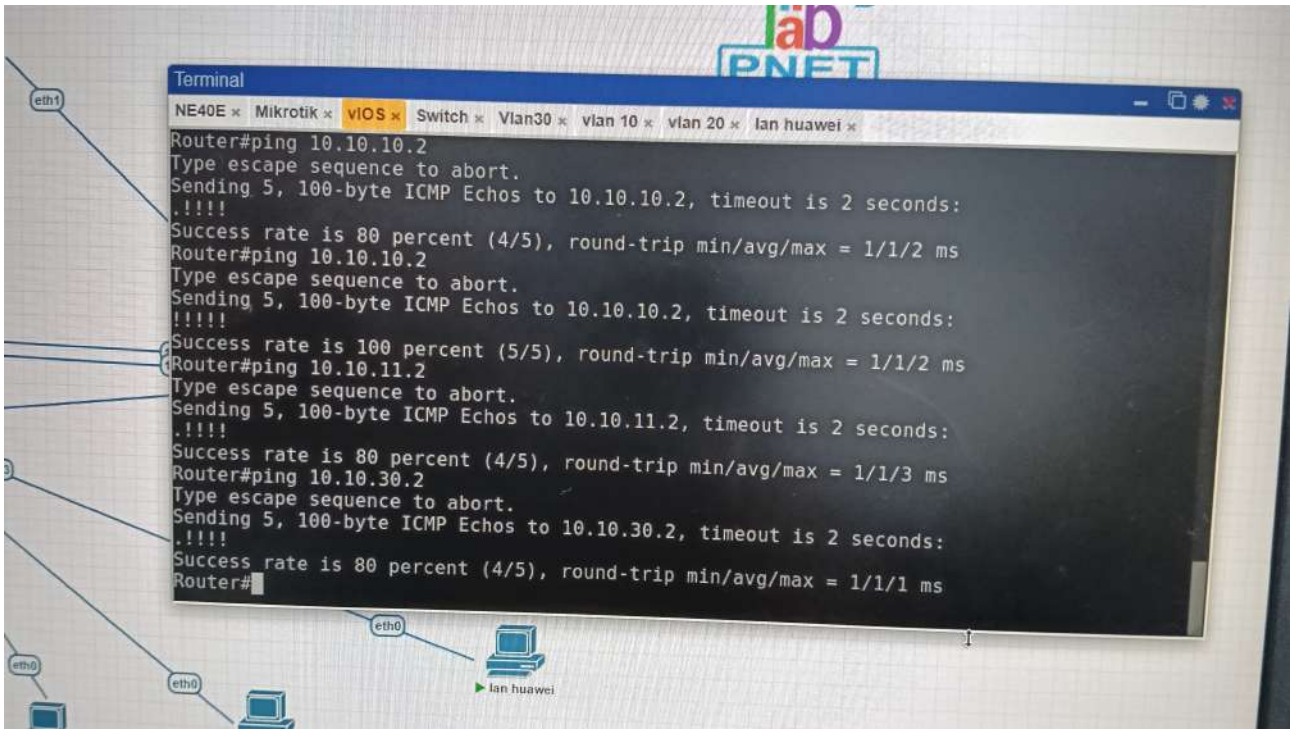

Terminal
vIOS
vlan 10
vlan 20
Vlan30
vlan40
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=4.006 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.964 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=7.898 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.132 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=3.669 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=7.948 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.077 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.918 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.614 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.345 ms
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.074 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.043 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.399 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.203 ms
Gi0/1
vIOS
Gi0/2
Gi0/0
Switch
Gi1/2
Gi1/3
Gi0/3
Gi1/1
Gi0/1
Gi0/2
1/0/2
NE40E

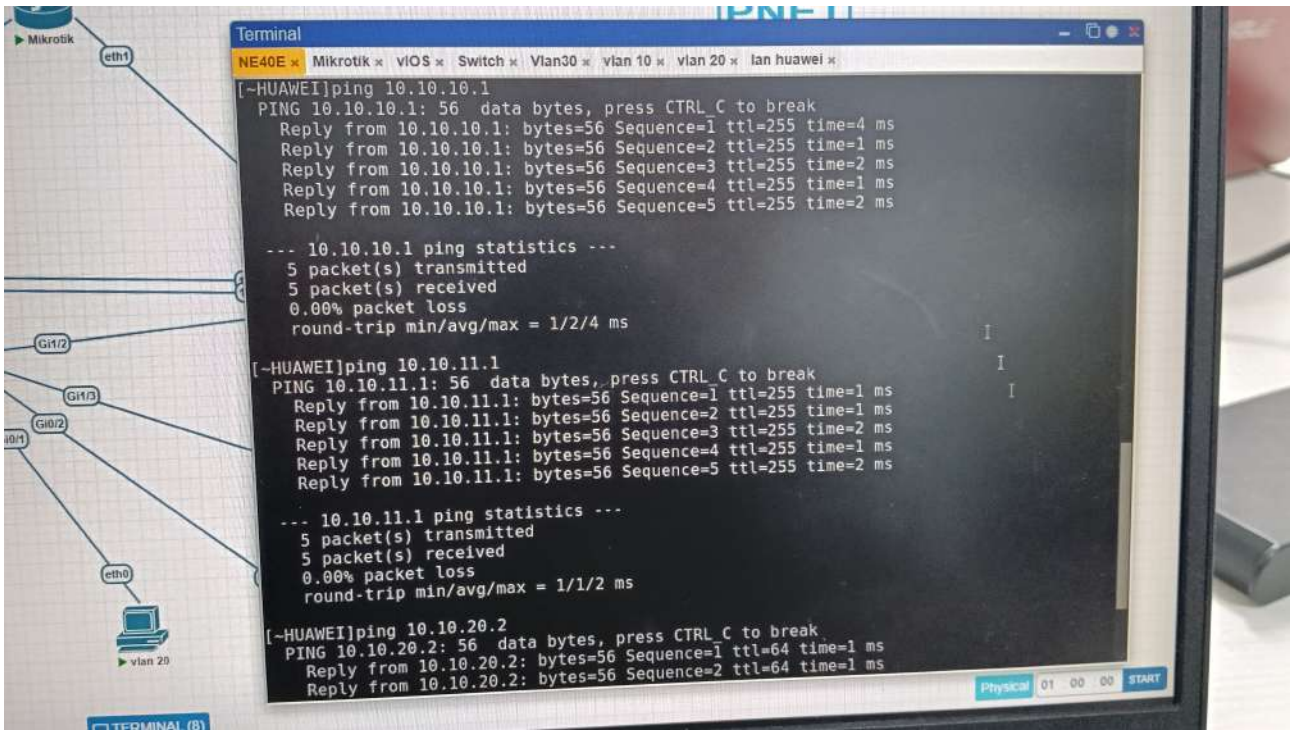

Terminal
lan huawei
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.50.10/24
GATEWAY : 192.168.50.1
DNS :
MAC : 00:50:79:66:68:97
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=5.421 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.998 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.035 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=2.734 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=2.840 ms
VPCS>
Gi0/1
vIOS
Gi0/2
Gi0/0
Gi1/2
1/0/2
NE40E

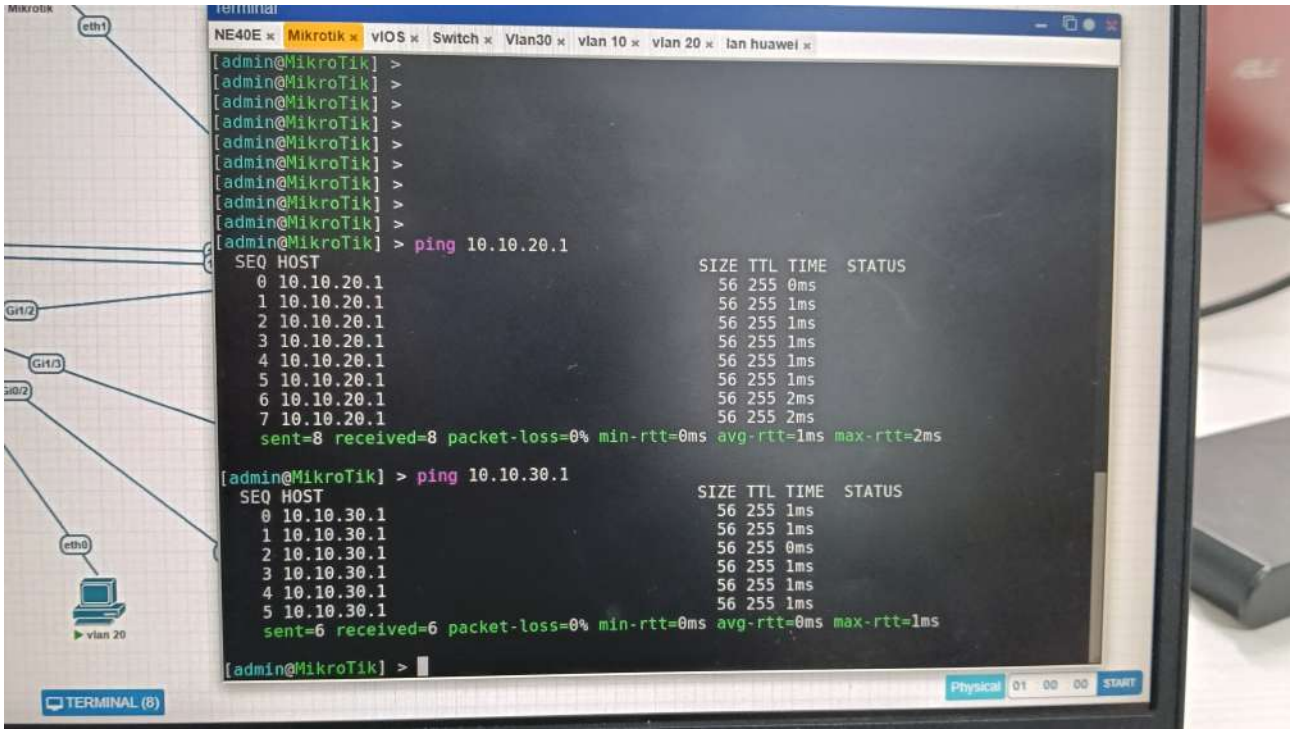

Terminal
NE40E × Mikrotik × vIOS ×
Router#show ip
*Jun 5 06:59:21.008: %SYS-5-CONFIG_I: Configured from console by consoleinterface brie
f
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
eth0
lan huawei

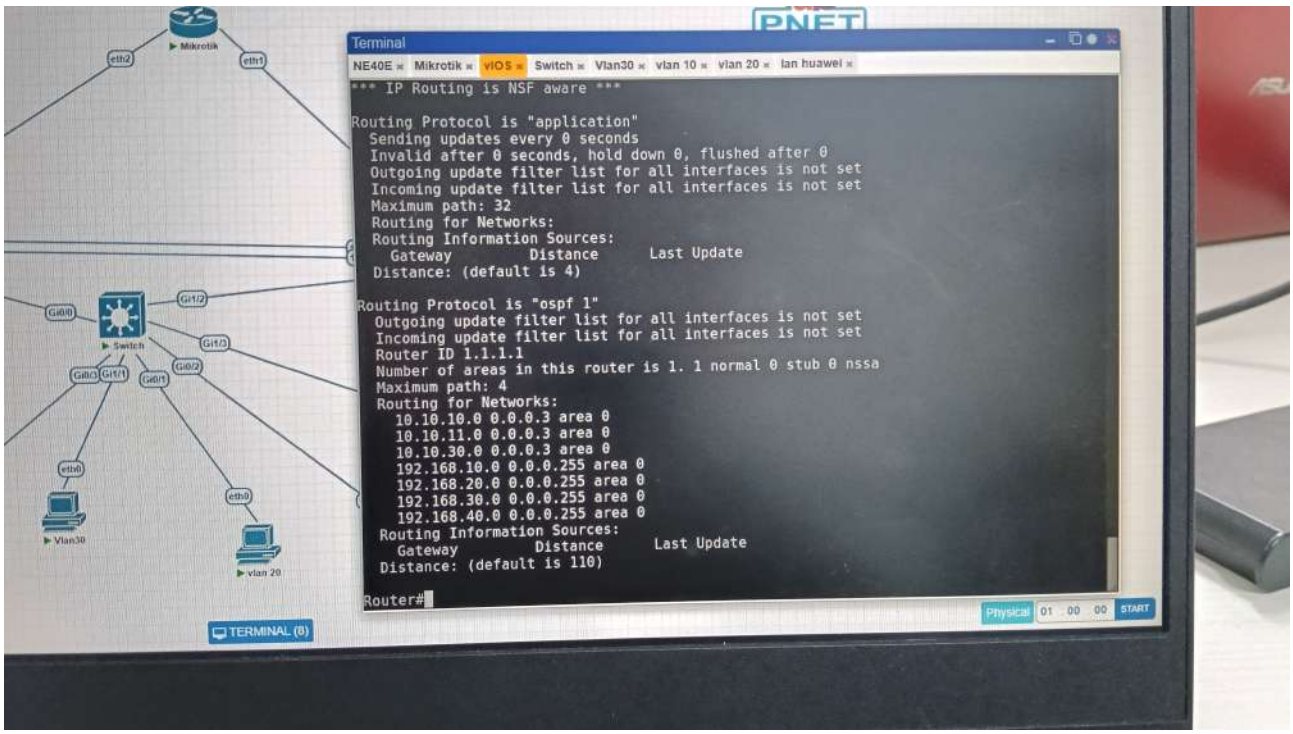

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7
Interface IP Address/Mask Physical Protocol VPN
Ethernet1/0/0 10.10.10.2/30 up up --
Ethernet1/0/0.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/1 unassigned up down --
Ethernet1/0/1.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/2 192.168.50.1/24 up up --
Ethernet1/0/2.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/3 10.10.20.1/30 up up --
Ethernet1/0/3.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/4 10.10.11.2/30 up up --
Ethernet1/0/4.4094 128.1.138.137/16 up up l3vpn
---- More ----
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
eth0
eth0
lan huawei
vlan 10
vlan 20

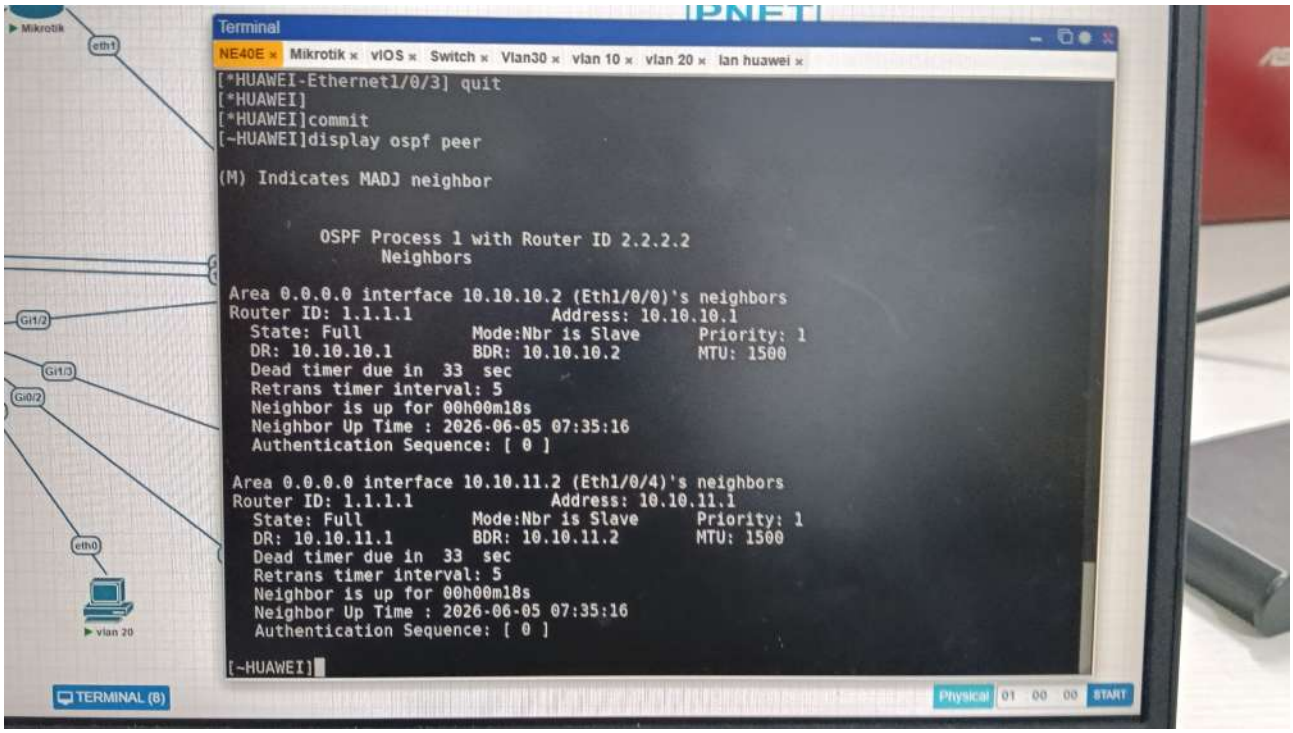

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] >
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
ADDRESS NETWORK INTERFACE
0 ;;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >
Mikrotik
Switch
Vlan30
vlan 20
vlan 10
lan huawei
Physical
START

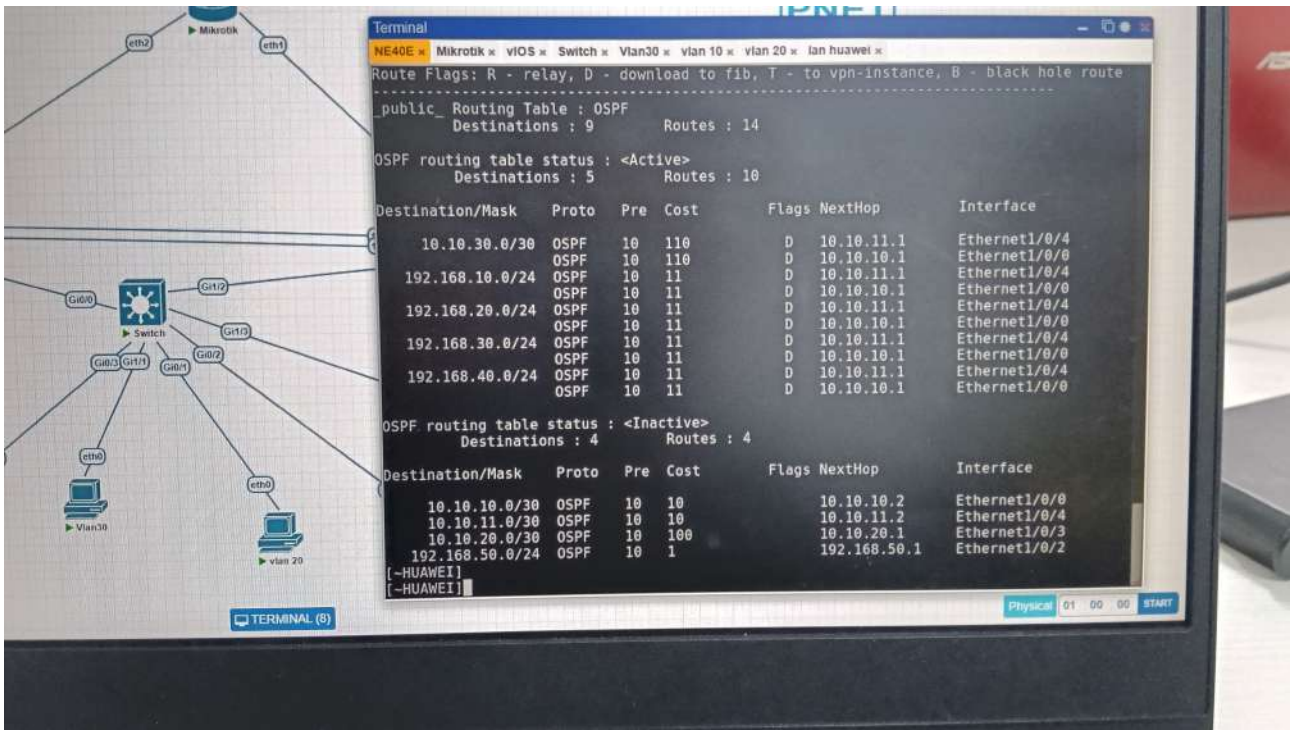

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#
lan huawei

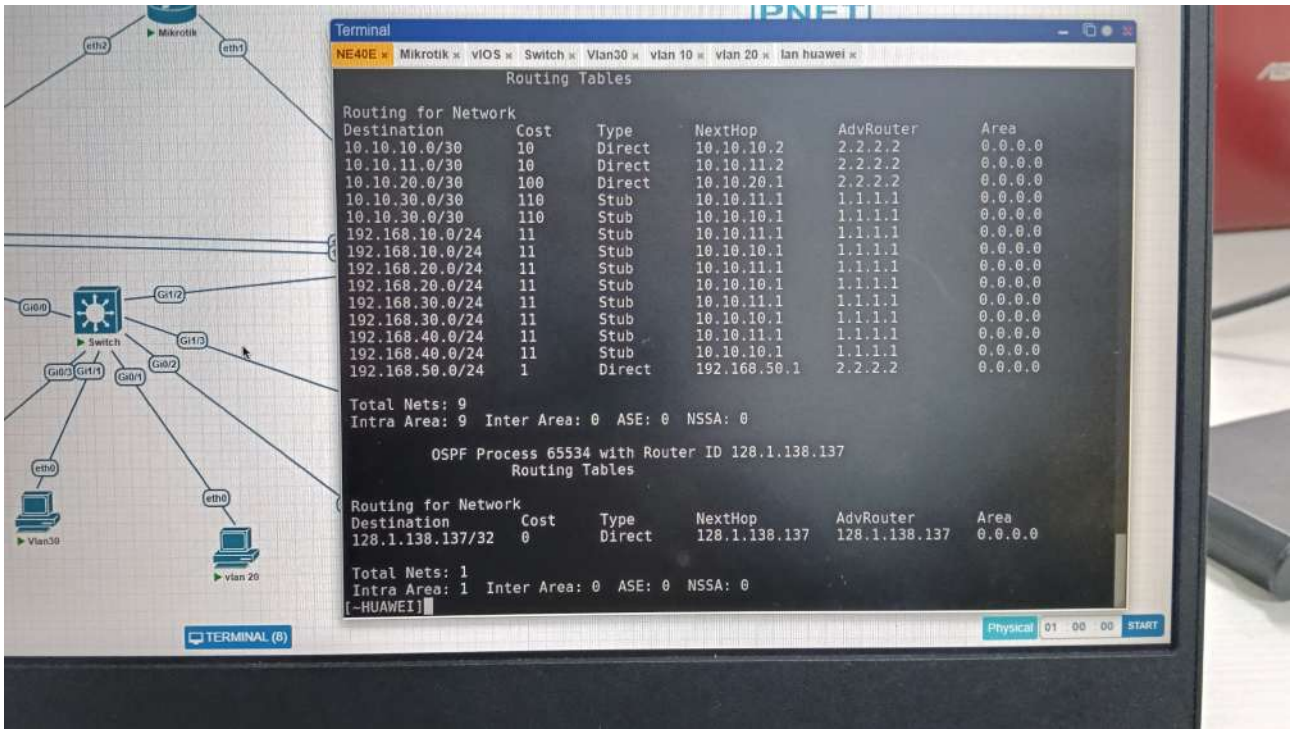

Mikrotik
eth1
Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=4 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/4 ms
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Gi1/2
Gi1/3
Gi0/2
eth0
vlan 20
Physical
START

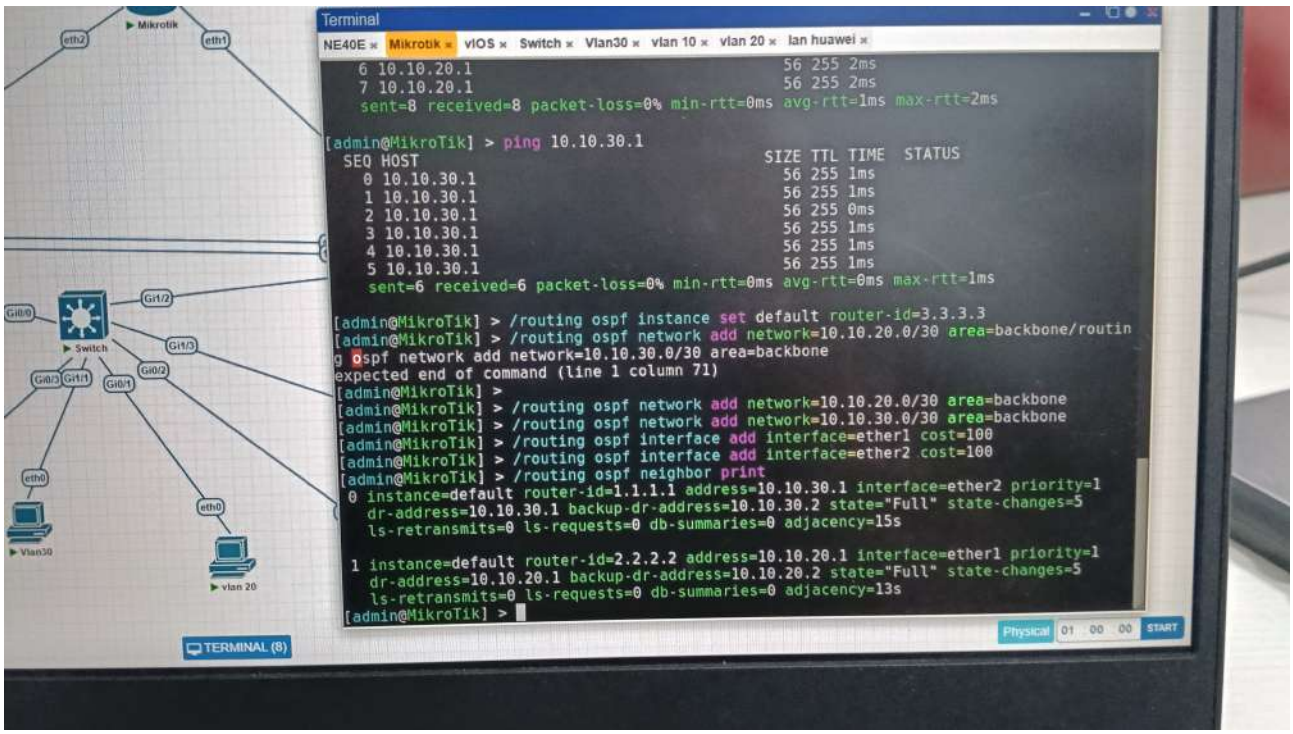

eth1
Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[admin@MikroTik] >
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 0ms
1 10.10.20.1 56 255 1ms
2 10.10.20.1 56 255 1ms
3 10.10.20.1 56 255 1ms
4 10.10.20.1 56 255 1ms
5 10.10.20.1 56 255 1ms
6 10.10.20.1 56 255 2ms
7 10.10.20.1 56 255 2ms
sent=8 received=8 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 0ms
3 10.10.30.1 56 255 1ms
4 10.10.30.1 56 255 1ms
5 10.10.30.1 56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms
[admin@MikroTik] >
Gi1/2
Gi1/3
eth0
vlan 20
TERMINAL (8)
Physical
START

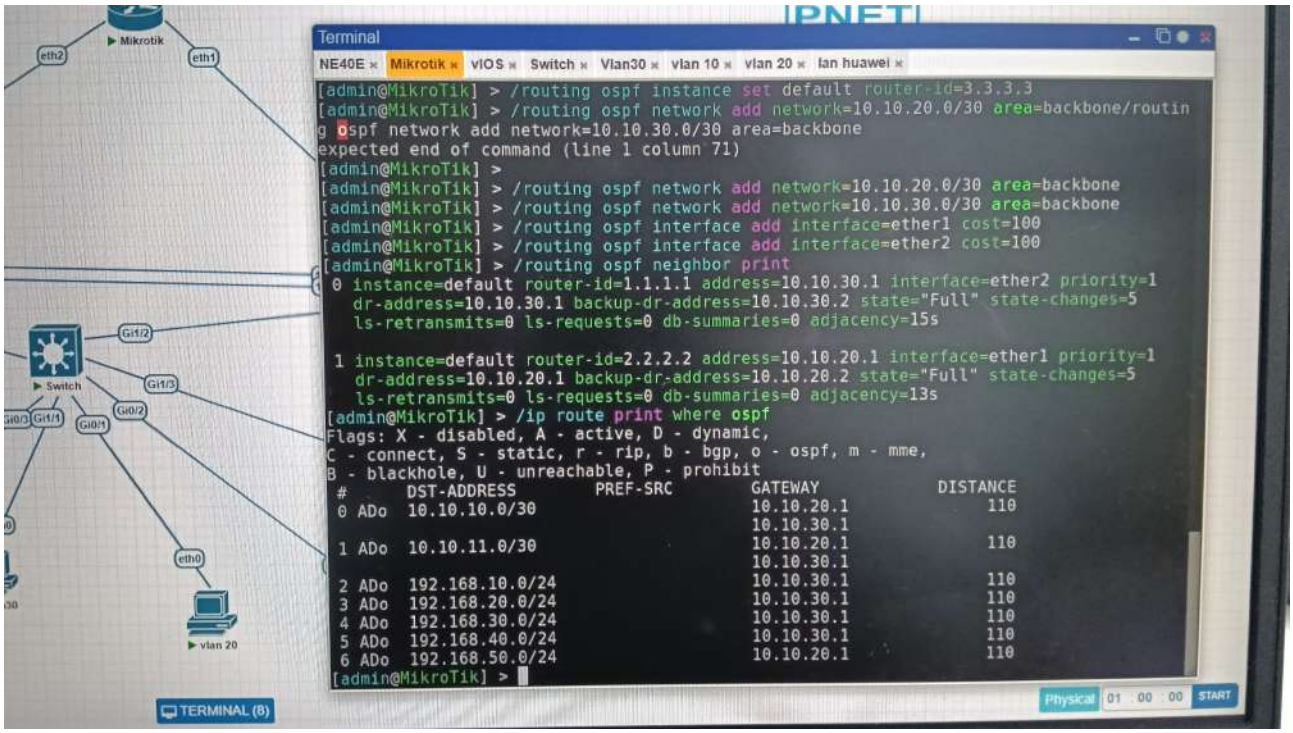

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
*** IP Routing is NSF aware ***
Routing Protocol is "application"
Sending updates every 0 seconds
Invalid after 0 seconds, hold down 0, flushed after 0
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Maximum path: 32
Routing for Networks:
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 4)
Routing Protocol is "ospf 1"
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Router ID 1.1.1.1
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Maximum path: 4
Routing for Networks:
10.10.10.0 0.0.0.3 area 0
10.10.11.0 0.0.0.3 area 0
10.10.30.0 0.0.0.3 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)
Router#
TERMINAL (8)
Physical
START

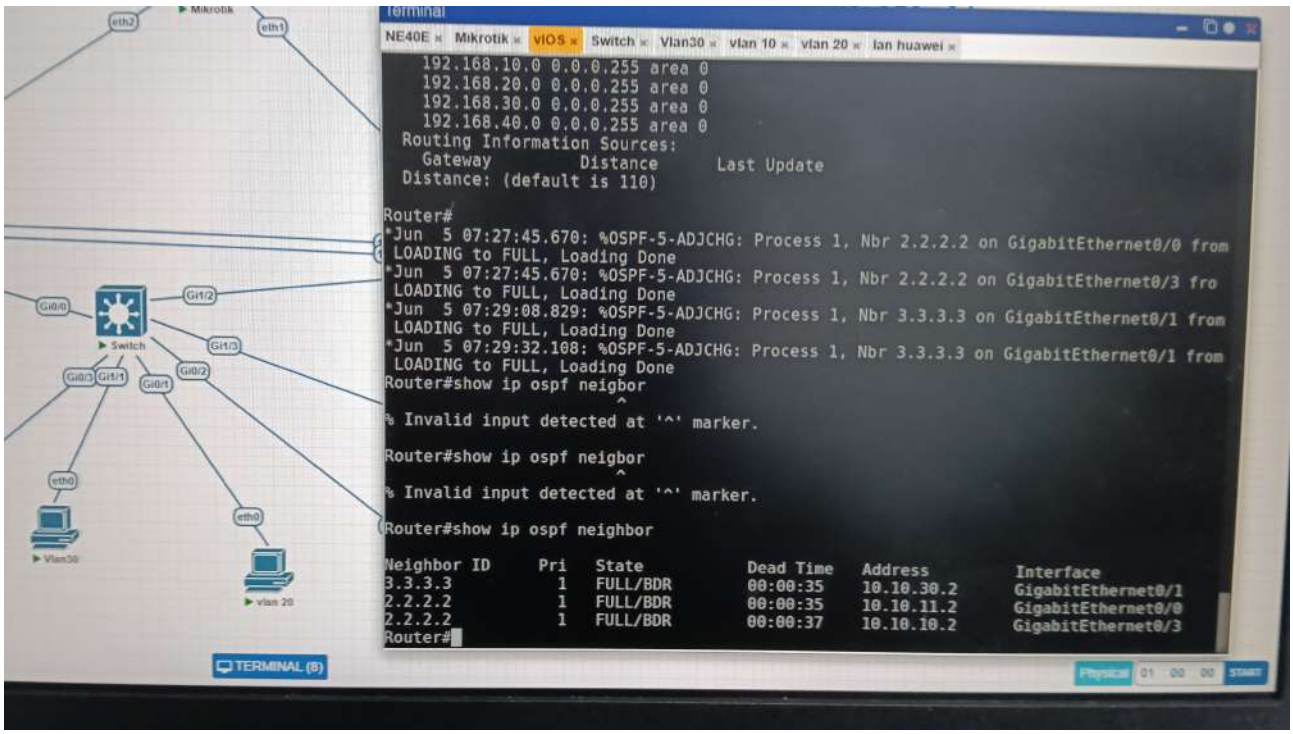

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor
OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h00m18s
Neighbor Up Time : 2026-06-05 07:35:16
Authentication Sequence: [0]
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h00m18s
Neighbor Up Time : 2026-06-05 07:35:16
Authentication Sequence: [0]
[~HUAWEI]
TERMINAL (8)
Physical
START

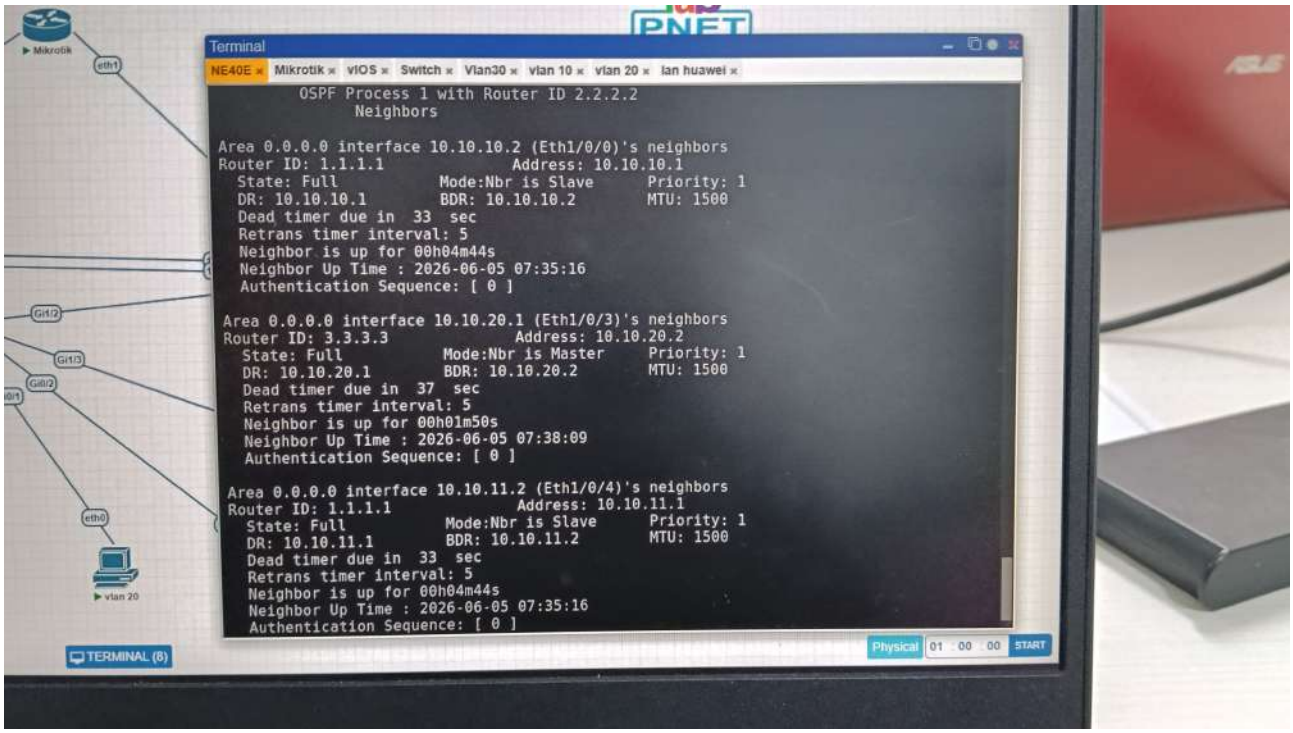

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
public Routing Table : OSPF
Destinations : 9 Routes : 14
OSPF routing table status : <Active>
Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
TERMINAL (8)
Physical
START

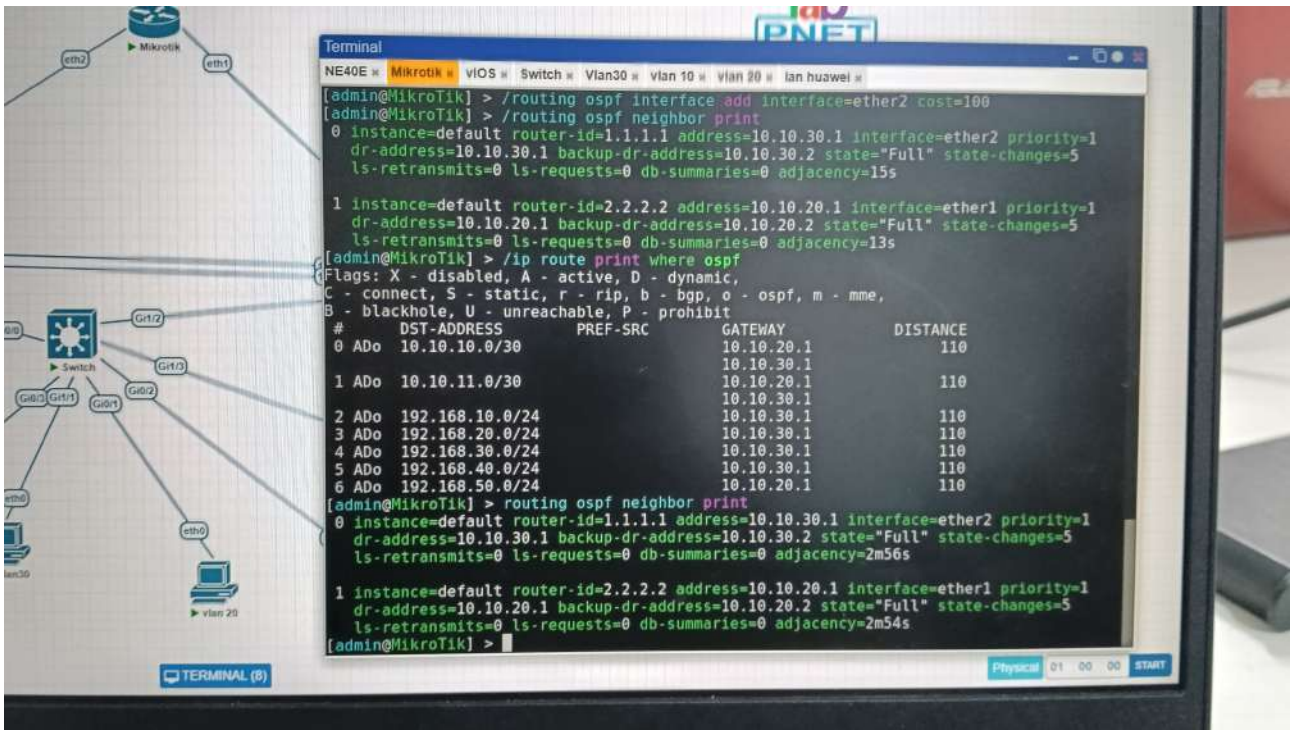

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
Routing Tables
Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0
Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables
Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0
Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]
TERMINAL (8)
Physical
START

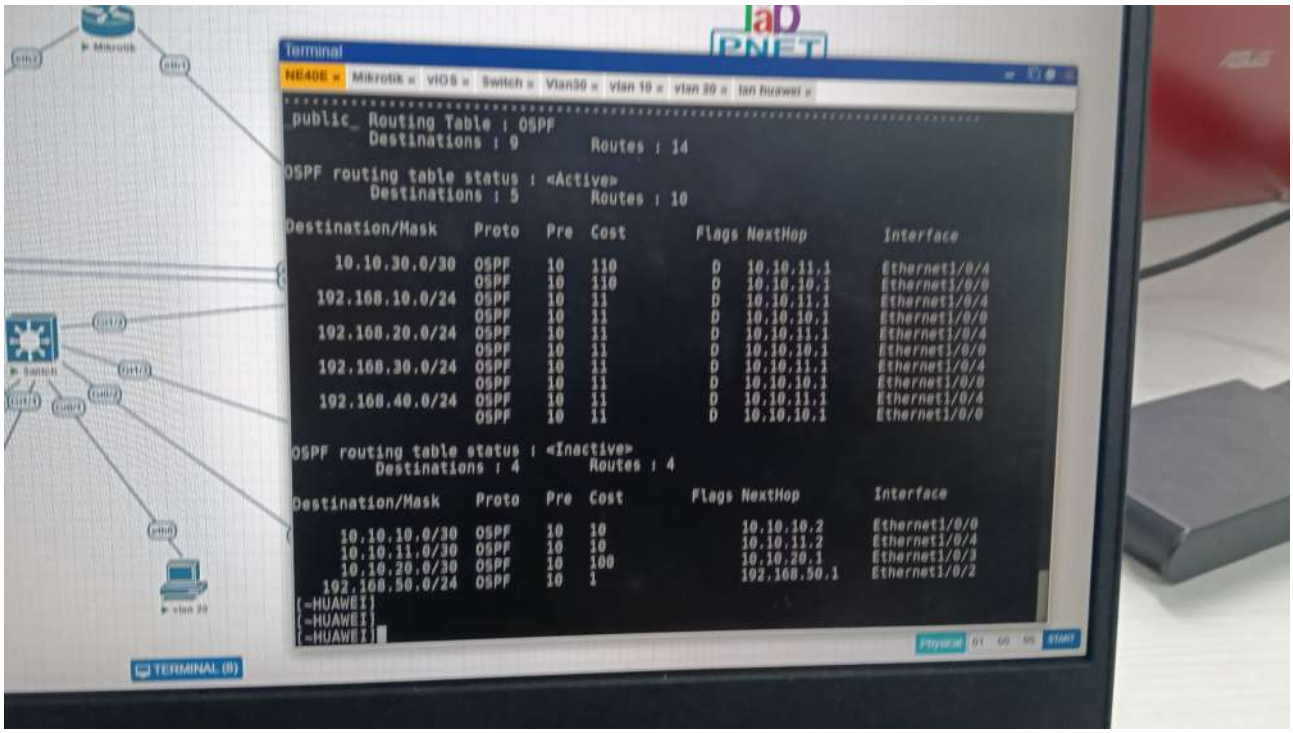

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
6 10.10.20.1 56 255 2ms
7 10.10.20.1 56 255 2ms
sent=8 received=8 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 0ms
3 10.10.30.1 56 255 1ms
4 10.10.30.1 56 255 1ms
5 10.10.30.1 56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms
[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routin
g ospf network add network=10.10.30.0/30 area=backbone
expected end of command (line 1 column 71)
[admin@MikroTik] >
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=15s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=13s
[admin@MikroTik] >
TERMINAL (8)
Physical
START

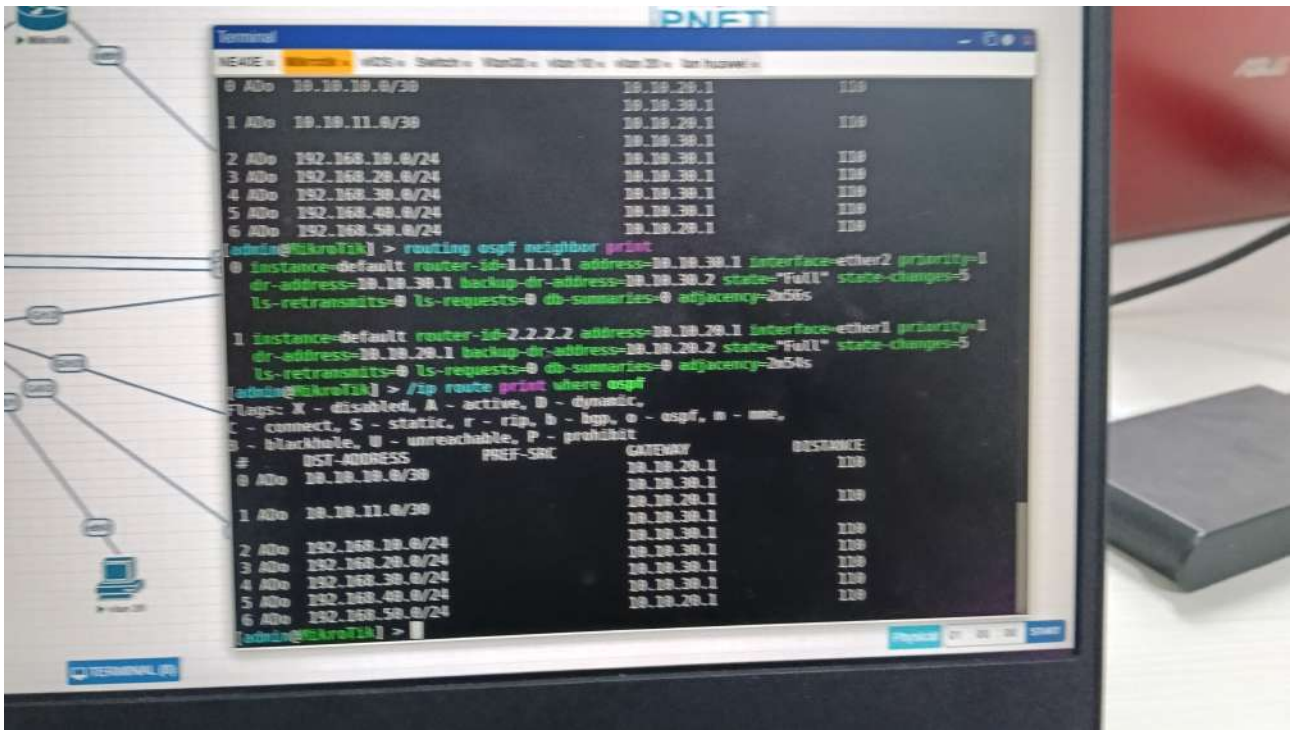

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone/routin
g ospf network add network=10.10.30.0/30 area=backbone
expected end of command (line 1 column 71)
[admin@MikroTik] >
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=15s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=13s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 ADo 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >
TERMINAL (8)
Physical
START

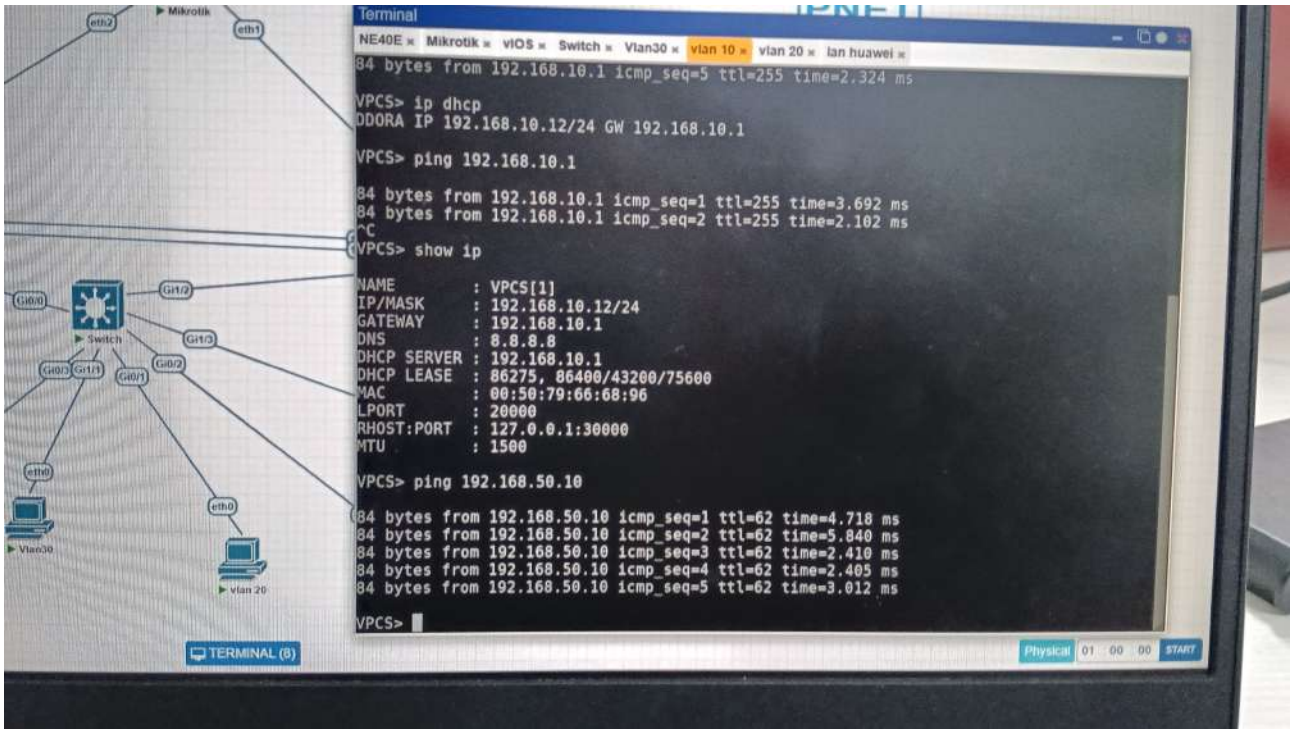

Terminal
NE40E Mikrotik vIOS Switch Vlan30 vlan 10 vlan 20 lan huawei
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)
Router#
*Jun 5 07:27:45.670: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
*Jun 5 07:27:45.670: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 fro
LOADING to FULL, Loading Done
*Jun 5 07:29:08.829: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
*Jun 5 07:29:32.108: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
Router#show ip ospf neigbor
% Invalid input detected at '^' marker.
Router#show ip ospf neigbor
% Invalid input detected at '^' marker.
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:35 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:35 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#
TERMINAL (8)

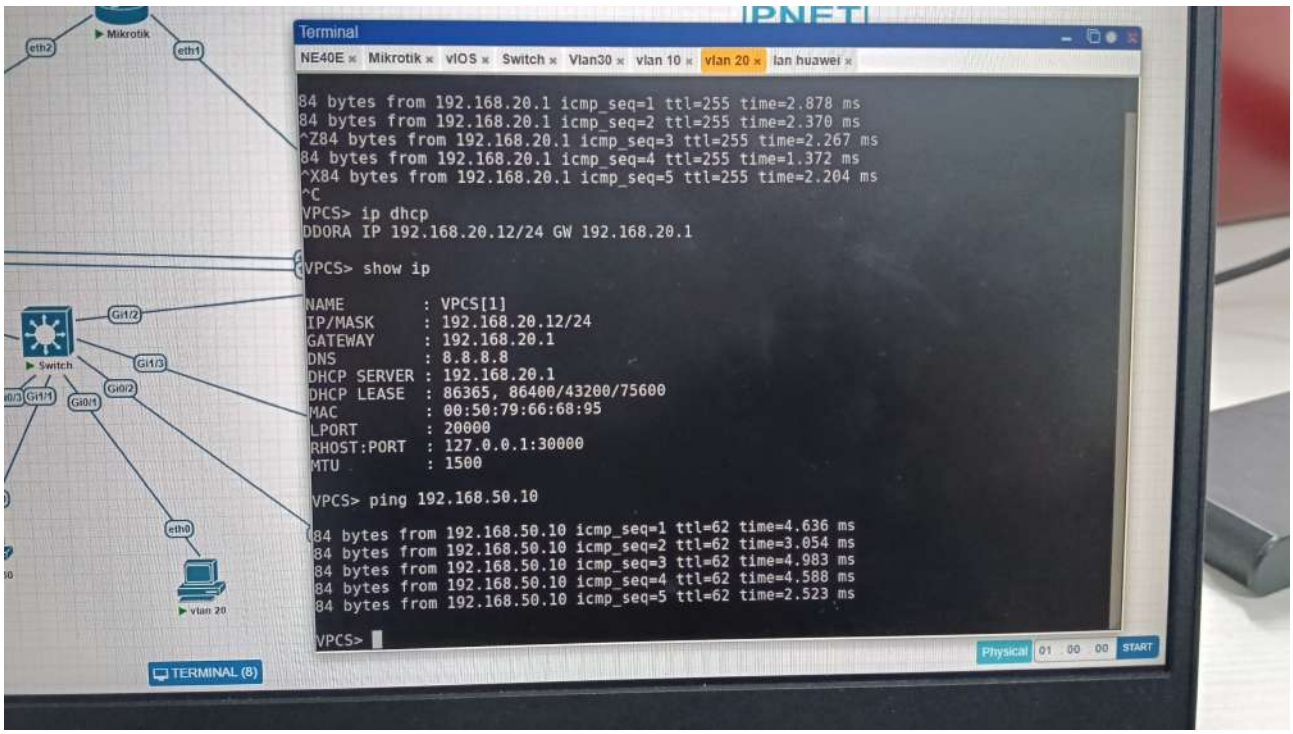

Terminal
NE40E Mikrotik vIOS Switch Vlan30 vlan 10 vlan 20 lan huawei
OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h04m44s
Neighbor Up Time : 2026-06-05 07:35:16
Authentication Sequence: [0]
Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h01m50s
Neighbor Up Time : 2026-06-05 07:38:09
Authentication Sequence: [0]
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h04m44s
Neighbor Up Time : 2026-06-05 07:35:16
Authentication Sequence: [0]
TERMINAL (8)

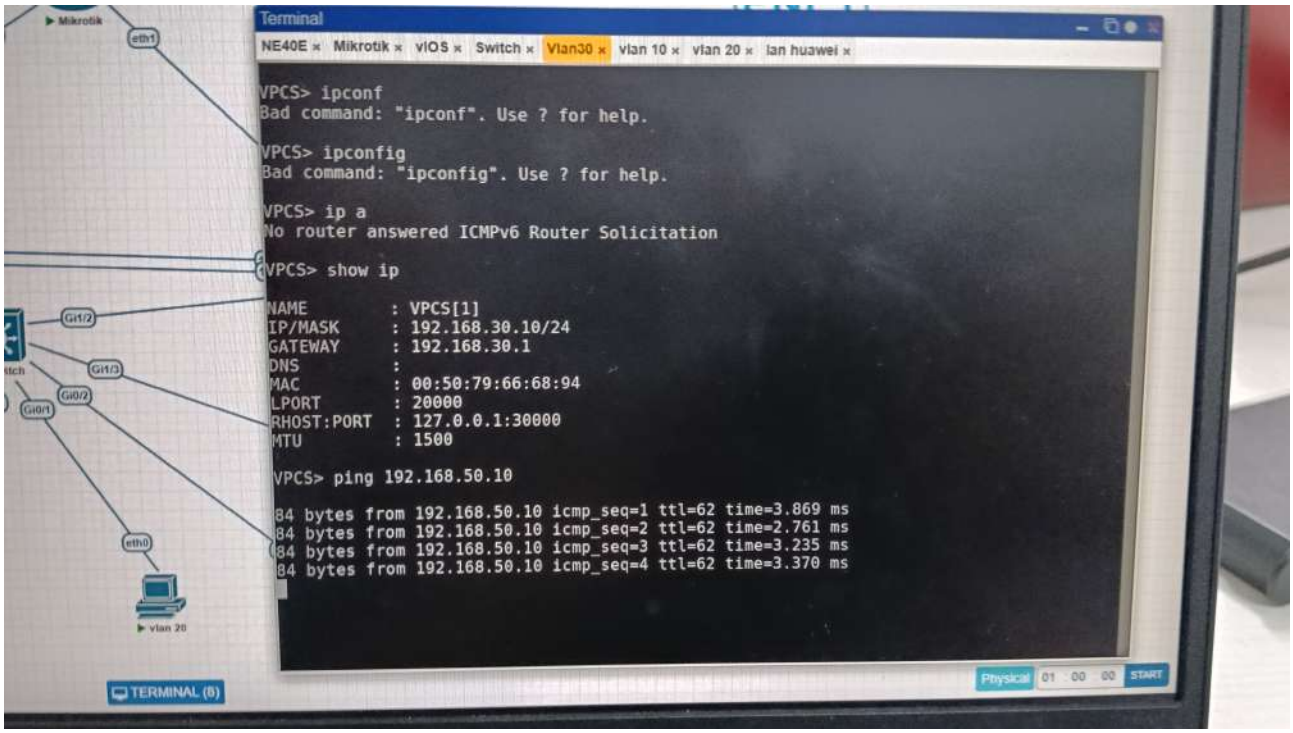

Terminal
NE40E
Mikrotik
vIOS
Switch
Vlan30
vlan 10
vlan 20
lan huawei
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=15s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=13s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 ADo 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m56s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m54s
[admin@MikroTik] >
TERMINAL (8)

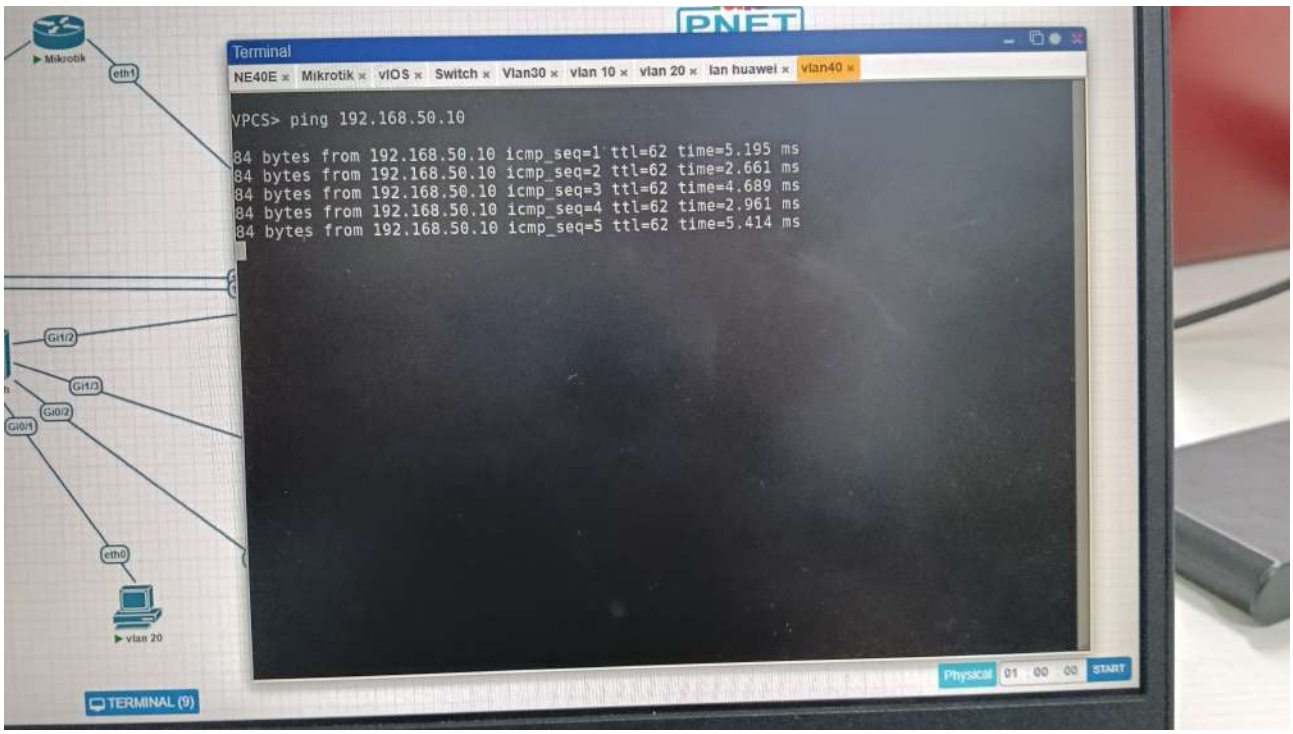

Terminal
NE40E
public_ Routing Table : OSPF
Destinations : 9 Routes : 14
OSPF routing table status : <Active>
Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
OSPF 10 110 D 10.10.10.1 Ethernet1/0/0
192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
TERMINAL (8)

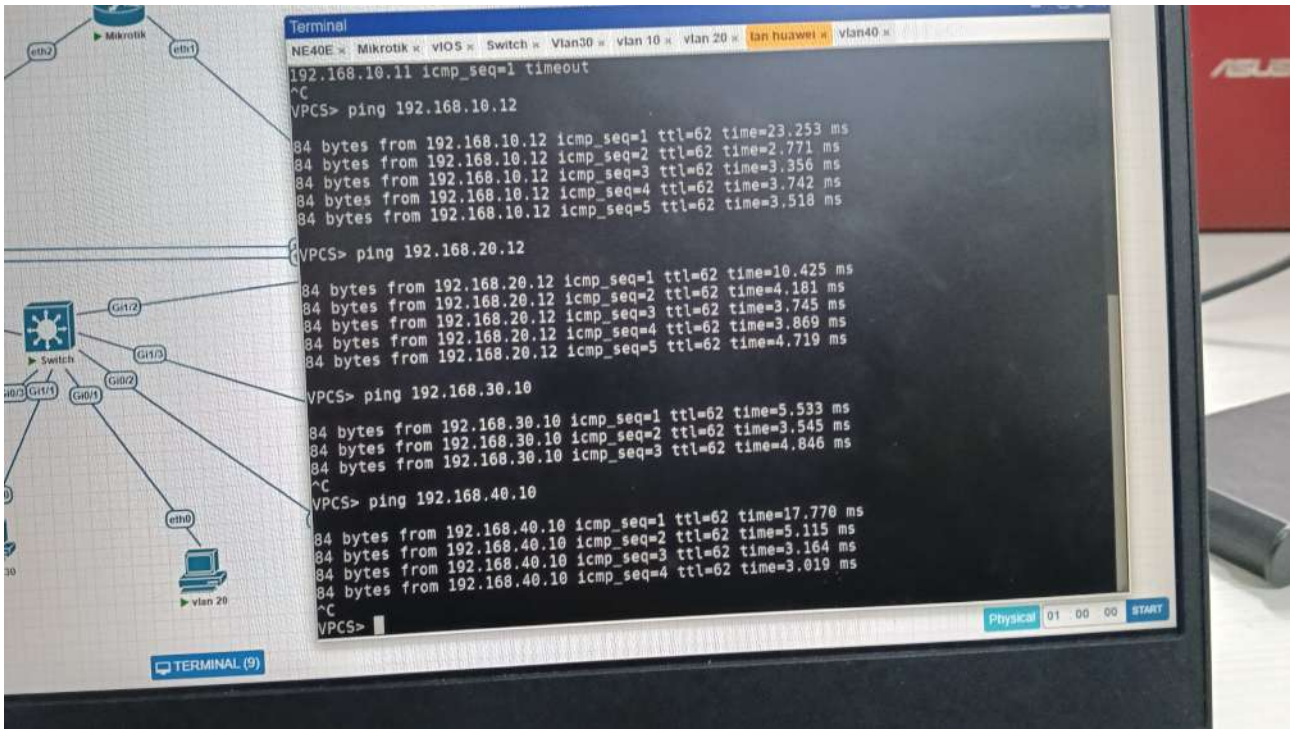

Terminal
NE40E
[admin@MikroTik] > routing ospf neighbor print
[admin@MikroTik] > /ip route print where ospf
[admin@MikroTik] >

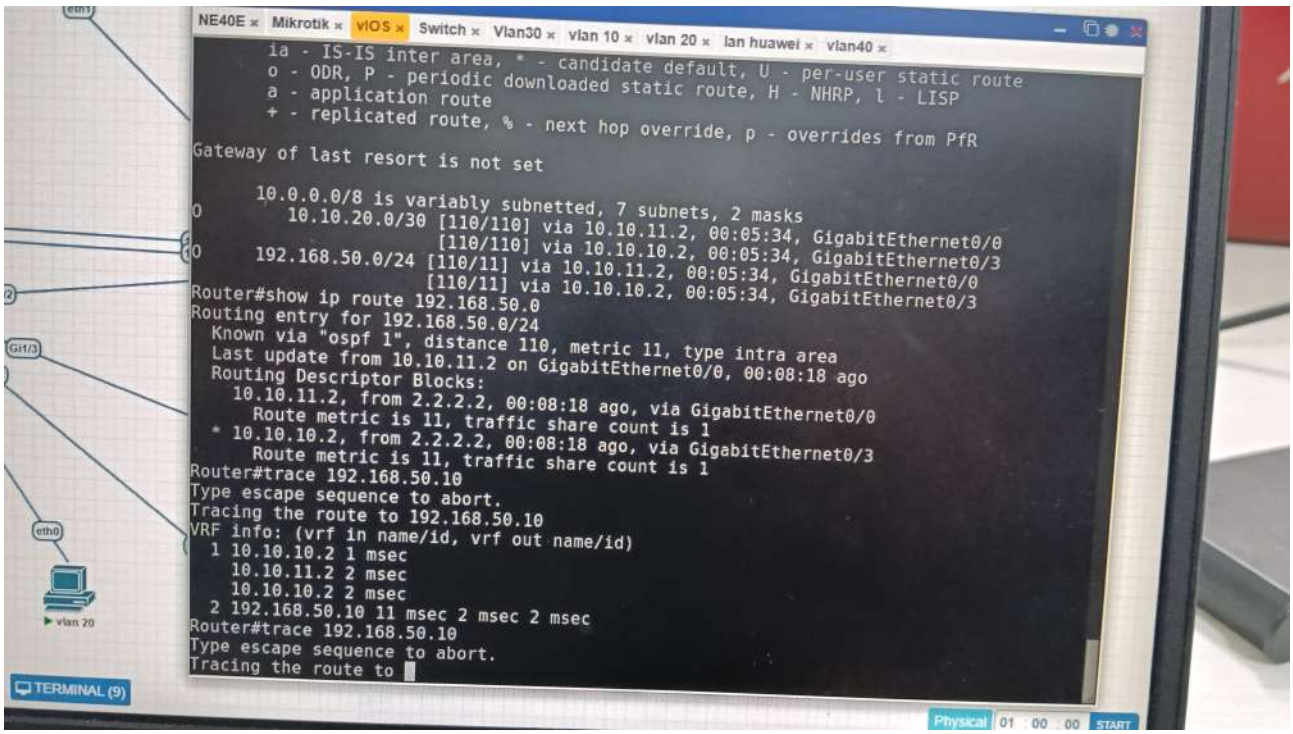

Terminal
NE40E
Mikrotik
vIOS
Switch
Vlan30
vlan 10
vlan 20
lan huawei
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.324 ms
VPCS> ip dhcp
DDORA IP 192.168.10.12/24 GW 192.168.10.1
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.692 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=2.102 ms
^C
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.10.12/24
GATEWAY : 192.168.10.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86275, 86400/43200/75600
MAC : 00:50:79:66:68:96
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.718 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.840 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.410 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=2.405 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.012 ms
VPCS>
TERMINAL (8)
Physical
START

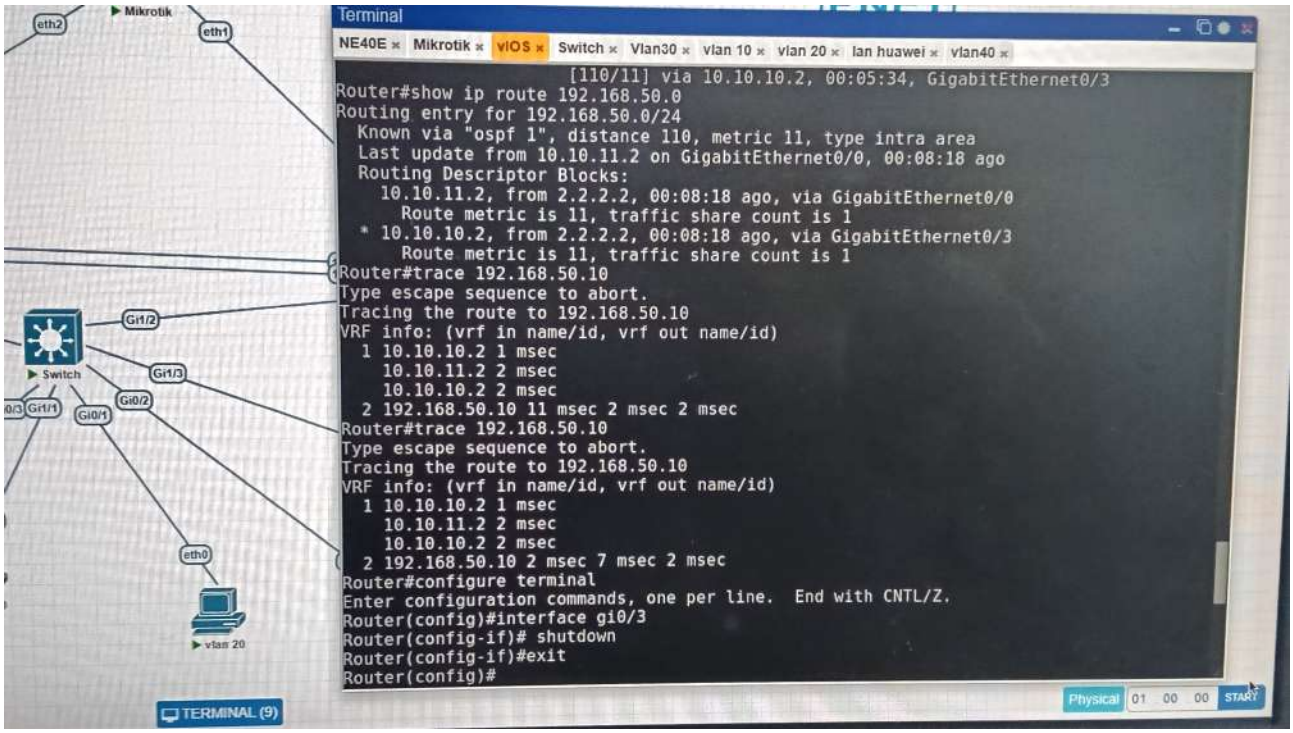

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.878 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.370 ms
^Z84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.267 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.372 ms
^X84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.204 ms
^C
VPCS> ip dhcp
DDORA IP 192.168.20.12/24 GW 192.168.20.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.20.12/24
GATEWAY : 192.168.20.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86365, 86400/43200/75600
MAC : 00:50:79:66:68:95
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.054 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.983 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.588 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.523 ms
VPCS>
Mikrotik
eth2
eth1
Switch
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (8)
Physical
START

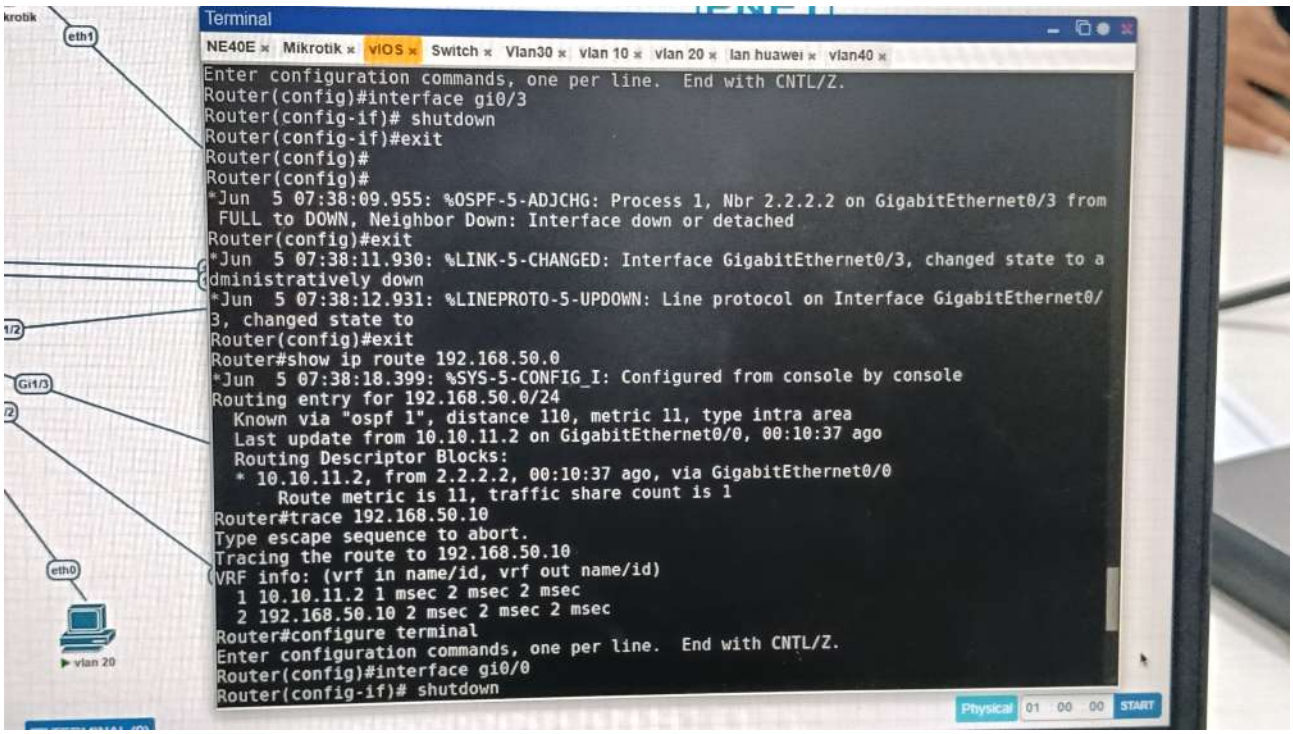

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei ×
VPCS> ipconf
Bad command: "ipconf". Use ? for help.
VPCS> ipconfig
Bad command: "ipconfig". Use ? for help.
VPCS> ip a
No router answered ICMPv6 Router Solicitation
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.30.10/24
GATEWAY : 192.168.30.1
DNS :
MAC : 00:50:79:66:68:94
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.869 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.761 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.235 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.370 ms
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (8)
Physical
START

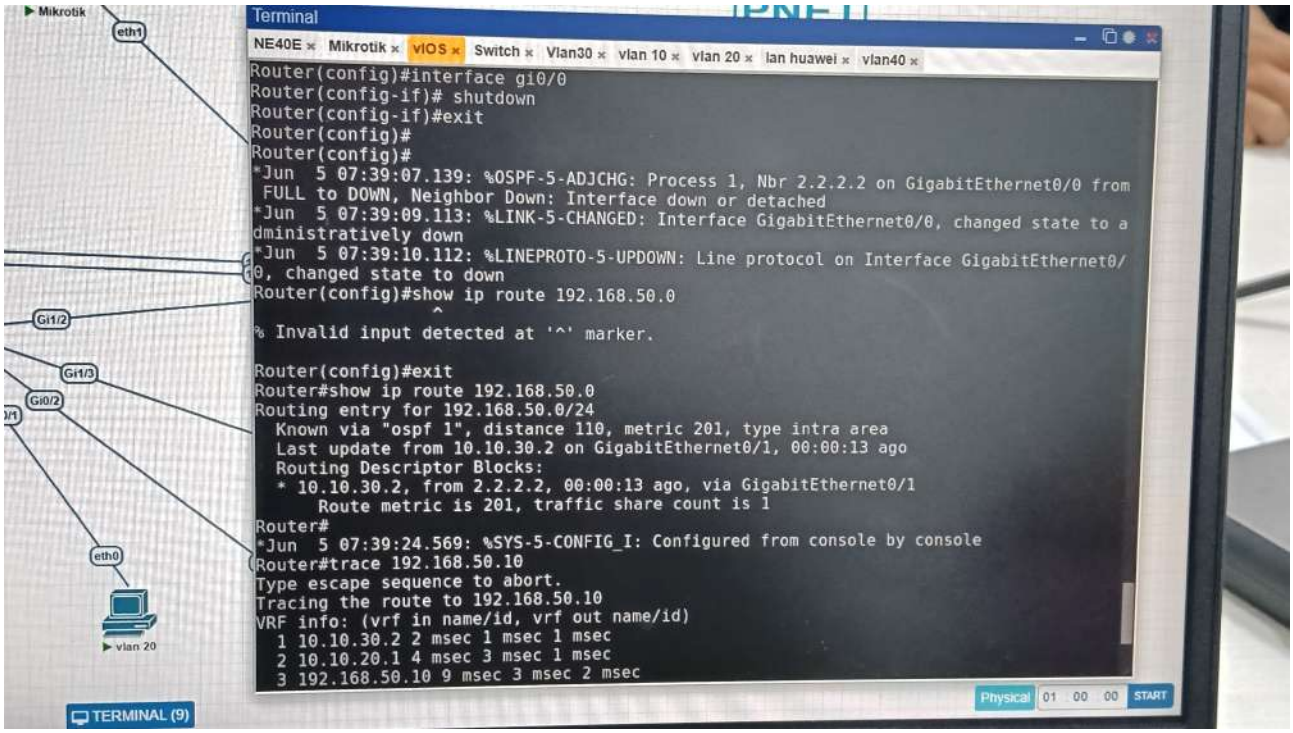

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.195 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.661 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.689 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=2.961 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.414 ms
Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (9)
Physical
START

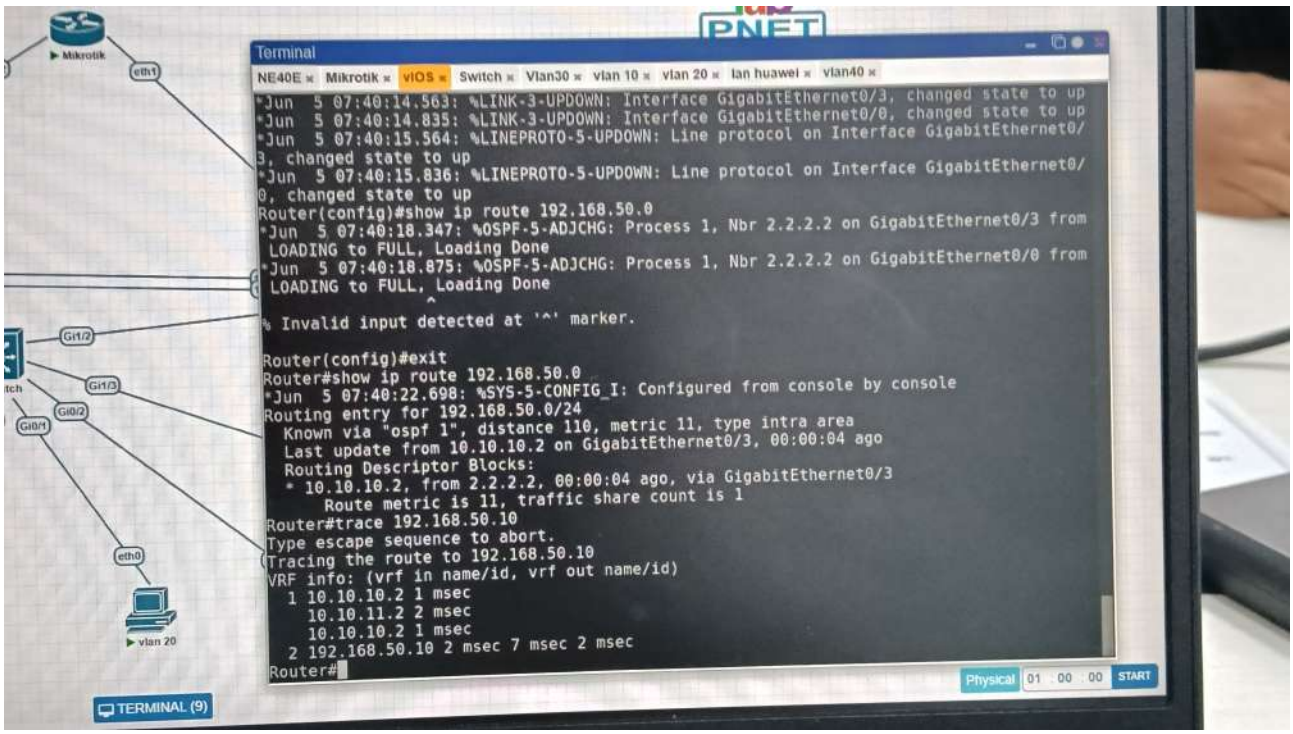

Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
192.168.10.11 icmp_seq=1 timeout
^C
VPCS> ping 192.168.10.12
84 bytes from 192.168.10.12 icmp_seq=1 ttl=62 time=23.253 ms
84 bytes from 192.168.10.12 icmp_seq=2 ttl=62 time=2.771 ms
84 bytes from 192.168.10.12 icmp_seq=3 ttl=62 time=3.356 ms
84 bytes from 192.168.10.12 icmp_seq=4 ttl=62 time=3.742 ms
84 bytes from 192.168.10.12 icmp_seq=5 ttl=62 time=3.518 ms
VPCS> ping 192.168.20.12
84 bytes from 192.168.20.12 icmp_seq=1 ttl=62 time=10.425 ms
84 bytes from 192.168.20.12 icmp_seq=2 ttl=62 time=4.181 ms
84 bytes from 192.168.20.12 icmp_seq=3 ttl=62 time=3.745 ms
84 bytes from 192.168.20.12 icmp_seq=4 ttl=62 time=3.869 ms
84 bytes from 192.168.20.12 icmp_seq=5 ttl=62 time=4.719 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=5.533 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=3.545 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.846 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=17.770 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=5.115 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=3.164 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=3.019 ms
^C
VPCS>
Mikrotik
eth2
eth1
Switch
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (9)
Physical
START


NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O 10.10.20.0/30 [110/110] via 10.10.11.2, 00:05:34, GigabitEthernet0/0
[110/110] via 10.10.10.2, 00:05:34, GigabitEthernet0/3
O 192.168.50.0/24 [110/11] via 10.10.11.2, 00:05:34, GigabitEthernet0/0
[110/11] via 10.10.10.2, 00:05:34, GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:18 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:08:18 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:08:18 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 1 msec
10.10.11.2 2 msec
10.10.10.2 2 msec
2 192.168.50.10 11 msec 2 msec 2 msec
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to
TERMINAL (9)
Physical 01 00 00 START


Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
[110/11] via 10.10.10.2, 00:05:34, GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:08:18 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:08:18 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:08:18 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 1 msec
10.10.11.2 2 msec
10.10.10.2 2 msec
2 192.168.50.10 11 msec 2 msec 2 msec
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 1 msec
10.10.11.2 2 msec
10.10.10.2 2 msec
2 192.168.50.10 2 msec 7 msec 2 msec
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#
Mikrotik
Switch
Gi1/2
Gi1/3
Gi0/2
Gi0/1
Gi1/1
eth0
eth1
eth2
vlan 20
TERMINAL (9)
Physical 01 00 00 START


Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#
Router(config)#
*Jun 5 07:38:09.955: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#exit
*Jun 5 07:38:11.930: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 5 07:38:12.931: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to
Router(config)#exit
Router#show ip route 192.168.50.0
*Jun 5 07:38:18.399: %SYS-5-CONFIG_I: Configured from console by console
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:10:37 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:10:37 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.11.2 1 msec 2 msec 2 msec
2 192.168.50.10 2 msec 2 msec 2 msec
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/0
Router(config-if)# shutdown
eth1
Gi1/3
eth0
vlan 20
Physical
START


Mikrotik
Terminal
NE40E × Mikrotik × vIOS × Switch × Vlan30 × vlan 10 × vlan 20 × lan huawei × vlan40 ×
Router(config)#interface gi0/0
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#
Router(config)#
*Jun 5 07:39:07.139: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jun 5 07:39:09.113: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
*Jun 5 07:39:10.112: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
Router(config)#show ip route 192.168.50.0
^
% Invalid input detected at '^' marker.
Router(config)#exit
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 201, type intra area
Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:13 ago
Routing Descriptor Blocks:
* 10.10.30.2, from 2.2.2.2, 00:00:13 ago, via GigabitEthernet0/1
Route metric is 201, traffic share count is 1
Router#
*Jun 5 07:39:24.569: %SYS-5-CONFIG_I: Configured from console by console
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.30.2 2 msec 1 msec 1 msec
2 10.10.20.1 4 msec 3 msec 1 msec
3 192.168.50.10 9 msec 3 msec 2 msec
eth1
Gi1/2
Gi1/3
Gi0/2
eth0
vlan 20
TERMINAL (9)
Physical
START


Mikrotik
eth1
Gi1/2
Gi1/3
Gi0/2
Gi0/1
eth0
vlan 20
TERMINAL (9)
Terminal
NE40E
Mikrotik
vIOS
Switch
Vlan30
vlan 10
vlan 20
lan huawei
vlan40
*Jun 5 07:40:14.563: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 5 07:40:14.835: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 5 07:40:15.564: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
*Jun 5 07:40:15.836: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#show ip route 192.168.50.0
*Jun 5 07:40:18.347: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun 5 07:40:18.875: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
^
% Invalid input detected at '^' marker.
Router(config)#exit
Router#show ip route 192.168.50.0
*Jun 5 07:40:22.698: %SYS-5-CONFIG_I: Configured from console by console
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.10.2 on GigabitEthernet0/3, 00:00:04 ago
Routing Descriptor Blocks:
* 10.10.10.2, from 2.2.2.2, 00:00:04 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 1 msec
10.10.11.2 2 msec
10.10.10.2 1 msec
2 192.168.50.10 2 msec 7 msec 2 msec
Router#
Physical
01 : 00 : 00
START